

Brazil — LGPD (Lei 13.709/2018)

Discovery and masking of Brazilian personal data: CPF, CNPJ, PIS, voter card and CNS with valid check digits, quasi-identifiers, the sensitive personal data of art. 5, II, financial data and criminal records.

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About this profile set

This profile set finds the personal data in a Brazilian database and masks it, so that a non-production copy — for testing, development, training or analysis — no longer identifies anyone.

Masking names and CPFs alone is not enough: a birth date, a sex and a small municipality already point at one person. The set works in the layers where the risk is:

- **Direct identifiers** — CPF, CNPJ, RG, CNH, voter card, PIS/NIS, work card, passport, foreigner documents, civil registry certificates, professional registrations, names, contact, address, accounts, cards and PIX key, devices, vehicles, property, free text.
- **Quasi-identifiers** — birth date, age, sex, municipality and IBGE code, CEP, neighbourhood, electoral zone, nationality, marital status, occupation, employer, education.
- **Sensitive personal data** — the categories of art. 5, II: racial or ethnic origin, religious conviction, political opinion, membership of a trade union or of an organization of religious, philosophical or political character, sexual life, genetic and biometric data — and, beyond the list, gender identity.
- **Health data** — sensitive under art. 5, II too, in a group of its own: National Health Card, medical records, ICD-10, medicines, clinical notes, tests, disability, occupational, reproductive and mental health.
- **Financial and credit data** — credit score and negative records, debt, income, assets, INSS benefits, social programmes and the Cadastro Único, income bracket, housing.
- **Criminal records and court cases** — records, prison status and case numbers.

On a Masking Engine, the set is a profile set with its classifiers, the domains they vote for, the algorithms those domains name — and the algorithms those lean on — and the lookup files they read. The appendices list each of them with its full configuration.

The assignment threshold is 60%. An unambiguous column name reaches it on its own; an ambiguous one — NOME , CARGO , RESULTADO , PROCESSO — needs the values to agree. A column whose name means nothing (CAMPO1 , OBS) is still found when the values have a recognizable format or vocabulary: a formatted CPF, a CNPJ, a CEP, a plate, a municipality.

Column names are recognized in Portuguese, in the prefixes of Brazilian data models (NR_CPF , NM_CLIENTE , DT_NASCIMENTO , CO_MUNICIPIO_IBGE , CS_RACA , NU_CNS , SG_PARTIDO) and in English.

Before masking, inventory the binary columns (BLOB, images, PDF). Onboarding and proof-of-life processes often store a photo of the RG or CNH and a selfie: the document is still readable in the image, and the selfie is biometric data. Report those columns as residual risk.

The LGPD in brief

The General Personal Data Protection Law — Lei 13.709 of 14 August 2018 — has been in force since September 2020, and its administrative sanctions since 1 August 2021 (art. 65). It was amended by Laws 13.853/2019, 14.010/2020 and 14.460/2022 and by Lei 15.352 of 25 February 2026, which turned the authority into the **Agência Nacional de Proteção de Dados (ANPD)**, a special-regime agency with functional, technical, decision-making, administrative and financial autonomy (art. 55-A). Personal data protection has been a fundamental right since Constitutional Amendment 115/2022 (art. 5, LXXIX, of the Constitution).

What matters for a copy of production data:

- **Personal data** is information related to an identified or identifiable natural person (art. 5, I). The law protects natural persons; the data of a legal entity is outside it — but the CNPJ of an individual micro-entrepreneur leads to a person.
- **Sensitive personal data** is personal data on racial or ethnic origin, religious conviction, political opinion, membership of a trade union or of an organization of religious, philosophical or political character, data concerning health or sexual life, genetic or biometric data, when linked to a natural person (art. 5, II). The regime also covers any processing that **reveals** sensitive data and may cause harm (art. 11, § 1) — an INSS incapacity benefit reveals an illness.
- **Anonymized data** is not personal data, unless the anonymization can be reversed by the controller's own means or with reasonable effort, considering cost, time and the available technology (art. 12). See the next section.
- **Principles.** Purpose, adequacy and necessity (art. 6, I to III): using identified production data to test a system hardly passes the necessity principle when the test works with masked data. Security and prevention (art. 6, VII and VIII), and security measures from the design stage on (art. 46, § 2).
- **Children and adolescents.** Processing in their best interest, with specific consent of a parent for children (art. 14). The Digital Statute of Children and Adolescents (Lei 15.211/2025), in force since 17 March 2026, is regulated and enforced by the ANPD.
- **Security incidents** that may cause relevant risk or harm are reported to the ANPD and to the data subjects (art. 48) within **three business days** (Resolução CD/ANPD 15/2024). Having made the data unintelligible to those without access counts in the assessment of severity (art. 48, § 3): a masked copy that leaks is not the production database leaking.
- **Sanctions.** Warning, a fine of up to 2% of revenue in Brazil capped at R\$ 50 million per infraction, daily fines, publicity, blocking and deletion of the data, suspension of the database or of the processing, and a ban on processing data (art. 52), calculated under Resolução CD/ANPD 4/2023.

Two subjects outside the text of the LGPD are in the set too. **Financial data** is not sensitive under the law, but the operations of financial institutions are under secrecy (Lei Complementar 105/2001), credit databases have a law of their own (Lei 12.414/2011, the positive credit register) and credit protection is a legal basis of its own (art. 7, X). **Criminal records** are not on the list of art. 5, II either, but the Superior Labour Court held that requiring a job candidate's criminal record certificate without a legal or job-related reason gives rise to moral damages (IRR Tema 1): it is data that exposes people to discrimination, and gets the care of sensitive data.

Anonymization, pseudonymization and non-production copies

The LGPD tells two results apart. **Anonymization** is the use of reasonable technical means by which data loses the possibility of being associated, directly or indirectly, with an individual (art. 5, XI); anonymized data is outside the law (art. 12). **Pseudonymization** is processing by which data loses that possibility *except through additional information kept separately by the controller in a controlled and secure environment* (art. 13, § 4); pseudonymized data is still personal data.

The algorithms of this set are deterministic for the key of the Masking Engine: the same CPF masks the same way in every table, which keeps joins working. So for whoever holds the key **and** production, the masked copy can be linked back — it is pseudonymized. For whoever holds only the copy, without the key or production, direct identifiers were replaced and quasi-identifiers generalized, and reversing the process is beyond reasonable effort.

- Keep the engine key out of reach of the users of the masked copy: it is the *additional information kept separately* of art. 13, § 4.

- Do not hand the copy to people who also access production without need.
- Review quasi-identifiers the set does not know. The ANPD recalls that postal code, sex and birth date were enough to identify 87% of the population of the United States in a 2000 study.

In January 2024 the ANPD put a preliminary study and a guide on anonymization and pseudonymization to public consultation; in December 2025 the subject was still on the Regulatory Agenda 2025–2026 (item 9). The study describes techniques — suppression, generalization, noise addition, permutation, masking, substitution, tokenization — and warns that partial masking of the CPF (`***.111.111-**`), common in publications, can be reconstructed when each source hides different parts. This set replaces the whole identifier with another valid one instead of hiding part of it.

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How sensitive data is discovered

A profiling job looks at every column of its rule set and asks each classifier of the profile set how confident it is that the column belongs to its domain. The set uses the four classifier frameworks of the Masking Engine:

- **Column name** (Path) — regular expressions on the column name. Each rule has a match strength: the confidence it gives when it matches.
- **Data type** (Type) — accepts the column type and length, or rules the domain out. It never assigns a domain on its own.
- **Value format** (Regex) — tests a sample of the values against patterns, some with a check digit. The confidence follows the share of values that match.
- **Known values** (List) — looks the sampled values up in lists; for names, word by word.

Evidence from the name and from the values is combined per domain, and the column is assigned to the domain with the highest confidence when it reaches the assignment threshold of the profile set — **60%** here. Values that do not match a format or a list count against their domain by the classifier's reject strength.

Column-name rules are shown as their alternatives. They ignore case and must not be glued to other letters:

`rut` finds `RUT_CLIENTE` and `CLIENTE_RUT` but not `RUTA`. In them, `_?` is an optional underscore, `(a|b)` either form, `s?` an optional letter and `[a-z_]*` any continuation.

Profiling proposes; it does not decide. Review the domains assigned in the inventory before running a masking job, and look for columns the set left unassigned.

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How data is masked

Each domain names one algorithm, which a masking job applies to every column the inventory assigns to the domain. Algorithms can be composed: a **Regex Decompose** splits a value into parts and gives each part its own action; a **String Algorithm Chain** applies algorithms one after another. The tree under each domain shows everything its algorithm uses; Appendix A has every configuration in full.

The algorithms are deterministic for their key: the same value masks the same way in every table, database and job, so joins between masked systems keep working — as long as every engine uses the same keys.

The masked examples in this document were produced with one particular key. On an engine with other keys the masked values differ; their shape does not.

Essential and extended packs

The set can be adopted whole or in part. The **extended pack** is everything this document describes. The **essential pack** is the minimum for the law — identity documents, names, contact details, address and birth date. To adopt only the essential pack, create on the engine just its domains, the classifiers that vote for them and the algorithms and files those use, and put only those classifiers in the profile set.

PACK	DOMAINS	CLASSIFIERS	ALGORITHMS	FILES
Essential	13	27	23	5
Extended	84	148	162	96

The essential pack holds these domains, marked **Essential** in the tables and chapters:

- CPF — BR_L1_CPF
- RG — BR_L1_RG
- CNH — BR_L1_CNH
- Passport — BR_L1_PASSAPORTE
- Given name — BR_L1_NOME
- Surname — BR_L1_SOBRENOME
- Full name — BR_L1_NOME_COMPLETO
- E-mail — BR_L1_EMAIL
- Phone — BR_L1_TELEFONE
- Address — BR_L1_ENDERECO
- Number and complement — BR_L1_ENDERECO_COMPLEMENTO
- Birth date — BR_L2_DATA_NASCIMENTO
- CEP (postal code) — BR_L2_CEP

With fewer domains competing, a column that a domain of the extended pack would claim by its name can be proposed for an essential domain instead — a numeric code, for instance, can look like a telephone number. Review the profiling result.

Domains at a glance

Every domain of the set, how it is found and the algorithm that masks it.

DOMAIN	FOUND BY			ALGORITHM
DIRECT IDENTIFIERS				
CPF ESSENTIAL BR_L1_CPF	COLUMN NAME	DATA TYPE	VALUE FORMAT	BR_CPF
CNPJ BR_L1_CNPJ	COLUMN NAME	DATA TYPE	VALUE FORMAT	BR_CNPJ
RG ESSENTIAL BR_L1_RG	COLUMN NAME	VALUE FORMAT		BR_CM_DIGITOS
CNH ESSENTIAL BR_L1_CNH	COLUMN NAME			BR_CM_ALFANUM
Voter card BR_L1_TITULO_ELEITOR	COLUMN NAME	VALUE FORMAT		BR_TITULO
PIS, PASEP, NIT and NIS BR_L1_PIS_NIS	COLUMN NAME	VALUE FORMAT		BR_PIS
Work card BR_L1_CTPS	COLUMN NAME			BR_CM_ALFANUM
Passport ESSENTIAL BR_L1_PASSAPORTE	COLUMN NAME	VALUE FORMAT		BR_CM_ALFANUM
Foreigner documents BR_L1_DOCUMENTO_ESTRANGEIRO	COLUMN NAME			BR_CM_ALFANUM
Civil registry certificates BR_L1_CERTIDAO	COLUMN NAME	VALUE FORMAT		BR_CM_ALFANUM
Professional registration BR_L1_REGISTRO_PROFISSIONAL	COLUMN NAME			BR_REGISTRO_PROFISSIONAL
Given name ESSENTIAL BR_L1_NOME	COLUMN NAME	KNOWN VALUES		BR_NOMES
Surname ESSENTIAL BR_L1_SOBRENOME	COLUMN NAME	KNOWN VALUES		BR_SOBRENOMES
Full name ESSENTIAL BR_L1_NOME_COMPLETO	COLUMN NAME	KNOWN VALUES		BR_NOME_COMPLETO
E-mail ESSENTIAL BR_L1_EMAIL	COLUMN NAME	DATA TYPE	VALUE FORMAT	BR_EMAIL
Phone ESSENTIAL BR_L1_TELEFONE	COLUMN NAME	VALUE FORMAT		BR_TELEFONE
Address ESSENTIAL BR_L1_ENDERECO	COLUMN NAME	DATA TYPE	VALUE FORMAT	BR_ENDERECO
Number and complement ESSENTIAL BR_L1_ENDERECO_COMPLEMENTO	COLUMN NAME			BR_CM_ALFANUM
Branch and account BR_L1_CONTA_BANCARIA	COLUMN NAME			BR_CM_DIGITOS

DOMAIN	FOUND BY			ALGORITHM
Payment card BR_L1_CARTAO	COLUMN NAME	VALUE FORMAT	BR_CARTAO	
PIX key BR_L1_CHAVE_PIX	COLUMN NAME	VALUE FORMAT	BR_CHAVE_PIX	
IP address BR_L1_IP	COLUMN NAME	VALUE FORMAT	BR_IP	
Device identifier BR_L1_DISPOSITIVO	COLUMN NAME	VALUE FORMAT	BR_CM_HEX	
Licence plate BR_L1_PLACA	COLUMN NAME	VALUE FORMAT	BR_PLACA	
RENAVAM BR_L1_RENAVAM	COLUMN NAME		BR_RENAVAM	
Chassis BR_L1_CHASSI	COLUMN NAME	VALUE FORMAT	BR_CM_ALFANUM	
Real estate BR_L1_IMOVEL	COLUMN NAME		BR_CM_ALFANUM	
Contract and enrolment numbers BR_L1_CONTRATO	COLUMN NAME		BR_CM_ALFANUM	
User and social media BR_L1_USUARIO	COLUMN NAME		BR_CM_ALFANUM	
Credentials BR_L1_CREDENCIAL	COLUMN NAME		BR_REDIGIR	
Geolocation BR_L1_GEOLOCALIZACAO	COLUMN NAME	VALUE FORMAT	BR_COORDENADA	
Free text BR_L1_TEXTO_LIVRE	COLUMN NAME	DATA TYPE	VALUE FORMAT	BR_TEXTO_LIVRE
QUASI-IDENTIFIERS				
Birth date ESSENTIAL BR_L2_DATA_NASCIMENTO	COLUMN NAME	DATA TYPE		BR_DATA_NASCIMENTO
Birth year BR_L2_ANO_NASCIMENTO	COLUMN NAME	DATA TYPE		BR_ANO
Age BR_L2_IDADE	COLUMN NAME	DATA TYPE		BR_IDADE
Personal event dates BR_L2_DATA_EVENTO	COLUMN NAME	DATA TYPE		BR_DATA_EVENTO
Sex BR_L2_SEX0	COLUMN NAME	KNOWN VALUES		BR_SEX0
Municipality BR_L2_MUNICIPIO	COLUMN NAME	KNOWN VALUES		BR_MUNICIPIO
Municipality code (IBGE) BR_L2_CODIGO_MUNICIPIO	COLUMN NAME	VALUE FORMAT		BR_CODIGO_MUNICIPIO
CEP (postal code) ESSENTIAL BR_L2_CEP	COLUMN NAME	VALUE FORMAT		BR_CEP
Neighbourhood BR_L2_BAIRRO	COLUMN NAME			BR_BAIRRO

DOMAIN	FOUND BY		ALGORITHM
Electoral zone and section BR_L2_ZONA_SECAO_ELEITORAL	COLUMN NAME		BR_CM_DIGITOS
Nationality BR_L2_NACIONALIDADE	COLUMN NAME	KNOWN VALUES	BR_NACIONALIDADE
Marital status BR_L2_ESTADO_CIVIL	COLUMN NAME	KNOWN VALUES	BR_ESTADO_CIVIL
Occupation BR_L2_OCUPACAO	COLUMN NAME		BR_OCUPACAO_OU_CBO
Employer BR_L2_EMPREGADOR	COLUMN NAME		BR_EMPREGADOR
Education BR_L2_ESCOLARIDADE	COLUMN NAME	KNOWN VALUES	BR_ESCOLARIDADE
School or educational institution BR_L2_ESCOLA	COLUMN NAME		BR_ESCOLA
Dependants and children BR_L2_DEPENDENTES	COLUMN NAME	DATA TYPE	BR_CONTAGEM_LIMITADA
SENSITIVE PERSONAL DATA			
Colour or race BR_L3_RACA_COR	COLUMN NAME	KNOWN VALUES	BR_RACA_COR
Indigenous people and traditional community BR_L3_ETNIA	COLUMN NAME	KNOWN VALUES	BR_ETNIA
Religion BR_L3_RELIGIAO	COLUMN NAME	KNOWN VALUES	BR_RELIGIAO
Convictions and religious, philosophical or political organizations BR_L3_CONVICCAO_ORGANIZACAO	COLUMN NAME		BR_CATEGORIA_SUPRIMIDA
Political opinion BR_L3_OPINIAO_POLITICA	COLUMN NAME	KNOWN VALUES	BR_OPINIAO_POLITICA
Party membership BR_L3_FILIACAO_PARTIDARIA	COLUMN NAME	KNOWN VALUES	BR_PARTIDO
Trade union membership BR_L3_FILIACAO_SINDICAL	COLUMN NAME	KNOWN VALUES	BR_CATEGORIA_SUPRIMIDA
Genetic data BR_L3_GENETICO	COLUMN NAME		BR_REDIGIR
Biometric data BR_L3_BIOMETRICO	COLUMN NAME	DATA TYPE	BR_REDIGIR
Sexual orientation BR_L3_ORIENTACAO_SEXUAL	COLUMN NAME	KNOWN VALUES	BR_ORIENTACAO_SEXUAL
Sexual life BR_L3_VIDA_SEXUAL	COLUMN NAME		BR_CATEGORIA_SUPRIMIDA
Gender identity BR_L3_IDENTIDADE_GENERO	COLUMN NAME	KNOWN VALUES	BR_IDENTIDADE_GENERO
HEALTH DATA			
National Health Card BR_SAUDE_CNS	COLUMN NAME	VALUE FORMAT	BR_CNS

DOMAIN	FOUND BY		ALGORITHM
Health identifiers BR_SAUDE_IDENTIFICADOR	COLUMN NAME		BR_CM_ALFANUM
Diagnosis and ICD BR_SAUDE_DIAGNOSTICO	COLUMN NAME	VALUE FORMAT	BR_DIAGNOSTICO
Medicines BR_SAUDE_MEDICAMENTO	COLUMN NAME	KNOWN VALUES	BR_MEDICAMENTO
Clinical text BR_SAUDE_TEXTO_CLINICO	COLUMN NAME	DATA TYPE	BR_TEXTO_LIVRE
Test results BR_SAUDE_RESULTADO_EXAME	COLUMN NAME		BR_CM_ALFANUM
Blood group BR_SAUDE_TIPO_SANGUINEO	COLUMN NAME	KNOWN VALUES	BR_TIPO_SANGUINEO
Disability BR_SAUDE_DEFICIENCIA	COLUMN NAME	KNOWN VALUES	BR_DEFICIENCIA
Occupational health BR_SAUDE_OCUPACIONAL	COLUMN NAME	KNOWN VALUES	BR_SAUDE_OCUPACIONAL
Sexual and reproductive health BR_SAUDE_REPRODUTIVA	COLUMN NAME		BR_CATEGORIA_SUPRIMIDA
Mental health BR_SAUDE_MENTAL	COLUMN NAME		BR_CATEGORIA_SUPRIMIDA
FINANCIAL AND CREDIT DATA			
Credit and negative records BR_FIN_CREDITO	COLUMN NAME		BR_CREDITO
Debt BR_FIN_DIVIDA	COLUMN NAME	DATA TYPE	BR_VALOR_DIVIDA
Income and salary BR_FIN_RENDA	COLUMN NAME	DATA TYPE	BR_RENDA
Assets BR_FIN_PATRIMONIO	COLUMN NAME	DATA TYPE	BR_PATRIMONIO
Benefit and pension amount BR_FIN_VALOR_BENEFICIO	COLUMN NAME	DATA TYPE	BR_VALOR_BENEFICIO
INSS benefit type BR_FIN_BENEFICIO_INSS	COLUMN NAME	KNOWN VALUES	BR_BENEFICIO_INSS
Benefit number BR_FIN_NUMERO_BENEFICIO	COLUMN NAME	VALUE FORMAT	BR_CM_DIGITOS
Social programmes BR_FIN_PROGRAMA_SOCIAL	COLUMN NAME	KNOWN VALUES	BR_PROGRAMA_SOCIAL
Income bracket and social class BR_FIN_FAIXA_RENDA	COLUMN NAME	KNOWN VALUES	BR_FAIXA_RENDA
Housing BR_FIN_MORADIA	COLUMN NAME	KNOWN VALUES	BR_MORADIA
CRIMINAL RECORDS AND COURT CASES			
Criminal records BR_PENAL_ANTECEDENTES	COLUMN NAME		BR_ANTECEDENTES

DOMAIN	FOUND BY		ALGORITHM
Case number	COLUMN NAME	VALUE FORMAT	BR_PROCESSO
BR_PENAL_PROCESSO			

Direct identifiers

Data that identifies a person on its own. The CPF is the key of almost every Brazilian record — and, under Lei 14.534/2023, it is enough to identify a citizen in public services and is the number of the new National Identity Card. It is masked the same way everywhere and keeps valid check digits, because applications validate them. The same goes for CNPJ, PIS, voter card and RENAVAM.

8.1 CPF

BR_L1_CPF

- ESSENTIAL
- COLUMN NAME
- DATA TYPE
- VALUE FORMAT
- REGEX DECOMPOSE

WHAT IT IS

The taxpayer number of the Cadastro de Pessoas Físicas of the Receita Federal: nine digits and two check digits, 529.982.247-25. The ninth digit is the tax region that issued the number. Under Lei 14.534/2023 it is enough to identify a citizen in public-service databases, and the National Identity Card uses the CPF as its number. It appears formatted, as bare digits or, in a numeric column, without its leading zeros.

CPF_CNPJ and NR_DOCUMENTO columns hold CPFs and CNPJs together, and fall here too.

HOW IT IS FOUND

COLUMN NAME BR_L1_CPF – Path

- 90%
cpf[a-z0-9]* c_?p_?f n(r|u|um|o|ro|umero)_?cpf cpf_?(cnpj|cgc) cadastro_?(de_?)? pessoas_?fisicas? documento_?fiscal (cliente|cli|clie|pessoa|pes|func|funcionario|colaborador|servidor|empregado|segurado|beneficiario|benef|titular|dependente|paciente|aluno|contribuinte|responsavel|resp|socio|motorista|condutor|tomador|devedor|avalista|fiador|consumidor|comprador|eleitor|candidato|usuario)_?cpf
- 55%
tax_?id taxpayer_?id documento_?(cliente|pessoa) doc_?(cliente|pessoa) nr_documento num_documento numero_documento n(r|u)_?doc

DATA TYPE BR_L1_CPF – Type

Accepts text (length 3 or more) or number (length 3 or more); any other type rules the domain out.

VALUE FORMAT BR_L1_CPF – Regex

PATTERN	STRENGTH	NOTES
\d{3}\.\d{3}\.\d{3}-\d{2}		100%
\d{11}		50%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

The first nine digits are mapped digit by digit — the tax region changes too — and the two check digits are recomputed: modulus 11, weights 10 to 2 for the first and 11 to 2 for the second, a remainder of 0 or 1 giving 0. Every masked CPF is valid, the format is kept and the mapping is one-to-one: a CPF used as a key does not collide.

A number of up to ten digits is taken for a CPF that lost its zeros and comes out with eleven. A value shaped like a CNPJ is masked as a CNPJ.

BR_CPF REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS.

BR_CPF_VALIDO CHECK DIGIT

Masks the digits with `BR_CPF_DV1` and recomputes a modulus 11 check digit with weights `11 10 9 8 7 6 5 4 3 2`.

`BR_CPF_DV1` **CHECK DIGIT**

Masks the digits with `BR_CM_DIGITOS` and recomputes a modulus 11 check digit with weights `10 9 8 7 6 5 4 3 2`.

`BR_CM_DIGITOS` **CHARACTER MAPPING**

Maps each character to another of its group (1 groups), keeping length and format.

`BR_CPF_VALIDO_ZEROS` **CHECK DIGIT**

Masks the digits with `BR_CPF_DV1` and recomputes a modulus 11 check digit with weights `11 10 9 8 7 6 5 4 3 2`.

`BR_CNPJ_VALIDO` **CHECK DIGIT**

Masks the digits with `BR_CNPJ_DV1` and recomputes a modulus 11 check digit with weights `6 5 4 3 2 9 8 7 6 5 4 3 2`.

`BR_CNPJ_DV1` **CHECK DIGIT**

Masks the digits with `BR_CNPJ_RAIZ` and recomputes a modulus 11 check digit with weights `5 4 3 2 9 8 7 6 5 4 3 2`.

`BR_CNPJ_RAIZ` **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: keep.

`BR_CM_DIGITOS_LETRAS` **CHARACTER MAPPING**

Maps each character to another of its group (2 groups), keeping length and format.

`BR_CNPJ_VALIDO_ZEROS` **CHECK DIGIT**

Masks the digits with `BR_CNPJ_DV1` and recomputes a modulus 11 check digit with weights `6 5 4 3 2 9 8 7 6 5 4 3 2`.

EXAMPLES

VALUE		MASKED
529.982.247-25	→	410.719.628-33
52998224725	→	41071962833
11.222.333/0001-81	→	21.137.125/0001-38

8.2

CNPJ

BR_L1_CNPJ

COLUMN NAME	DATA TYPE	VALUE FORMAT	REGEX DECOMPOSE
-------------	-----------	--------------	-----------------

WHAT IT IS

The number of the National Register of Legal Entities: an eight-character root, a four-character establishment order (`0001` is the head office) and two check digits, `11.222.333/0001-81`. Since July 2026 root and order may hold letters (Instrução Normativa RFB 2.229/2024): `12.ABC.345/01DE-35`.

Legal-entity data is outside the LGPD, but the CNPJ of an individual micro-entrepreneur (MEI) or sole trader leads to a person: the public CNPJ lookup shows their name.

HOW IT IS FOUND

COLUMN NAME	BR_L1_CNPJ – Path
90%	<code>cnpj[a-z0-9]* c_?n_?p_?j cgc n(r u um o ro umero)?cnpj cadastro_?nacional_?(da_?)? pessoa_?juridica (empresa emp fornecedor forn cliente cli empregador estabelecimento estab prestador tomador emitente emissor destinatario dest transportadora operadora mei matriz filial)?cnpj</code>
DATA TYPE	BR_L1_CNPJ – Type
	Accepts text (length 3 or more) or number (length 3 or more); any other type rules the domain out.

PATTERN	STRENGTH	NOTES
<code>[0-9A-Z]{2}\.[0-9A-Z]{3}\.[0-9A-Z]{3}/[0-9A-Z]{4}-\d{2}</code>	100%	
<code>[0-9A-Z]{12}\d{2}</code>	60%	

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

The root is mapped — digits to digits, letters to letters — and the establishment order stays, so the branches of a company remain branches of the same masked company. Both check digits are recomputed with weights 5 to 2 and 9 to 2, modulus 11, each character worth its ASCII code minus 48 — the rule of the alphanumeric CNPJ, which for digits is the usual one. Every masked CNPJ is valid, and the mapping is one-to-one.

BR_CNPJ REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

BR_CNPJ_VALIDO CHECK DIGIT

Masks the digits with BR_CNPJ_DV1 and recomputes a modulus 11 check digit with weights 6 5 4 3 2 9 8 7 6 5 4 3 2 .

BR_CNPJ_DV1 CHECK DIGIT

Masks the digits with BR_CNPJ_RAIZ and recomputes a modulus 11 check digit with weights 5 4 3 2 9 8 7 6 5 4 3 2 .

BR_CNPJ_RAIZ REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

BR_CM_DIGITOS_LETRAS CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

BR_CPF_VALIDO CHECK DIGIT

Masks the digits with BR_CPF_DV1 and recomputes a modulus 11 check digit with weights 11 10 9 8 7 6 5 4 3 2 .

BR_CPF_DV1 CHECK DIGIT

Masks the digits with BR_CM_DIGITOS and recomputes a modulus 11 check digit with weights 10 9 8 7 6 5 4 3 2 .

BR_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

BR_CNPJ_VALIDO_ZEROS CHECK DIGIT

Masks the digits with BR_CNPJ_DV1 and recomputes a modulus 11 check digit with weights 6 5 4 3 2 9 8 7 6 5 4 3 2 .

EXAMPLES

VALUE		MASKED
11.222.333/0001-81	→	21.137.125/0001-38
12.ABC.345/01DE-35	→	82.XPZ.728/01DE-28

8.3 RG

BR_L1_RG

ESSENTIAL COLUMN NAME VALUE FORMAT CHARACTER MAPPING

WHAT IT IS

The identity card number issued by each state's identification institute, with a format and check digit of its own — in São Paulo 12.345.678-X , with X as a digit. It appears with the issuing body (SSP/SP) or the state (MG-12.345.678).

HOW IT IS FOUND

COLUMN NAME BR_L1_RG - Path

90% rg[a-z0-9_]* r_?g n(r|u|um|o|ro|umero)?rg registro_?geral carteira_?(de_)?identidade
cedula_?(de_)?identidade identidade (except when followed by *genero*, *visual*, *de_?genero*) doc_?
identidade documento_?(de_)?identidade ci_?(numero|num|nr) rg_
(cliente|pessoa|func|titular|dependente|mae|pai|responsavel|segurado)
(cliente|pessoa|func|titular|responsavel)?rg identity_?(card|document|no|number) id_
card

VALUE FORMAT BR_L1_RG - Regex

PATTERN	STRENGTH	NOTES
\d{1,2}\.\d{3}\.\d{3}-[\dxX]		70%
[A-Z]{2}-?\d{1,2}\.?\d{3}\.?\d{3}		60%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Digits are mapped; letters, issuing body and separators stay. The check digit is not recomputed: each state has its own rule.

BR_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
12.345.678-X	→	46.916.122-X
MG-12.345.678	→	MG-46.916.122

8.4 CNH

BR_L1_CNH

ESSENTIAL **COLUMN NAME** **CHARACTER MAPPING**

WHAT IT IS

The registration number of the national driving licence: eleven digits, the last two check digits. RENACH and learner permit columns fall here too.

HOW IT IS FOUND

COLUMN NAME BR_L1_CNH - Path

90% cnh[a-z0-9_]* c_?n_?h carteira_?(nacional_)?(de_)?habilitacao habilitacao(_?
(numero|num|nr|registro))? registro_?cnh n(r|u|um|o|ro|umero)?(registro_)?cnh renach
pgu permissao_?(para_)?dirigir drivers_?licen[cs]e(_?(no|num|number))?

HOW IT IS MASKED

Digit by digit, keeping the length. The second check digit of the CNH depends on whether the first went past 9, a rule Check Digit cannot follow; it is not recomputed.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

02650306461

→ 66573572582

8.5 Voter card

BR_L1_TITULO_ELEITOR

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

The voter registration number: eight sequential digits, two for the state (01 São Paulo, 02 Minas Gerais ... 28 abroad) and two check digits, 1023 8501 0671 .

HOW IT IS FOUND

COLUMN NAME

BR_L1_TITULO_ELEITOR – Path

90%

titulo?(de_)?eleitor(_?(numero|num|nr))? titulo?eleitoral tit?eleitor
n(r|u|um|o|ro|umero)?titulo(_?eleitor)? inscricao?eleitoral voter?
(id|card|registration)

VALUE FORMAT

BR_L1_TITULO_ELEITOR – Regex

PATTERN

STRENGTH

NOTES

\d{4}\s?\d{4}\s?\d{4}

40%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

The state stays, the sequence is mapped and both check digits are recomputed, including the São Paulo and Minas Gerais rule in which a remainder of 0 gives 1. Every masked voter card is valid, and a number that lost its zeros in a numeric column gets them back before the computation. See *Check digits computed by the set*.

BR_TITULO

REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

BR_TITULO_VALIDO STRING ALGORITHM CHAIN

Applies BR_TITULO_CORPO → BR_TITULO_DV , in this order.

BR_TITULO_CORPO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

BR_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

BR_TITULO_DV STRING ALGORITHM CHAIN

Applies BR_TITULO_DV_P1 → BR_TITULO_DV_P2 → BR_TITULO_DV_P3 → BR_TITULO_DV_P4 → BR_TITULO_DV_P5 , in this order.

BR_TITULO_DV_P1 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

BR_TITULO_DV_T1 DATA CLEANSING

Replaces each value found in br-titulo-dv-1.txt (100 pairs) by its counterpart; values not in the file stay as they are.

BR_TITULO_DV_P2 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

BR_TITULO_DV_T2 DATA CLEANSING

Replaces each value found in br-titulo-dv-2.txt (1,100 pairs) by its counterpart; values not in the file stay as they are.

BR_TITULO_DV_P3 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

BR_TITULO_DV_T3	DATA CLEANSING
Replaces each value found in <code>br-titulo-dv-3.txt</code> (1,100 pairs) by its counterpart; values not in the file stay as they are.	
BR_TITULO_DV_P4	REGEX DECOMPOSE
Matches one pattern and gives each part of the value its own action; when none matches: keep.	
BR_TITULO_DV_T4	DATA CLEANSING
Replaces each value found in <code>br-titulo-dv-4.txt</code> (1,100 pairs) by its counterpart; values not in the file stay as they are.	
BR_TITULO_DV_P5	REGEX DECOMPOSE
Matches one pattern and gives each part of the value its own action; when none matches: keep.	
BR_TITULO_DV_T5	DATA CLEANSING
Replaces each value found in <code>br-titulo-dv-5.txt</code> (1,100 pairs) by its counterpart; values not in the file stay as they are.	
BR_TITULO_VALIDO_1_ZERO	STRING ALGORITHM CHAIN
Applies <code>BR_TITULO_1_ZERO</code> → <code>BR_TITULO_VALIDO</code> , in this order.	
BR_TITULO_1_ZERO	REGEX DECOMPOSE
Matches one pattern and gives each part of the value its own action; when none matches: keep.	
BR_TITULO_1_ZERO_TABELA	DATA CLEANSING
Replaces each value found in <code>br-zeros-1.txt</code> (10 pairs) by its counterpart; values not in the file stay as they are.	
BR_TITULO_VALIDO_2_ZEROS	STRING ALGORITHM CHAIN
Applies <code>BR_TITULO_2_ZEROS</code> → <code>BR_TITULO_VALIDO</code> , in this order.	
BR_TITULO_2_ZEROS	REGEX DECOMPOSE
Matches one pattern and gives each part of the value its own action; when none matches: keep.	
BR_TITULO_2_ZEROS_TABELA	DATA CLEANSING
Replaces each value found in <code>br-zeros-2.txt</code> (10 pairs) by its counterpart; values not in the file stay as they are.	

EXAMPLES

VALUE		MASKED
102385010671	→	330614820671
004356870906	→	512854140990

8.6 PIS, PASEP, NIT and NIS

BR_L1_PIS_NIS

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The worker's number in PIS (Caixa) or PASEP (Banco do Brasil), which is also the INSS NIT and the NIS of the Cadastro Único and Bolsa Família: ten digits and a check digit, `120.56412.54-7` .

HOW IT IS FOUND

COLUMN NAME BR_L1_PIS_NIS - Path

90% `pis[a-z0-9_]* pasep pis_?pasep nit nis n(r|u|um|o|ro|umero)?(pis|nit|nis|pasep) nis?(titular|responsavel|beneficiario) numero_?identificacao_?(do_)?trabalhador inscricao_?(inss|previdencia) nit_?(inss|contribuinte)`

VALUE FORMAT BR_L1_PIS_NIS – Regex

PATTERN	STRENGTH	NOTES
<code>\d{3}\.\d{5}\.\d{2}-\d</code>		90%
<code>\d{11}</code>		50%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

The first digit stays, the other nine are mapped and the check digit is recomputed with weights 3 2 9 8 7 6 5 4 3 2, modulus 11. Every masked PIS is valid and the mapping is one-to-one. The same algorithm masks the PIS inside the National Health Card.

BR_PIS REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

BR_PIS_VALID0 CHECK DIGIT

Masks the digits with BR_PIS_CORPO and recomputes a modulus 11 check digit with weights 3 2 9 8 7 6 5 4 3 2 .

BR_PIS_CORPO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

BR_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

BR_PIS_VALID0_ZERO5 CHECK DIGIT

Masks the digits with BR_PIS_CORPO and recomputes a modulus 11 check digit with weights 3 2 9 8 7 6 5 4 3 2 .

EXAMPLES

VALUE	MASKED
120.56412.54-7	→ 113.81104.27-9

8.7 Work card

BR_L1_CTPS

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

Number and series of the paper Carteira de Trabalho e Previdência Social. The digital card uses the CPF.

HOW IT IS FOUND

COLUMN NAME BR_L1_CTPS – Path

90% `ctps[a-z0-9]*` `carteira_?(de_)?trabalho` `serie_?ctps` `n(r|u|um|o|ro|umero)_?ctps` `work_?`
`permit`

HOW IT IS MASKED

Character by character, keeping length and format.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
1234567/001	→ 7162897/064

8.8 Passport

BR_L1_PASSAPORTE

ESSENTIAL

COLUMN NAME

VALUE FORMAT

CHARACTER MAPPING

WHAT IT IS

Brazilian passport numbers — two letters and six digits, FZ123456 — and foreign ones.

HOW IT IS FOUND

COLUMN NAME BR_L1_PASSAPORTE – Path

90% `passaporte[a-z0-9_]*` `passport[a-z0-9_]*` `n(r|u|um|o|ro|umero)?passaporte` `pasaporte`

VALUE FORMAT BR_L1_PASSAPORTE – Regex

PATTERN

STRENGTH

NOTES

`[A-Z]{2}\d{6}`

40%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Character by character, keeping length and format.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

FZ123456

→

RS124760

8.9 Foreigner documents

BR_L1_DOCUMENTO_ESTRANGEIRO

COLUMN NAME

CHARACTER MAPPING

WHAT IT IS

The National Migration Registry number (RNM, formerly RNE) of the migration card, the refugee protocol and the visa number.

HOW IT IS FOUND

COLUMN NAME BR_L1_DOCUMENTO_ESTRANGEIRO – Path

85% `rnm` `crnm` `rne` `registro?nacional?(migratorio|estrangeiro)` `carteira?(de_)?registro?(nacional_)?migratorio` `protocolo?(de_)?refugio` `refugiado?(protocolo|numero)` `visto(_?(numero|num|nr))?` `n(r|u|um|o|ro|umero)?visto` `visa?(no|number)` `dpri`

HOW IT IS MASKED

Character by character, keeping length and format.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

V123456-7

→

P220047-2

8.10 Civil registry certificates

BR_L1_CERTIDAO

COLUMN NAME VALUE FORMAT CHARACTER MAPPING

WHAT IT IS

The registration number (matrícula) of birth, marriage and death certificates: 32 digits with the registry office, collection, year, book type, book, page, entry and two check digits. Also the Declaration of Live Birth and the Death Declaration.

HOW IT IS FOUND

COLUMN NAME BR_L1_CERTIDAO - Path

85% certidao_(de_)?(nascimento|casamento|obito)(_?(numero|num|nr|matricula))? matricula_(da_)?certidao certidao?matricula termo?(de_)?nascimento registro?(de_)?nascimento livro?folha?termo dnv declaracao?(de_)?nascido?vivo declaracao?(de_)?obito n(r|u)?(dnv|do)

VALUE FORMAT BR_L1_CERTIDAO - Regex

PATTERN	STRENGTH	NOTES
<code>\d{6}\s?\d{2}\s?\d{2}\s?\d{4}\s?\d\s?\d{5}\s?\d{3}\s?\d{7}\s?\d{2}</code>	80%	

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Digit by digit, keeping the length. The check digit is not recomputed.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
10453901552013100012021000012321	→ 82431996553020383891084411419170

8.11 Professional registration

BR_L1_REGISTRO_PROFISSIONAL

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

The registration number with a professional council — CRM, OAB, CREA, COREN, CRO, CRP, CRC and others —, CRM-SP 123456 . The councils publish these registers with the professional's name.

HOW IT IS FOUND

COLUMN NAME BR_L1_REGISTRO_PROFISSIONAL - Path

85% crm cro crp crf crn crefito coren crea cau crc oab cress crmv cfm conselho?(de_)?classe(_?(numero|num|nr))? registro?(profissional|conselho|de_?classe) n(r|u|um|o|ro|umero)?(crm|oab|crea|coren|cro|crp|conselho|registro?profissional) inscricao?(oab|crm|crea|conselho)

HOW IT IS MASKED

Digits are mapped; council, state and separators stay.

BR_REGISTRO_PROFISSIONAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

BR_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE

CRM-SP 123456

MASKED

→ CRM-SP 904292

8.12 Given name

BR_L1_NOME

ESSENTIAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The given name, one or more: Maria , João Pedro , Maria de Fátima . The detection list holds the most frequent names of the 2022 census.

HOW IT IS FOUND

COLUMN NAME

BR_L1_NOME - Path

85%

primeiro_?nome prenome nome_?proprio nomes_?(de_)?batismo nome_?1 first_?names?
given_?names? fname middle_?names?

50%

nome (except when followed by list E1) nm (except when followed by list E2) name (except when followed by list E3)

KNOWN VALUES

BR_L1_NOME - List

LIST

VALUES

STRENGTH

FIRST VALUES

br-detectar-nomes.txt

1,046

70%

Maria, Ana, Francisca, Júlia, Julia, Antônia

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Word by word, each name replaced with one from the 2022 census list. de , da , dos and e stay in place. More than three words are masked character by character.

BR_NOMES REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: BR_CM_ALFANUM .

BR_PALAVRA_NOME REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: BR_NOME .

BR_NOME NAME

Replaces the name with one of the 895 names in br-nomes.txt .

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

Maria de Fátima

JOÃO PEDRO

MASKED

→ Joelson de Camila

→ ALEXANDRE VANESSA

8.13 Surname

BR_L1_SOBRENOME

ESSENTIAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The surnames, usually two — the mother's and the father's —, with particles: `da Silva Santos` . Also the maiden name. The detection list holds the most frequent surnames of the 2022 census.

HOW IT IS FOUND

COLUMN NAME

BR_L1_SOBRENOME – Path

85%

`sobrenomes?` `ultimo_?nome` `nome_?(de_)?familia` `last_?names?` `surnames?` `family_?names?`
`lname` `apelido_?familia` `sobrenome_?(pai|mae|paterno|materno|solteira|casada)` `nome_?(de_)?`
`solteira` `sobrenome_?(de_)?casada`

KNOWN VALUES

BR_L1_SOBRENOME – List

LIST

VALUES

STRENGTH

FIRST VALUES

br-detector-sobrenomes.txt

636

70%

Silva, Santos, Oliveira, Souza, Pereira, Ferreira

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Word by word, each surname replaced with one from the list. Particles and agnomens — Júnior, Filho, Neto, Sobrinho — stay in place.

BR_SOBRENOMES REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM` .

BR_PALAVRA_SOBRENOME REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SOBRENOME` .

BR_SOBRENOME NAME

Replaces the name with one of the 535 names in `br-sobrenomes.txt` .

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

da Silva Santos

→

da Brás Fortunato

8.14 Full name

BR_L1_NOME_COMPLETO

ESSENTIAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Given names and surnames in one column: `José Carlos da Silva Júnior` . Also the mother's and father's names — asked in almost every Brazilian register and used as a security question —, the guardian's, the spouse's, and the **social name**, which reveals a trans person's gender identity.

A column called just `NOME` may hold given names or full names: the values decide between the two domains.

HOW IT IS FOUND

COLUMN NAME

BR_L1_NOME_COMPLETO – Path

90%

nome_?(completo|compl|civil|social|registro|guerra|abreviado) nomecompleto
(nm|no|nome|ds)_?
(cliente|cli|paciente|pac|funcionario|func|empregado|colaborador|servidor|segurado|beneficiario|benef|titular|dependente|dep|mae|pai|responsavel|resp|conjugue|companheiro|contato|avalista|fiador|devedor|tomador|requerente|solicitante|testemunha|representante|procurador|proprietario|locatario|locador|inquilino|vitima|reu|autor|acusado|investigado|doador|herdeiro|pessoa|pes|motorista|condutor|cidadao|eleitor|contribuinte|aluno|estudante|crianca|medico|enfermeiro|professor|socio|tutor|curador|pagador|favorecido|destinatario|remetente|candidato|associado|membro|cooperado|consumidor|comprador|corretor|interessado|parte|advogado|usuario|atendente|vendedor|gerente|supervisor|avo|filho|filha|irmao|parente|customer|patient|employee|member|holder|mother|father|spouse|contact|person|student|driver|owner)
(cliente|cli|paciente|pac|funcionario|func|empregado|colaborador|servidor|segurado|beneficiario|benef|titular|dependente|dep|mae|pai|responsavel|resp|conjugue|companheiro|contato|avalista|fiador|devedor|tomador|requerente|solicitante|testemunha|representante|procurador|proprietario|locatario|locador|inquilino|vitima|reu|autor|acusado|investigado|doador|herdeiro|pessoa|pes|motorista|condutor|cidadao|eleitor|contribuinte|aluno|estudante|crianca|medico|enfermeiro|professor|socio|tutor|curador|pagador|favorecido|destinatario|remetente|candidato|associado|membro|cooperado|consumidor|comprador|corretor|interessado|parte|advogado|usuario|atendente|vendedor|gerente|supervisor|avo|filho|filha|irmao|parente|customer|patient|employee|member|holder|mother|father|spouse|contact|person|student|driver|owner)?(nome|nm) nome_?(d[aoe]_)?
(cliente|cli|paciente|pac|funcionario|func|empregado|colaborador|servidor|segurado|beneficiario|benef|titular|dependente|dep|mae|pai|responsavel|resp|conjugue|companheiro|contato|avalista|fiador|devedor|tomador|requerente|solicitante|testemunha|representante|procurador|proprietario|locatario|locador|inquilino|vitima|reu|autor|acusado|investigado|doador|herdeiro|pessoa|pes|motorista|condutor|cidadao|eleitor|contribuinte|aluno|estudante|crianca|medico|enfermeiro|professor|socio|tutor|curador|pagador|favorecido|destinatario|remetente|candidato|associado|membro|cooperado|consumidor|comprador|corretor|interessado|parte|advogado|usuario|atendente|vendedor|gerente|supervisor|avo|filho|filha|irmao|parente|customer|patient|employee|member|holder|mother|father|spouse|contact|person|student|driver|owner) filiacao(_?[12])? full_?name person_?name

60%

mae pai genitora? conjugue responsavel_?legal representante_?legal procurador
avalista fiador testemunha beneficiario herdeiro curador tutor titular

50%

nome (except when followed by list E1) nm (except when followed by list E2) name (except when followed by list E3)

KNOWN VALUES

BR_L1_NOME_COMPLETO – List

LIST	VALUES	STRENGTH	FIRST VALUES
br-detectar-nomes.txt	1,046	60%	Maria, Ana, Francisca, Júlia, Julia, Antônia
br-detectar-sobrenomes.txt	636	60%	Silva, Santos, Oliveira, Souza, Pereira, Ferreira

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

In Brazilian order: two words are a given name and a surname; three, one given name and two surnames (Maria da Silva too); from four on, two given names and the rest surnames. Particles and agnomens stay. SURNAMES, GIVEN NAMES with a comma, as in lists, is recognized too. The result is a plausible Brazilian name, never the original.

BR_NOME_COMPLETO REGEX DECOMPOSE

Tries 7 patterns in order and gives each part of the value its own action; when none matches: BR_CM_ALFANUM .

BR_SOBRENOMES REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: BR_CM_ALFANUM .

BR_PALAVRA_SOBRENOME REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: BR_SOBRENOME .

BR_SOBRENOME NAME

Replaces the name with one of the 535 names in br-sobrenomes.txt .

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

BR_NOMES REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: BR_CM_ALFANUM .

BR_PALAVRA_NOME REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: BR_NOME .

BR_NOME NAME

Replaces the name with one of the 895 names in br-nomes.txt .

EXAMPLES

VALUE		MASKED
José Carlos da Silva Júnior	→	Renato Neusa da Brás Júnior
Ana Paula Souza Lima	→	Cristiane Gerson Cipriano Tobias
SILVA, MARIA	→	BRÁS, JOELSON

8.15 E-mail

BR_L1_EMAIL

ESSENTIAL COLUMN NAME DATA TYPE VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Personal and corporate e-mail addresses.

HOW IT IS FOUND

COLUMN NAME BR_L1_EMAIL - Path

85% e_?mail[a-z0-9_]* mail[a-z0-9_]* correio_?eletronico endereco_?eletronico (ds|de|tx)? email

DATA TYPE BR_L1_EMAIL - Type

Accepts text (length 5 or more); any other type rules the domain out.

VALUE FORMAT BR_L1_EMAIL - Regex

PATTERN	STRENGTH	NOTES
[A-Za-z0-9._%+'\-\-]+@[A-Za-z0-9.\-\-]+\.[A-Za-z]{2,24}	100%	

A value that matches none counts 20% against the domain.

HOW IT IS MASKED

User and domain are mapped character by character, and the top-level domain — .com , .com.br — becomes .test , reserved for testing: a masked address never delivers a message.

BR_EMAIL REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: BR_CM_ALFANUM .

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
maria.silva@gmail.com	→ cazua.zotro@pgoeb.test

8.16 Phone

BR_L1_TELEPHONE

ESSENTIAL COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Numbers with the country code (55), the area code (DDD) and eight digits for a landline (starting 2 to 5) or nine for a mobile (starting 9): (11) 98765-4321 , +55 21 3456-7890 , 11987654321 .

HOW IT IS FOUND

COLUMN NAME BR_L1_TELEPHONE - Path

85% tel telephone[a-z0-9]* fone[a-z0-9]* celular[a-z0-9]* cel movel whats?app zap
contato?(tel|telephone|celular) (nr|nu|num|numero|ds|tx)?(tel|telephone|fone|celular|cel)
ddd?(tel|telephone|celular|fone) fax ramal phone[a-z0-9]* mobile[a-z0-9]*

VALUE FORMAT BR_L1_TELEPHONE - Regex

PATTERN	STRENGTH	NOTES
(\+?55\s?)?(?0?[1-9]\d)?\s?9\d{4}[\s-]?d{4}	100%	
(\+?55\s?)?(?0?[1-9]\d)?\s?[2-5]\d{3}[\s-]?d{4}	80%	
9\d{4}-\d{4}	80%	

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Country code, DDD and the first digit stay — the DDD covers a whole region, and the first digit tells a mobile from a landline —; the other digits are mapped, keeping length and separators.

BR_TELEPHONE REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

BR_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
(11) 98765-4321	→ (11) 93134-0400
+55 21 3456-7890	→ +55 21 3047-9607

8.17 Address

BR_L1_ENDERECO

ESSENTIAL

COLUMN NAME

DATA TYPE

VALUE FORMAT

SECURE LOOKUP

WHAT IT IS

The address line: street type and name, number, complement, sometimes the neighbourhood. Av. Paulista, 1578, 12º andar – Bela Vista .

HOW IT IS FOUND

COLUMN NAME

BR_L1_ENDERECO – Path

80%

endereco[a-z0-9]* (except when preceded by mac , ip , e_?mail , web , url , eletronico ; except when followed by list E4) end_?(residencial|comercial|cobranca|entrega|correspondencia|completo) (except when preceded by mac , ip , e_?mail , web , url) logradouro[a-z0-9]* lograd (ds|no|nm|tx)? (logradouro|endereco|rua) rua residencia domicilio[a-z0-9]* address[a-z0-9]* (except when preceded by mac , ip , e_?mail , web , url ; except when followed by list E5) street(_?(address|name))?

DATA TYPE

BR_L1_ENDERECO – Type

Accepts text (length 6 or more); any other type rules the domain out.

VALUE FORMAT	BR_L1_ENDERECO – Regex	
PATTERN		STRENGTH NOTES
(?i)(rua r\. avenida av\.? travessa tv\.? alameda al\. estrada est\. rodovia rod\. praca praça pc\. largo viela servidao quadra qd\.? setor conjunto beco ladeira vila)\s+.		80%
(?i).*,\s*\d+.*		30%

A value that matches none counts 20% against the domain.

HOW IT IS MASKED

The whole line is replaced with a fictitious Brazilian address — a common street name, number, apartment or house, neighbourhood. Municipality, state and CEP in their own columns are handled by their domains.

BR_ENDERECO

SECURE LOOKUP

Replaces the value with one of the 4,000 values in br-enderecos.txt , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Av. Paulista, 1578, 12º andar – Bela Vista	→	Rua do Comércio, 980

8.18 Number and complement

BR_L1_ENDERECO_COMPLEMENTO

ESSENTIAL

COLUMN NAME

CHARACTER MAPPING

WHAT IT IS

House number, complement, apartment, block and lot in their own columns.

HOW IT IS FOUND

COLUMN NAME BR_L1_ENDERECO_COMPLEMENTO – Path

70% complemento(_?endereco)? compl n(r|u|um|o|ro|umero)?
(endereco|casa|residencia|imovel|logradouro) numero?(end|lograd) num?casa apto
apartamento bloco lote quadra casa?(numero|num|nr) house?number apartment unit?
number

HOW IT IS MASKED

Character by character, keeping length and format.

BR_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

Apto 42 Bloco B

→

Offi 87 Nsaso W

8.19 Branch and account

BR_L1_CONTA_BANCARIA

COLUMN NAME

CHARACTER MAPPING

WHAT IT IS

Bank branch and account, with each bank's check digits (sometimes X). The bank code (COMPE, ISPB) is not personal data and has no domain.

HOW IT IS FOUND

COLUMN NAME BR_L1_CONTA_BANCARIA – Path

85% conta?(corrente|poupanca|bancaria|banco|salario|pagamento|deposito|credito|debito) cc?
(numero|num|nr) n(r|u|um|o|ro|umero)?conta num?cta nr?cta cta?(corrente|bancaria)
agencia(_?(conta|bancaria|numero|num|nr|dv))? ag?(conta|bancaria) dv?(conta|agencia)
digito?(conta|agencia) iban account?(no|num|number) bank?account operacao?conta

HOW IT IS MASKED

Digits are mapped; X and separators stay. Check digits are not recomputed: each bank has its own rule.

BR_CM_DIGITOS **CHARACTER MAPPING**

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

12345–X

→

31564–X

8.20 Payment card

BR_L1_CARTAO

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

Credit and debit card numbers, recognized by the Luhn algorithm.

HOW IT IS FOUND

COLUMN NAME BR_L1_CARTAO – Path

85% `cartao(_?(credito|debito|numero|num|nr|pan))?` `n(r|u|um|o|ro|umero)?cartao` `pan` `card_?`
`(no|num|number)` `credit_?card` `cc_?number`

VALUE FORMAT BR_L1_CARTAO – Regex

PATTERN	STRENGTH	NOTES
---------	----------	-------

<code>\d{13,19}</code>	90%	LUHN check digit; after removing <code>[\s\~]</code>
------------------------	-----	--

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Payment Card: another number that passes the Luhn check. Only values shaped like a card reach it.

BR_CARTAO **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: keep.

BR_CARTAO_LUHN **PAYMENT CARD**

Replaces the card number with another that passes the Luhn check.

EXAMPLES

VALUE	MASKED
5162 3062 1937 8829	→ 0913 6733 7729 0366

8.21 PIX key

BR_L1_CHAVE_PIX

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

A PIX key is a CPF, a CNPJ, an e-mail, a phone number in the `+5511987654321` format or a random key (a UUID).

HOW IT IS FOUND

COLUMN NAME BR_L1_CHAVE_PIX – Path

90% `chave_?pix` `pix(_?(chave|key|destino|recebedor|pagador|favorecido))?` `chave_?`
`(aleatoria|evp)` `evp` `pix_?key`

VALUE FORMAT BR_L1_CHAVE_PIX – Regex

PATTERN	STRENGTH	NOTES
<code>[0-9a-f]{8}-[0-9a-f]{4}-4[0-9a-f]{3}-[89ab][0-9a-f]{3}-[0-9a-f]{12}</code>	30%	

A value that matches none counts 20% against the domain.

HOW IT IS MASKED

Each kind of key is masked as its own domain: CPF and CNPJ with valid check digits, e-mail with `.test`, phone keeping the DDD, random key keeping its hexadecimal format. A CPF key and the same person's CPF still agree.

BR_CHAVE_PIX **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: **BR_CM_ALFANUM**.

BR_EMAIL **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: **BR_CM_ALFANUM**.

BR_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

BR_TELEFONE REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

BR_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

BR_CPF REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

BR_CPF_VALIDO CHECK DIGIT

Masks the digits with BR_CPF_DV1 and recomputes a modulus 11 check digit with weights 11 10 9 8 7 6 5 4 3 2 .

BR_CPF_DV1 CHECK DIGIT

Masks the digits with BR_CM_DIGITOS and recomputes a modulus 11 check digit with weights 10 9 8 7 6 5 4 3 2 .

BR_CPF_VALIDO_ZEROS CHECK DIGIT

Masks the digits with BR_CPF_DV1 and recomputes a modulus 11 check digit with weights 11 10 9 8 7 6 5 4 3 2 .

BR_CNPJ_VALIDO CHECK DIGIT

Masks the digits with BR_CNPJ_DV1 and recomputes a modulus 11 check digit with weights 6 5 4 3 2 9 8 7 6 5 4 3 2 .

BR_CNPJ_DV1 CHECK DIGIT

Masks the digits with BR_CNPJ_RAIZ and recomputes a modulus 11 check digit with weights 5 4 3 2 9 8 7 6 5 4 3 2 .

BR_CNPJ_RAIZ REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

BR_CM_DIGITOS_LETRAS CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

BR_CNPJ_VALIDO_ZEROS CHECK DIGIT

Masks the digits with BR_CNPJ_DV1 and recomputes a modulus 11 check digit with weights 6 5 4 3 2 9 8 7 6 5 4 3 2 .

BR_CNPJ REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

BR_CM_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
529.982.247-25	→	410.719.628-33
123e4567-e89b-12d3-a456-426614174000	→	951b0298-d33a-44c2-b413-745894460718

8.22 IP address

BR_L1_IP

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

IPv4 and IPv6 addresses. The Marco Civil da Internet requires providers to keep connection and access logs, with the IP, which lead to the subscriber.

HOW IT IS FOUND

COLUMN NAME

BR_L1_IP - Path

85%

ip ip?(address|addr|origem|destino|cliente|usuario|acesso|login|remoto|publico)
endereco_?ip ipv[46] remote_?addr(ess)? client_?ip host_?ip

VALUE FORMAT BR_L1_IP - Regex

PATTERN	STRENGTH	NOTES
((25[0-5] 2[0-4]\d 1?\d?\d)\.){3}(25[0-5] 2[0-4]\d 1?\d?\d)	100%	
[0-9A-Fa-f]{1,4}(:[0-9A-Fa-f]{0,4}){2,7}	90%	

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Each IPv4 octet is mapped to 1-254; an IPv6 is mapped keeping its hexadecimal characters.

BR_IP REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: BR_CM_HEX .

BR_OCTETO NUMERIC MAPPING

Maps the number to another between 1 and 254.

BR_CM_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
200.147.67.142	→ 146.137.44.231

8.23 Device identifier

BR_L1_DISPOSITIVO

COLUMN NAME VALUE FORMAT CHARACTER MAPPING

WHAT IT IS

IMEI, IMSI, ICCID, MAC address, advertising identifiers, cookies and sessions.

HOW IT IS FOUND

COLUMN NAME BR_L1_DISPOSITIVO - Path

80% imei imsi iccid mac mac_address endereco_mac device_(id|uuid|serial) id_(dispositivo|aparelho) dispositivo_id advertising_id idfa gaid cookie(_id?) session_id id_sessao serial_(aparelho|celular)

VALUE FORMAT BR_L1_DISPOSITIVO - Regex

PATTERN	STRENGTH	NOTES
([0-9A-Fa-f]{2}[:-]){5}[0-9A-Fa-f]{2}	100%	
\d{15}	80%	LUHN check digit

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Character by character, with hexadecimal digits kept hexadecimal: a MAC stays a MAC.

BR_CM_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
3c:22:fb:9a:10:e4	→ 5e:83:bb:3c:31:f9

8.24 Licence plate

BR_L1_PLACA

- COLUMN NAME
- VALUE FORMAT
- REGEX DECOMPOSE

WHAT IT IS

Plates in the old pattern, three letters and four digits (ABC-1234), and in the Mercosur pattern, with a letter in the fifth position (BRA2E19).

HOW IT IS FOUND

COLUMN NAME

BR_L1_PLACA – Path

85%

placa(_?(veiculo|carro|moto|numero))?n(r|u|um|o|ro|umero)?placaplaca_?mercosul
license_?plateplate(_?(no|number))?

VALUE FORMAT

BR_L1_PLACA – Regex

PATTERN	STRENGTH	NOTES
[A-Z]{3}\d[A-Z]\d{2}		90%
[A-Z]{3}-?\d{4}		80%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Letters become letters and digits become digits, position by position, with the hyphen: the plate pattern stays.

BR_PLACA REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: BR_CM_ALFANUM .

- BR_CM_DIGITOS_LETRAS CHARACTER MAPPING
Maps each character to another of its group (2 groups), keeping length and format.
- BR_CM_DIGITOS CHARACTER MAPPING
Maps each character to another of its group (1 groups), keeping length and format.
- BR_CM_ALFANUM CHARACTER MAPPING
Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
BRA2E19	→	WHM3Z25
ABC-1234	→	VLD-6328

8.25 RENAVAM

BR_L1_RENAVAM

- COLUMN NAME
- REGEX DECOMPOSE

WHAT IT IS

The code of the National Motor Vehicle Registry: eleven digits, the last a check digit. Old nine-digit numbers are padded with zeros.

HOW IT IS FOUND

COLUMN NAME

BR_L1_RENAVAM – Path

90%

renavamcod_?renavamcodigo_?renavamn(r|u|um|o|ro|umero)?renavamregistro_?nacional_?
(de_)?veiculos

HOW IT IS MASKED

The first ten digits are mapped and the check digit recomputed with weights 3 2 9 8 7 6 5 4 3 2, modulus 11. Every masked RENAVAM is valid, with eleven digits.

BR_RENAVAM REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

BR_RENAVAM_VALIDO CHECK DIGIT

Masks the digits with BR_CM_DIGITOS and recomputes a modulus 11 check digit with weights 3 2 9 8 7 6 5 4 3 2 .

BR_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
00639884962	→	05594949808

8.26 Chassis

BR_L1_CHASSI

COLUMN NAME VALUE FORMAT CHARACTER MAPPING

WHAT IT IS

The 17-character vehicle identification number (VIN) engraved on the chassis, and the engine number.

HOW IT IS FOUND

COLUMN NAME BR_L1_CHASSI – Path

85% chassis(_?(numero|num|nr))? n(r|u|um|o|ro|umero)?chassi vin numero?motor n(r|u)?motor

VALUE FORMAT BR_L1_CHASSI – Regex

PATTERN	STRENGTH	NOTES
[A-HJ-NPR-Z0-9]{17}		70%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Character by character, keeping length and format.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
9BWZZZ377VT004251	→	3KVRMX584XM353424

8.27 Real estate

BR_L1_IMOVEL

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

The property registration number (matrícula), the municipal property tax (IPTU) number and the rural property codes — CAR, CCIR, NIRF, SNCR. Each leads to the owner through public registers.

HOW IT IS FOUND

COLUMN NAME BR_L1_IMOVEL – Path

85% matricula_?(do_)?imovel matricula_?(cartorio|cri|registro) inscricao_?
(imobiliaria|municipal_?imovel|iptu|cadastral) iptu(_?(numero|num|nr|inscricao))?
indicacao_?fiscal codigo_?(car|ccir|nirf|snrc|incra) n(r|u)_?(car|ccir|nirf) car_?
(imovel|rural) ccir nirf cib snrc

HOW IT IS MASKED

Character by character, keeping length and format.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

123.456.7890-1

→

061.766.2692-2

8.28 Contract and enrolment numbers

BR_L1_CONTRATO

COLUMN NAME

CHARACTER MAPPING

WHAT IT IS

Contract, policy, proposal, protocol, consumer unit and installation numbers, employee, civil servant (SIAPE) or student enrolment, member cards and customer codes.

HOW IT IS FOUND

COLUMN NAME BR_L1_CONTRATO – Path

70% n(r|u|um|o|ro|umero)_?
(contrato|apolice|proposta|matricula|carteirinha|protocolo|instalacao|uc|unidade_?
consumidora|cliente|pedido_?cliente) contrato_?(numero|num|nr) matricula(_?
(funcionario|func|empregado|servidor|aluno|associado))? *(except when followed by imovel, certidao, cartorio, cri)* matr cod_?
(cliente|funcionario|servidor|aluno|associado|segurado|beneficiario) codigo_?
(cliente|funcionario|servidor|aluno|associado) id_?
(cliente|funcionario|servidor|aluno|associado) unidade_?consumidora instalacao apolice
carteirinha ra_?aluno registro_?academico siape

HOW IT IS MASKED

Character by character, keeping length and format. The mapping is deterministic, so joins keep working.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

CT-123456

→

NG-557411

8.29 User and social media

BR_L1_USUARIO

COLUMN NAME **CHARACTER MAPPING**

WHAT IT IS

Login, nickname and social media profiles.

HOW IT IS FOUND

COLUMN NAME BR_L1_USUARIO – Path

70% usuario user_?name username login(_?(nome|usuario))? apelido nick(_?name)? alias
perfil?(instagram|facebook|twitter|tiktok|linkedin) instagram facebook twitter tiktok
linkedin rede_?social arroba handle

HOW IT IS MASKED

Character by character, keeping length and format.

BR_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

@maria_silva

→

@cazua_zotro

8.30 Credentials

BR_L1_CREDENCIAL

COLUMN NAME **REGEX DECOMPOSE**

WHAT IT IS

Passwords and their hashes, PINs, security codes, CVV, tokens, API keys and secret answers.

HOW IT IS FOUND

COLUMN NAME BR_L1_CREDENCIAL – Path

90% senha password passwd pwd pass hash_?senha senha_?hash pin codigo_
(seguranca|verificacao|acesso) cvv cvc token(_?(acesso|refresh|api))? access_?token
refresh_?token api_?key chave_?api segredo secret otp frase_?secreta pergunta_
secreta resposta_?secreta

HOW IT IS MASKED

Every character becomes X. A credential has no use in a test copy, and a hash masked character by character could still be attacked.

BR_REDIGIR **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE

MASKED

Senha#2024

→

XXXXXXXXXX

8.31 Geolocation

BR_L1_GEOLOCALIZACAO

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Latitude and longitude, together or in separate columns, from apps, deliveries and visits.

HOW IT IS FOUND

COLUMN NAME BR_L1_GEOLOCALIZACAO – Path

85% lat latitude lon lng longitude coordenadas? coord_?[xy] geo_
(lat|lon|lng|loc|localizacao|posicao) geolocalizacao gps(_?
(localizacao|posicao|coordenadas))? localizacao?gps

50% long

VALUE FORMAT BR_L1_GEOLOCALIZACAO – Regex

PATTERN	STRENGTH	NOTES
-(?([0-2]?[d 3[0-3]])\.\d{3,}\s*,\s*-(3[4-9] [4-6]\d 7[0-4])\.\d{3,})	90%	
-(3[4-9] [4-6]\d 7[0-4])\.\d{4,}	60%	
-(?([0-2]?[d 3[0-3]])\.\d{4,})	30%	

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

The integer part and the first decimal stay — about 11 km — and the other decimals are mapped: the place is generalized, not moved far away.

BR_COORDENADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
-23.561414,-46.655881	→ -23.522235,-46.651532

8.32 Free text

BR_L1_TEXTO_LIVRE

COLUMN NAME DATA TYPE VALUE FORMAT FREE TEXT REDACTION

WHAT IT IS

Remarks, service histories, complaints, justifications and messages where anything gets written — CPF, phone and name included.

HOW IT IS FOUND

COLUMN NAME BR_L1_TEXTO_LIVRE – Path

60% observac(ao|oes) obs comentarios? notas? anotac(ao|oes) descricao_
(ocorrencia|atendimento|chamado|reclamacao|solicitacao|caso|fato) historico_
(atendimento|chamado|contato) relato parecer justificativa mensagem (except when preceded by list
E6; except when followed by list E7) texto(_?livre)? detalhes resumo_(atendimento|caso) reclamacao
ocorrencia_(descricao|texto) remarks? comments? notes? free?text memo

DATA TYPE BR_L1_TEXTO_LIVRE – Type

Accepts text (length 30 or more); any other type rules the domain out.

VALUE FORMAT BR_L1_TEXTO_LIVRE – Regex

PATTERN	STRENGTH	NOTES
<code>\d{3}\.\d{3}\.\d{3}-\d{2}</code>	70%	anywhere in the value
<code>[\w.+\\-]+@[\\w\\-]+\\. [A-Za-z]{2,}</code>	70%	anywhere in the value
<code>\\(?:\\d{2})?\\s?9\\d{4}-\\d{4}</code>	60%	anywhere in the value

A value that matches none does not count against the domain.

HOW IT IS MASKED

Free Text Redaction: CPF and CNPJ (formatted or not), e-mail, phones, cards, CEP, plates and IPs become [DADO] ; given names and surnames of the 2022 census list become [NOME] . The rest of the text stays.

BR_TEXTO_LIVRE FREE TEXT REDACTION

Replaces the text matching 12 patterns with [DADO] , and the 3,302 terms of `br-texto-nomes.txt` with [NOME] .

EXAMPLES

VALUE	MASKED
Cliente Souza, CPF 529.982.247-25, e-mail m.souza@gmail.com, tel (11) 98765-4321.	→ Cliente [NOME], CPF [DADO], e-mail [DADO], tel (11) [DADO].

Quasi-identifiers

Data that identifies no one on its own, but does in combination. Brazil has 5,570 municipalities, and 3,860 of them had fewer than 20,000 inhabitants in the 2022 census; the smallest, Serra da Saudade (MG), had 833. A birth date, a sex and one of those municipalities point at a handful of people. These attributes are generalized or shifted rather than replaced, and the data keeps its analytical value.

9.1 Birth date

BR_L2_DATA_NASCIMENTO

ESSENTIAL **COLUMN NAME** **DATA TYPE** **DATE SHIFT**

WHAT IT IS

The birth date, as a date, timestamp or text.

HOW IT IS FOUND

COLUMN NAME BR_L2_DATA_NASCIMENTO – Path

90% dt_?nasc(imento)? data_?(de_)?nasc(imento)? dat_?nasc d_?nasc nascimento dn dta_? nasc dtnasc datanasc(imento)? aniversario dob date_?of_?birth birth_?date birthday

DATA TYPE BR_L2_DATA_NASCIMENTO – Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the domain out.

HOW IT IS MASKED

Shifted up to 60 days either way. Ages and school years change for few people; the date stays plausible and the same birth shifts the same way in every table.

BR_DATA_NASCIMENTO **DATE SHIFT**

Moves the date by -60 to 60 days.

EXAMPLES

VALUE		MASKED
1985-07-20	→	1985-08-05T00:00

9.2 Birth year

BR_L2_ANO_NASCIMENTO

COLUMN NAME **DATA TYPE** **REGEX DECOMPOSE**

WHAT IT IS

The birth year in a column of its own.

HOW IT IS FOUND

COLUMN NAME BR_L2_ANO_NASCIMENTO – Path

90% ano_?(de_)?nasc(imento)? an_?nasc nu_?ano_?nasc year_?of_?birth birth_?year yob

DATA TYPE BR_L2_ANO_NASCIMENTO – Type

Accepts number (length up to 4) or text (length up to 4); any other type rules the domain out.

HOW IT IS MASKED

The decade stays and the last digit is mapped: 1985 becomes a year from 1980 to 1989.

BR_ANO **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: keep.

BR_CM_DIGITOS **CHARACTER MAPPING**

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

1985

→

1980

9.3

Age

BR_L2_IDADE

COLUMN NAME

DATA TYPE

REGEX DECOMPOSE

WHAT IT IS

Age in years, also age group.

HOW IT IS FOUND

COLUMN NAME

BR_L2_IDADE – Path

85%

idade(_?(anos|atual|paciente|cliente|obito|diagnostico|admissao|na_?data))? nu_?idade qt_?
idade faixa_?etaria anos_?de_?idade age(_?(years|group))?

DATA TYPE

BR_L2_IDADE – Type

Accepts number (length up to 6) or text (length up to 6); any other type rules the domain out.

HOW IT IS MASKED

The tens digit stays and the unit is mapped (37 becomes 30 to 39). A hundred or more becomes 90: rare ages identify.

BR_IDADE

REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_CM_DIGITOS **CHARACTER MAPPING**

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

37

→

35

104

→

90

9.4

Personal event dates

BR_L2_DATA_EVENTO

COLUMN NAME

DATA TYPE

DATE SHIFT

WHAT IT IS

Dates that say something about the person: death, marriage, hiring and dismissal, retirement, admission and discharge, diagnosis, childbirth, arrest, document issuance.

HOW IT IS FOUND

COLUMN NAME BR_L2_DATA_EVENTO - Path

80% dt_?
(obito|falecimento|casamento|divorcio|admissao|demissao|desligamento|contratacao|aposentadoria|internacao|alta|diagnostico|atendimento|consulta|parto|prisao|condenacao|expedicao|emissao_?rg|emissao_?
(cnh|documento)|naturalizacao|vacinacao|cirurgia|acidente|afastamento|posse|exoneracao)
data_?(do_|de_|)?
(obito|falecimento|casamento|divorcio|admissao|demissao|desligamento|contratacao|aposentadoria|internacao|alta|diagnostico|atendimento|consulta|parto|prisao|condenacao|expedicao|naturalizacao|vacinacao|cirurgia|acidente|afastamento|posse|exoneracao) date_?of_?
(death|marriage|hire|admission|discharge)

DATA TYPE BR_L2_DATA_EVENTO - Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the domain out.

HOW IT IS MASKED

Shifted up to 30 days either way.

BR_DATA_EVENTO **DATE SHIFT**

Moves the date by -30 to 30 days.

EXAMPLES

VALUE	MASKED
2021-03-15	→ 2021-03-05T00:00

9.5

Sex

BR_L2_SEX0

COLUMN NAME **KNOWN VALUES** **REGEX DECOMPOSE**

WHAT IT IS

Sex as the record holds it: F / M , Feminino / Masculino , Mulher / Homem or the codes 1 and 2 of IBGE, SUS and eSocial.

HOW IT IS FOUND

COLUMN NAME BR_L2_SEX0 - Path

85% sexo(_?(biologico|nascimento|registro|paciente|cliente))? (tp|cs|co|cd|ds|sg|in|fl|id)?
sexo genero (except when followed by identidade , autodeclarado , social) sex gender (except when followed by identity)

KNOWN VALUES BR_L2_SEX0 - List

LIST	VALUES	STRENGTH	FIRST VALUES
br-detector-sexo.txt	10	50%	f, m, feminino, masculino, fem, masc

A value that matches none counts 50% against the domain.

HOW IT IS MASKED

Replaced within its own vocabulary: F stays F or M , 1 stays 1 or 2 .

BR_SEX0 **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_SEX0_FEMININO_MASCULINO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-feminino-masculino.txt`, chosen by a keyed hash.

BR_SEX0_MULHER_HOMEM SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-mulher-homem.txt`, chosen by a keyed hash.

BR_SEX0_FEM_MASC SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-fem-masc.txt`, chosen by a keyed hash.

BR_SEX0_FEMALE_MALE SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-female-male.txt`, chosen by a keyed hash.

BR_SEX0_F_M SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-f-m.txt`, chosen by a keyed hash.

BR_SEX0_1_2 SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-1-2.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
F	→	M
Masculino	→	Feminino

9.6 Municipality

BR_L2_MUNICIPIO

COLUMN NAME

KNOWN VALUES

DATA CLEANSING

WHAT IT IS

The municipality of residence, birth (naturalidade) or occurrence, by name — on its own or with the state, `Bonito – MS`, `Bonito/MS`.

HOW IT IS FOUND

COLUMN NAME

BR_L2_MUNICIPIO – Path

85%

```
municipio(_?  
(residencia|nascimento|endereço|domicilio|paciente|cliente|ocorrencia|notificacao|atendimento)  
)? (except when followed by ibge, cod, codigo, id) (no|nm|ds|nome)?(municipio|cidade) cidade(_?  
(residencia|nascimento|natal|endereço|cliente))? naturalidade local_(de_)?nascimento  
mun?(res|resid|nasc|nascimento) city town localidade
```

KNOWN VALUES

BR_L2_MUNICIPIO – List

LIST

VALUES

STRENGTH

FIRST VALUES

<code>br-detectar-municipios.txt</code>	7,431	80%	Acrelândia, Acrelandia, Assis Brasil, Brasília, Brasileira, Bujari
---	-------	-----	--

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A municipality with fewer than 20,000 inhabitants becomes the nearest municipality of the same state with at least 20,000; larger ones stay. See *Where people live*.

BR_MUNICIPIO DATA CLEANSING

Replaces each value found in `br-municipios-generalizados.txt` (55,203 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

VALUE		MASKED
Serra da Saudade	→	São Gotardo
Borá – SP	→	Paraguaçu Paulista – SP
Campinas	→	Campinas

9.7 Municipality code (IBGE)

BR_L2_CODIGO_MUNICIPIO

COLUMN NAME VALUE FORMAT DATA CLEANSING

WHAT IT IS

The IBGE municipality code, with seven digits, or six without the check digit, as in DATASUS.

HOW IT IS FOUND

COLUMN NAME BR_L2_CODIGO_MUNICIPIO – Path

85% (cd|co|cod|codigo|id|nu)?(municipio|mun|cidade)(?
(ibge|res|resid|residencia|nasc|nascimento|ocorrencia|notificacao))? (cd|co|cod|codigo)?
ibge?(?municipio|mun|cidade)? ibge?(?municipio|mun|cidade|codigo|cod)? municipio_ibge
geocodigo id_mn_resi id_municip mun_res_ibge

VALUE FORMAT BR_L2_CODIGO_MUNICIPIO – Regex

PATTERN	STRENGTH	NOTES
(1[1-7] 2[1-9] 3[1-35] 4[1-3] 5[0-3])\d{4,5}	30%	

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

The same generalization as the name: a small municipality's code becomes the code of the municipality that replaces its name. The two still agree.

BR_CODIGO_MUNICIPIO DATA CLEANSING

Replaces each value found in br-codigos-municipio-generalizados.txt (7,720 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

VALUE		MASKED
3166600	→	3162104
350720	→	353550

9.8 CEP (postal code)

BR_L2_CEP

ESSENTIAL COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The postal code: region, sub-region, sector, sub-sector, sub-sector division and a three-digit suffix that identifies the street or the locality. In a large city, a CEP is a stretch of street.

HOW IT IS FOUND

COLUMN NAMEBR_L2_CEP – Path

90%cep(_?(residencia|residencial|comercial|cobranca|entrega|cliente|endereco|paciente|origem|destino))?(nr|nu|num|numero|cd|co|cod)?cep codigo_?(de_)?endecamento_postal codigo_postal zip_code zip postal_code

VALUE FORMATBR_L2_CEP – Regex

PATTERN	STRENGTH	NOTES
\d{5}-\d{3}		80%
\d{2}\.\d{3}-\d{3}		90%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

The first three digits — the postal sector — stay, and the others become zero: 01310-100 becomes 01300-000. In a numeric column, without the leading zero, too.

BR_CEP REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE	MASKED
01310-100	→ 01300-000

9.9

Neighbourhood

BR_L2_BAIRRO

COLUMN NAME

SECURE LOOKUP

WHAT IT IS

The neighbourhood of residence.

HOW IT IS FOUND

COLUMN NAMEBR_L2_BAIRRO – Path

85%bairro(_?(residencia|endereco|cliente|entrega|cobranca))?(no|nm|ds|nome)?bairro distrito_residencia localidade_bairro vila_residencia neighbou?rhood

HOW IT IS MASKED

Replaced with a neighbourhood name common in Brazil — Centro, Jardim América, Vila Nova —, unrelated to the original.

BR_BAIRRO SECURE LOOKUP

Replaces the value with one of the 40 values in br-bairros.txt, chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Vila Madalena	→ Jardim Esperança

9.10 Electoral zone and section

BR_L2_ZONA_SECAO_ELEITORAL

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

The zone and section where the person votes. A section gathers a few hundred voters of one polling place.

HOW IT IS FOUND

COLUMN NAME BR_L2_ZONA_SECAO_ELEITORAL - Path

80% zona_?(eleitoral)? secas_?(eleitoral)? zona?secao nr_?(zona|secao) nu_?(zona|secao) local_?(de_?)?votacao

HOW IT IS MASKED

Digit by digit, keeping the length.

BR_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
0245	→ 5097

9.11 Nationality

BR_L2_NACIONALIDADE

COLUMN NAME KNOWN VALUES SECURE LOOKUP

WHAT IT IS

Nationality or country of birth. Among immigrants, Venezuelans, Haitians, Bolivians and Portuguese are the largest groups.

HOW IT IS FOUND

COLUMN NAME BR_L2_NACIONALIDADE - Path

80% nacionalidade pais_?(de_?)?(nascimento|origem|nacionalidade) cidadania nationality citizenship country_of_birth

KNOWN VALUES BR_L2_NACIONALIDADE - List

LIST	VALUES	STRENGTH	FIRST VALUES
br-detectar-nacionalidades.txt	58	70%	Brasileira, Venezuelana, Haitiana, Boliviana, Portuguesa, Paraguaia

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced with a nationality from the list: Brazilian and the largest foreign communities.

BR_NACIONALIDADE SECURE LOOKUP

Replaces the value with one of the 16 values in br-nacionalidades.txt, chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Senegalesa	→ Venezuelana

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Marital status as IBGE and eSocial record it — single, married, stable union, divorced, separated, widowed —, in words or as a code.

HOW IT IS FOUND

COLUMN NAME

BR_L2_ESTADO_CIVIL - Path

85%

estado_?civil est_?civil (cd|co|tp|ds|in)_?estado_?civil situacao_?conjugal estciv

marital_?status civil_?status

KNOWN VALUES

BR_L2_ESTADO_CIVIL - List

LIST	VALUES	STRENGTH	FIRST VALUES
br-detectar-estado-civil.txt	23	70%	Solteiro(a), Casado(a), União estável, Uniao estavel, Divorciado(a), Separado(a)

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced with another status from the list; a code from 1 to 5 stays a code from 1 to 5.

BR_ESTADO_CIVIL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_ESTADO_CIVIL_LISTA .

BR_FLAG REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-sim-nao.txt , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-true-false.txt , chosen by a keyed hash.

BR_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-yes-no.txt , chosen by a keyed hash.

BR_FLAG_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-s-n.txt , chosen by a keyed hash.

BR_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-y-n.txt , chosen by a keyed hash.

BR_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-0-1.txt , chosen by a keyed hash.

BR_ESTADO_CIVIL_CODIGO SECURE LOOKUP

Replaces the value with one of the 5 values in br-estado-civil-codigos.txt , chosen by a keyed hash.

BR_ESTADO_CIVIL_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in br-estado-civil.txt , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Separado judicialmente	→	União estável
2	→	4

COLUMN NAME

REGEX DECOMPOSE

WHAT IT IS

Occupation, profession or position, in words or by the code of the Brazilian Classification of Occupations (CBO), 2235-10 . A rare occupation in a small municipality identifies.

HOW IT IS FOUND

COLUMN NAME

BR_L2_OCUPACAO - Path

80%

ocupacao profissao cbo (cd|co|cod|codigo)?(cbo|ocupacao|profissao) cargo_
 (funcionario|atual|ocupado) funcao?(exercida|funcionario) atividade?profissional
 occupation profession job?title

50%

cargo funcao

HOW IT IS MASKED

A text is replaced with a common occupation; a six-digit CBO code with a common CBO code; 2235-10 is mapped digit by digit.

BR_OCUPACAO_OU_CBO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: BR_OCUPACAO .

BR_CBO SECURE LOOKUP

Replaces the value with one of the 34 values in br-cbo.txt , chosen by a keyed hash.

BR_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

BR_OCUPACAO SECURE LOOKUP

Replaces the value with one of the 40 values in br-ocupacoes.txt , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Coordenadora de operações regionais

→

Do lar

2235-10

→

0208-89

COLUMN NAME

SECURE LOOKUP

WHAT IT IS

The name of the company or public body the person works for.

HOW IT IS FOUND

COLUMN NAME

BR_L2_EMPREGADOR - Path

80%

empregador?(?(nome|razao?social))?(nome|nm|no)?empregador empresa?(onde?
 trabalha|trabalho|empregadora|atual) local?(de?)?trabalho empregadora razao?social?
 empregador employer(?(name)? workplace

HOW IT IS MASKED

Replaced with a fictitious company.

BR_EMPREGADOR SECURE LOOKUP

Replaces the value with one of the 24 values in br-empregadores.txt , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Petróleo Brasileiro S.A.

→ Associação Mãos Solidárias

9.15 Education

BR_L2_ESCOLARIDADE

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Education level as IBGE and eSocial record it, in words or as a code.

HOW IT IS FOUND

COLUMN NAME

BR_L2_ESCOLARIDADE - Path

80%

escolaridade grau_(de_)?instrucao grau_instr nivel_(de_)?
(escolaridade|instrucao|ensino) (cd|co|tp|ds)_escolaridade instrucao formacao_
academica education(_?level)?

KNOWN VALUES

BR_L2_ESCOLARIDADE - List

LIST

VALUES STRENGTH FIRST VALUES

br-detectar-escolaridade.txt	32	60%	Sem instrução, Sem instrucao, Fundamental incompleto, Fundamental completo, Médio incompleto, Medio incompleto
------------------------------	----	-----	--

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced with another level from the list; a code stays a code.

BR_ESCOLARIDADE REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_ESCOLARIDADE_LISTA .

BR_FLAG REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-sim-nao.txt , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-true-false.txt , chosen by a keyed hash.

BR_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-yes-no.txt , chosen by a keyed hash.

BR_FLAG_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-s-n.txt , chosen by a keyed hash.

BR_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-y-n.txt , chosen by a keyed hash.

BR_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-0-1.txt , chosen by a keyed hash.

BR_ESCOLARIDADE_CODIGO SECURE LOOKUP

Replaces the value with one of the 12 values in br-escolaridade-codigos.txt , chosen by a keyed hash.

BR_ESCOLARIDADE_LISTA SECURE LOOKUP

Replaces the value with one of the 10 values in br-escolaridade.txt , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Pós-doutorado

→

Sem instrução

9.16 School or educational institution

BR_L2_ESCOLA

COLUMN NAME

SECURE LOOKUP

WHAT IT IS

The school, college or university, by name or by its INEP code.

HOW IT IS FOUND

COLUMN NAME

BR_L2_ESCOLA – Path

75%

escola(_?(nome|atual|origem))? (nome|nm|no)?(escola|instituicao_?ensino|faculdade|universidade) instituicao_?(de_)?ensino faculdade universidade colegio (cd|co)?(inep|entidade|escola) school(_?name)? university

HOW IT IS MASKED

Replaced with a fictitious institution.

BR_ESCOLA

SECURE LOOKUP

Replaces the value with one of the 12 values in `br-escolas.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Escola Estadual Doutor Arnaldo

→

Escola Estadual Professora Maria José

9.17 Dependants and children

BR_L2_DEPENDENTES

COLUMN NAME

DATA TYPE

REGEX DECOMPOSE

WHAT IT IS

Number of dependants, children or household members.

HOW IT IS FOUND

COLUMN NAME

BR_L2_DEPENDENTES – Path

80%

dependentes qt?(dependentes|filhos|dep) qtd?(dependentes|filhos) quantidade_(de_)?(dependentes|filhos) n(r|u|um|o|ro|umero)?(de_)?(dependentes|filhos) filhos moradores_(no_)?domicilio qt?moradores number_of_children dependents

DATA TYPE

BR_L2_DEPENDENTES – Type

Accepts number (length up to 4) or text (length up to 4); any other type rules the domain out.

HOW IT IS MASKED

Values from 0 to 5 stay; above 5 they become 5. Very large families are rare, and a rare count identifies.

BR_CONTAGEM_LIMITADA

REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE

MASKED

9

→

5

Sensitive personal data

The categories of art. 5, II that are not health data — which has a group of its own: racial or ethnic origin, religious conviction, political opinion, membership of a trade union or of an organization of religious, philosophical or political character, sexual life, genetic and biometric data — and, beyond the list, sexual orientation and gender identity, which expose people to the same discrimination. Categorical columns keep their shape: an S/N flag stays a flag, a numeric code is replaced by another code of the same catalog and any other value is replaced by one from the list of its category. Where the values share no vocabulary, they are suppressed to `Não informado`.

10.1 Colour or race

BR_L3_RACA_COR

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Self-declared colour or race in the IBGE categories — in the 2022 census, brown (parda) 45.3%, white 43.5%, black 10.2%, indigenous 0.8% and yellow 0.4% —, in words or as the codes 1 to 5 and 9 used in SUS and eSocial. Racial origin is sensitive data (art. 5, II).

HOW IT IS FOUND

COLUMN NAME BR_L3_RACA_COR – Path

95% `raca(_?cor)? cor(_?raca)? raca?etnia (cd|co|cs|tp|ds|id|in)?(raca|cor|raca?cor) autodeclaracao?(racial|raca|cor) cor?da?pele etnia?racial race ethnicity`

KNOWN VALUES BR_L3_RACA_COR – List

LIST	VALUES	STRENGTH	FIRST VALUES
<code>br-detectar-raca-cor.txt</code>	23	90%	Branca, Preta, Parda, Amarela, Indígena, Indígena
<code>br-detectar-codigos-raca-cor.txt</code>	7	30%	1, 2, 3, 4, 5, 6

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A code is replaced with another code of the catalog; any other value with an IBGE category.

BR_RACA_COR REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_RACA_COR_LISTA .

BR_FLAG REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt` , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt` , chosen by a keyed hash.

BR_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt` , chosen by a keyed hash.

BR_FLAG_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt` , chosen by a keyed hash.

BR_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt` , chosen by a keyed hash.

BR_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt` , chosen by a keyed hash.

BR_RACA_COR_CODIGO **SECURE LOOKUP**

Replaces the value with one of the 6 values in `br-raca-cor-codigos.txt` , chosen by a keyed hash.

BR_RACA_COR_LISTA **SECURE LOOKUP**

Replaces the value with one of the 6 values in `br-raca-cor.txt` , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Parda	→	Não declarada
3	→	5

10.2 Indigenous people and traditional community

BR_L3_ETNIA

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The indigenous people — the 2022 census counted 1,694,836 indigenous people of 391 ethnicities; the most numerous are the Tikúna, Kokama, Makuxí and Guaraní Kaiowá —, the quilombola or traditional community and the village, often as an S/N flag.

HOW IT IS FOUND

COLUMN NAME

BR_L3_ETNIA - Path

90%

etnia (except when followed by *racial*) povo(?:indigena)? etnia?indigena (cd|co|ds|no)?etnia
comunidade?(tradicional|quilombola|indigena) quilombola aldeia terra?indigena povo?
tradicional indigena lingua?indigena ethnic?group indigenous

KNOWN VALUES

BR_L3_ETNIA - List

LIST

VALUES

STRENGTH

FIRST VALUES

br-detectar-etnias.txt

63

90%

Tikúna, Tikuna, Kokama, Makuxí, Makuxi, Guaraní Kaiowá

br-detectar-flag.txt

8

30%

s, n, sim, não, nao, y

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A flag stays a flag in its vocabulary and any other value is replaced with one from the list.

BR_ETNIA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_ETNIA_LISTA` .

BR_FLAG **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt` , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-true-false.txt` , chosen by a keyed hash.

BR_FLAG_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-yes-no.txt` , chosen by a keyed hash.

BR_FLAG_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-s-n.txt` , chosen by a keyed hash.

BR_FLAG_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-y-n.txt` , chosen by a keyed hash.

BR_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt` , chosen by a keyed hash.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

BR_ETNIA_LISTA SECURE LOOKUP

Replaces the value with one of the 30 values in `br-etnias.txt` , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Krenak	→	Guarani Nhandeva
S	→	S

10.3 Religion

BR_L3_RELIGIAO

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Religion or church. In the 2022 census, among people aged 10 or over, 56.7% declared themselves Catholic, 26.9% Evangelical, 9.3% of no religion, 1.8% Spiritist and 1.0% of Umbanda or Candomblé.

HOW IT IS FOUND

COLUMN NAME

BR_L3_RELIGIAO - Path

95%

religiao crenca_?religiosa convicc?ao_?religiosa confissao_?religiosa (cd|co|ds|tp)?
religiao igreja(_?(frequenta|membro|nome))? denominacao_?religiosa culto religion
church

KNOWN VALUES

BR_L3_RELIGIAO - List

LIST

VALUES

STRENGTH

FIRST VALUES

br-detectar-religioes.txt	67	90%	Católica Apostólica Romana, Catolica Apostolica Romana, Evangélica, Evangelica, Evangélica pentecostal, Evangelica pentecostal
---------------------------	----	-----	--

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A flag stays a flag and any other value is replaced with a religion from the list.

BR_RELIGIAO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_RELIGIAO_LISTA .

BR_FLAG REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt` , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt` , chosen by a keyed hash.

BR_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt` , chosen by a keyed hash.

BR_FLAG_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt` , chosen by a keyed hash.

BR_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt` , chosen by a keyed hash.

BR_FLAG_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

BR_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

BR_RELIGIAO_LISTA **SECURE LOOKUP**

Replaces the value with one of the 13 values in `br-religoes.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Adventista do Sétimo Dia

→ **Candomblé**

10.4 Convictions and religious, philosophical or political organizations

BR_L3_CONVICCAO_ORGANIZ
ACAO

COLUMN NAME

REGEX DECOMPOSE

WHAT IT IS

Philosophical and moral convictions, conscientious objection and membership of an organization of religious, philosophical or political character — a religious order, freemasonry, a social movement —, which the LGPD lists apart from religion and party.

HOW IT IS FOUND

COLUMN NAME

BR_L3_CONVICCAO_ORGANIZACAO - Path

90%

```
convicc?ao?(filosofica|moral|politica)  crenca?(filosofica|pessoal)  escusa?(de_)?  
consciencia  objecao?(de_)?consciencia  maconaria  loja?maconica  ordem?  
(religiosa|filosofica)  organizacao?(religiosa|filosofica|politica)  filiacao?  
(religiosa|filosofica)  associacao?(religiosa|filosofica)  movimento?  
(social|politico|religioso)  grupo?(religioso|filosofico)  beliefs?  philosophical?  
beliefs?
```

HOW IT IS MASKED

The values share no vocabulary: they are suppressed to `Não informado`. Flags and codes keep their shape.

BR_CATEGORIA_SUPRIMIDA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SUPRIMIR`.

BR_FLAG **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: `keep`.

BR_FLAG_SIM_NAO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

BR_FLAG_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

BR_FLAG_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

BR_FLAG_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

BR_FLAG_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

BR_FLAG_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

BR_SUPRIMIR SECURE LOOKUP

Replaces the value with one of the 1 values in `br-nao-informado.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Recusa transfusão de sangue

→

Não informado

10.5 Political opinion

BR_L3_OPINIAO_POLITICA

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Political position, ideology and voting intention, from surveys and registers.

HOW IT IS FOUND

COLUMN NAME

BR_L3_OPINIAO_POLITICA - Path

90%

opinioao_?politica posicionamento_?politico orientacao_?politica preferencia_?
(politica|partidaria) ideologia(_?politica)? intencao_?(de_?)?voto voto_
(pretendido|intencao) espectro_?politico political_?(opinion|view|orientation) voting_
intention

KNOWN VALUES

BR_L3_OPINIAO_POLITICA - List

LIST

VALUES

STRENGTH

FIRST VALUES

br-detectar-opinioes-
politicas.txt

23

70%

Esquerda, Centro-esquerda, Centro, Centro-direita, Direita, Sem
preferência

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced with a position from the list — left to right, no preference.

BR_OPINIAO_POLITICA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_OPINIAO_POLITICA_LISTA

BR_FLAG REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

BR_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

BR_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

BR_FLAG_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

BR_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

BR_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

BR_OPINIAO_POLITICA_LISTA **SECURE LOOKUP**

Replaces the value with one of the 7 values in `br-opinioes-politicas.txt` , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Conservador	→	Centro-esquerda

10.6

Party membership

BR_L3_FILIACAO_PARTIDARIA

COLUMN NAME **KNOWN VALUES** **REGEX DECOMPOSE**

WHAT IT IS

The party the person is a member of, by acronym, name or ballot number. The TSE had 30 registered parties in September 2026.

HOW IT IS FOUND

COLUMN NAME	BR_L3_FILIACAO_PARTIDARIA - Path		
95%	filiacao_?partidaria partido(_?politico)?(_?(filiacao sigla nome numero))? sigla_?partido (cd co sg nm nr)_?partido filiado_?(a_)?partido militancia militante political_?party party_?membership		
KNOWN VALUES	BR_L3_FILIACAO_PARTIDARIA - List		
LIST	VALUES	STRENGTH	FIRST VALUES
br-detectar-partidos.txt	81	80%	MDB, Movimento Democrático Brasileiro, Movimento Democratico Brasileiro, PDT, Partido Democrático Trabalhista, Partido Democratico Trabalhista

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

An acronym becomes another acronym, a ballot number another number and a name another name from the list of registered parties; a flag stays a flag.

BR_PARTIDO **REGEX DECOMPOSE**

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_PARTIDO_NOME` .

BR_FLAG **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt` , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-true-false.txt` , chosen by a keyed hash.

BR_FLAG_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-yes-no.txt` , chosen by a keyed hash.

BR_FLAG_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-s-n.txt` , chosen by a keyed hash.

BR_FLAG_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-y-n.txt` , chosen by a keyed hash.

BR_FLAG_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-0-1.txt` , chosen by a keyed hash.

BR_PARTIDO_NUMERO **SECURE LOOKUP**

Replaces the value with one of the 30 values in `br-partidos-numeros.txt` , chosen by a keyed hash.

BR_PARTIDO_SIGLA **SECURE LOOKUP**

Replaces the value with one of the 30 values in `br-partidos-siglas.txt`, chosen by a keyed hash.

BR_PARTIDO_NOME **SECURE LOOKUP**

Replaces the value with one of the 31 values in `br-partidos-nomes.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
PT	→	MISSÃO
Partido Trabalhista Brasileiro	→	Democracia Cristã

10.7 Trade union membership

BR_L3_FILIACAO_SINDICAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The union, being a union member, the union contribution or fee deducted from payroll and the union confederation.

HOW IT IS FOUND

COLUMN NAME

BR_L3_FILIACAO_SINDICAL - Path

95%

```
sindicato(_?(nome|filiado|filiacao|categoria|sigla|cnpj))? sindicalizado filiacao_?
sindical filiacao_?sindicato contribuicao_?(sindical|assistencial|confederativa)
mensalidade_?sindical desconto_?sindical imposto_?sindical central_?sindical entidade_?
sindical delegado_?sindical dirigente_?sindical trade_?union union_?member(ship)?
```

KNOWN VALUES

BR_L3_FILIACAO_SINDICAL - List

LIST

VALUES

STRENGTH

FIRST VALUES

br-detectar-sindicatos.txt

29

60%

CUT, Força Sindical, Forca Sindical, UGT, CTB, NCST

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

A flag stays a flag; any other value becomes `Não informado`.

BR_CATEGORIA_SUPRIMIDA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SUPRIMIR`.

BR_FLAG **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

BR_FLAG_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

BR_FLAG_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

BR_FLAG_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

BR_FLAG_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

BR_FLAG_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

BR_SUPRIMIR SECURE LOOKUP

Replaces the value with one of the 1 values in `br-nao-informado.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Sindicato dos Bancários de SP	→	Não informado
S	→	S

10.8 Genetic data

BR_L3_GENETICO

COLUMN NAME

REGEX DECOMPOSE

WHAT IT IS

Genetic and DNA test results, genetic profiles, mutations, karyotype, newborn screening.

HOW IT IS FOUND

COLUMN NAME BR_L3_GENETICO - Path

90%

dna adn genetic[oa]s? dados_?geneticos? genoma genotip[a-z]* exame_?genetico
teste_?(genetico|de_?dna|paternidade) perfil_?genetico mutacao(_?genetica)? brca[12]?
cariotipo triagem_?neonatal teste_?do_?pezinho painel_?genetico snp haplotipo
genetic[a-z]* genome

HOW IT IS MASKED

Every character becomes X .

BR_REDIGIR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE		MASKED
BRCA1 positivo	→	XXXXXXXXXXXXXX

10.9 Biometric data

BR_L3_BIOMETRICO

COLUMN NAME

DATA TYPE

REGEX DECOMPOSE

WHAT IT IS

Fingerprint, face, iris and voice templates from time clocks, proof of life, account opening and turnstiles.

HOW IT IS FOUND

COLUMN NAME BR_L3_BIOMETRICO - Path

90%

biometri[a-z]* digita(l|is)(_?(template|polegar|indicador|hash|imagem))? impressao_?
digital template_?(facial|digital|biometrico) face_?(template|hash|id|encoding)
reconhecimento_?facial facial_?(template|hash|embedding) iris(_?(template|scan))? retina
voz_?(template|biometrica) assinatura_?(biometrica|digitalizada) palma_?mao
fingerprints? biometric[a-z]*

DATA TYPE BR_L3_BIOMETRICO – Type

Accepts text; any other type rules the domain out.

HOW IT IS MASKED

Every character becomes `X`. A masked template authenticates no one.

BR_REDIGIR **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE		MASKED
A1B2C3D4E5F6	→	XXXXXXXXXXXX

10.10 Sexual orientation

BR_L3_ORIENTACAO_SEXUAL

COLUMN NAME **KNOWN VALUES** **REGEX DECOMPOSE**

WHAT IT IS

Self-declared sexual orientation, in health and diversity registers.

HOW IT IS FOUND

COLUMN NAME BR_L3_ORIENTACAO_SEXUAL – Path

95% `orientacao?sexual` `orient?sexual` `(cd|co|ds|tp)?orientacao?sexual` `preferencia?sexual`
`sexual?orientation`

KNOWN VALUES BR_L3_ORIENTACAO_SEXUAL – List

LIST	VALUES	STRENGTH	FIRST VALUES
<code>br-detectar-orientacoes-sexuais.txt</code>	18	80%	Heterossexual, Gay, Lésbica, Lesbica, Bissexual, Pansexual

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced with a category from the list.

BR_ORIENTACAO_SEXUAL **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_ORIENTACAO_SEXUAL_LISTA .

BR_FLAG **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt` , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-true-false.txt` , chosen by a keyed hash.

BR_FLAG_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-yes-no.txt` , chosen by a keyed hash.

BR_FLAG_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-s-n.txt` , chosen by a keyed hash.

BR_FLAG_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-y-n.txt` , chosen by a keyed hash.

BR_FLAG_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-0-1.txt` , chosen by a keyed hash.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

BR_ORIENTACAO_SEXUAL_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in `br-orientacoes-sexuais.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Homossexual

→

Pansexual

10.11 Sexual life

BR_L3_VIDA_SEXUAL

COLUMN NAME

REGEX DECOMPOSE

WHAT IT IS

Sexual activity and behaviour, number of partners, sexual debut — data concerning sexual life, sensitive under art. 5, II.

HOW IT IS FOUND

COLUMN NAME

BR_L3_VIDA_SEXUAL - Path

90%

vida_?sexual atividade_?sexual comportamento_?sexual pratica_?sexual parceiros_?
sexua(l|is) n(r|u|um)_?parceiros sexualmente_?ativ[oa] inicio_?(da_?)?vida_?sexual
relacao_?sexual sex(ual)?_?life sexual?(activity|behavio(u)?r|partners)

HOW IT IS MASKED

A flag stays a flag; any other value becomes `Não informado`.

BR_CATEGORIA_SUPRIMIDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SUPRIMIR`.

BR_FLAG REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: `keep`.

BR_FLAG_SIM_NAO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

BR_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

BR_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

BR_FLAG_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

BR_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

BR_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

BR_SUPRIMIR SECURE LOOKUP

Replaces the value with one of the 1 values in `br-nao-informado.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Ativa

→

Não informado

10.12 Gender identity

BR_L3_IDENTIDADE_GENERO

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Self-declared gender identity. Not on the list of art. 5, II, but it exposes people to the same discrimination. The social name falls under the full-name domain.

HOW IT IS FOUND

COLUMN NAME

BR_L3_IDENTIDADE_GENERO - Path

95%

identidade_(de_)?genero (cd|co|ds|tp)_?identidade?genero genero_?
(autodeclarado|social) transgenero pessoa?trans pronomes? gender?identity

KNOWN VALUES

BR_L3_IDENTIDADE_GENERO - List

LIST

VALUES STRENGTH FIRST VALUES

VALUES	STRENGTH	FIRST VALUES
br-detectar- identidades-genero.txt	26	70% Mulher cisgênero, Mulher cisgenero, Homem cisgênero, Homem cisgenero, Mulher trans, Homem trans

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced with a category from the list.

BR_IDENTIDADE_GENERO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_IDENTIDADE_GENERO_LISTA .

BR_FLAG REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-sim-nao.txt , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-true-false.txt , chosen by a keyed hash.

BR_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-yes-no.txt , chosen by a keyed hash.

BR_FLAG_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-s-n.txt , chosen by a keyed hash.

BR_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-y-n.txt , chosen by a keyed hash.

BR_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-0-1.txt , chosen by a keyed hash.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

BR_IDENTIDADE_GENERO_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in br-identidades-genero.txt , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Transgênero



Prefere não informar

Health data

Data concerning health is sensitive personal data (art. 5, II). The law forbids sharing health data between controllers to obtain an economic advantage, except in health services and pharmaceutical care, and forbids health plan operators to use it for risk selection (art. 11, §§ 4 and 5). Public health studies must anonymize or pseudonymize the data whenever possible (art. 13). The group covers what health systems, health plans, laboratories and HR (occupational health) keep.

11.1 National Health Card

BR_SAUDE_CNS

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The number of the Cartão Nacional de Saúde (SUS card): fifteen digits whose weighted sum, with weights 15 to 1, is a multiple of 11. A definitive number starts with 1 or 2 and is the person's PIS followed by 000 or 001 and a digit; a provisional one starts with 7, 8 or 9.

HOW IT IS FOUND

COLUMN NAME BR_SAUDE_CNS Path

90% `cns[a-z0-9]*    cartao_?(nacional_)?(de_)?saude    cartao_?sus    n(r|u|um|o|ro|umero)?`
`(cns|cartao_?sus)    (co|cd|nu)?cns?(paciente|usuario|cidadao|profissional)?    cns_?`
`(paciente|usuario|cidadao|profissional)    carteira_?sus`

VALUE FORMAT BR_SAUDE_CNS Regex

PATTERN	STRENGTH	NOTES
<code>[12789]\d{2}\s?\d{4}\s?\d{4}\s?\d{4}</code>		50%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

In a definitive number, the first eleven digits are masked by the PIS algorithm — the masked CNS still carries the masked PIS — and the tail is recomputed. Every masked definitive CNS is valid. A provisional one is masked digit by digit, keeping the first.

BR_CNS REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

BR_CNS_VALIDO STRING ALGORITHM CHAIN

Applies BR_CNS_CORPO → BR_CNS_DV , in this order.

BR_CNS_CORPO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

BR_PIS_VALIDO CHECK DIGIT

Masks the digits with BR_PIS_CORPO and recomputes a modulus 11 check digit with weights 3 2 9 8 7 6 5 4 3 2 .

BR_PIS_CORPO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

BR_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

BR_CNS_DV STRING ALGORITHM CHAIN

Applies BR_CNS_DV_P1 → BR_CNS_DV_P2 → BR_CNS_DV_P3 → BR_CNS_DV_P4 → BR_CNS_DV_P5 → BR_CNS_DV_P6 , in this order. It is made of 12 algorithms that repeat the same step position by position; each one is in Appendix A.

EXAMPLES

VALUE

MASKED

120564125470008

→

113811042790001

11.2 Health identifiers

BR_SAUDE_IDENTIFICADOR

COLUMN NAME

CHARACTER MAPPING

WHAT IT IS

Medical record, visit, hospitalization (AIH), TISS form, health plan card, prescription and SINAN notification numbers.

HOW IT IS FOUND

COLUMN NAME

BR_SAUDE_IDENTIFICADOR – Path

85%

```
prontuario(?: (numero|num|nr|eletronico) )? n(r|u|um|o|ro|umero)?  
(prontuario|atendimento|internacao|aih|guia|carteirinha|beneficiario_?plano|registro_?  
paciente) aih autorizacao_?(de_)?internacao numero_?(da_)?guia guia_  
(tiss|sadt|internacao) carteirinha_?(plano|convenio|saude) matricula_?(convenio|plano)  
cod_?(beneficiario|paciente)_?(ans|plano|convenio) registro_?(do_)?paciente id_?paciente  
codigo_?paciente n(r|u)?receita receita_?(numero|controlada) notificacao_  
(sinan|numero) nu_?notific laudo_?(numero|nr)
```

55%

```
atendimento internacao guia episodio
```

HOW IT IS MASKED

Character by character, keeping length and format.

BR_CM_ALFANUM

CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

PRONT-2024-00123

→

TVAQH-6080-27071

11.3 Diagnosis and ICD

BR_SAUDE_DIAGNOSTICO

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

Diagnoses in words or by ICD-10 code, with the dot (F33.1) or without it, as in DATASUS (F331); causes of sick leave, hospitalization and death, comorbidities and allergies.

HOW IT IS FOUND

COLUMN NAME BR_SAUDE_DIAGNOSTICO – Path

85% diagnostico[a-z]* cid(_?
(10|11|principal|secundario|obito|afastamento|atestado|internacao))?((cd|co|ds)?cid cid?
(cd|co|ds) hipotese?diagnostica doenca[a-z]* agravo comorbidade[a-z]* patologia
condicao?(de_)?saude causa?(do_)?(obito|afastamento|internacao|basica) motivo?(do_)?
(afastamento|atestado|internacao|consulta) queixa?principal alergias? doencas?_pre?
existentes? historico?(medico|clinico|familiar) diagnos(is|es) icd?(10|11)?

VALUE FORMAT BR_SAUDE_DIAGNOSTICO – Regex

PATTERN	STRENGTH	NOTES
[A-TV-Za-tv-z]\d{2}(\.\d{1,2} \d)?		50%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

An ICD code becomes another common code in the same format, and a text a common diagnosis: F33.1 may become J06.9 . So an HIV or mental disorder diagnosis does not survive as a category.

BR_DIAGNOSTICO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches:

BR_DIAGNOSTICO_TEXT0

BR_CID10_DECIMAL SECURE LOOKUP

Replaces the value with one of the 33 values in br-cid10-decimal.txt , chosen by a keyed hash.

BR_CID10 SECURE LOOKUP

Replaces the value with one of the 37 values in br-cid10.txt , chosen by a keyed hash.

BR_FLAG REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-sim-nao.txt , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-true-false.txt , chosen by a keyed hash.

BR_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-yes-no.txt , chosen by a keyed hash.

BR_FLAG_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-s-n.txt , chosen by a keyed hash.

BR_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-y-n.txt , chosen by a keyed hash.

BR_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-0-1.txt , chosen by a keyed hash.

BR_DIAGNOSTICO_TEXT0 SECURE LOOKUP

Replaces the value with one of the 30 values in br-diagnostics.txt , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
F33.1	→	K21.9
B20	→	S60

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Medicines prescribed, dispensed or taken continuously. An antiretroviral, an antipsychotic or a contraceptive reveals a diagnosis.

HOW IT IS FOUND

COLUMN NAME

BR_SAUDE_MEDICAMENTO – Path

85%

medicamentos?(?_?(uso|prescrito|continuo|nome|dispensado))?

remedios?

farmacos?

principio_?ativo

prescricao(?_medicamento)?

posologia

receita_?(medicamento|item)

uso_?

continuo

(cd|co|ds|no)_?medicamento

medications?

drugs?_?prescribed

KNOWN VALUES

BR_SAUDE_MEDICAMENTO – List

HOW IT IS MASKED

Replaced with a common medicine from the list; a flag stays a flag.

BR_MEDICAMENTO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_MEDICAMENTO_LISTA .

BR_FLAG REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-sim-nao.txt , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-true-false.txt , chosen by a keyed hash.

BR_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-yes-no.txt , chosen by a keyed hash.

BR_FLAG_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-s-n.txt , chosen by a keyed hash.

BR_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-y-n.txt , chosen by a keyed hash.

BR_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-0-1.txt , chosen by a keyed hash.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

BR_MEDICAMENTO_LISTA SECURE LOOKUP

Replaces the value with one of the 25 values in br-medicamentos.txt , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Sertralina 50 mg	→ Captopril

11.5 Clinical text

BR_SAUDE_TEXTO_CLINICO

COLUMN NAME **DATA TYPE** **FREE TEXT REDACTION**

WHAT IT IS

Medical history, progress notes, physical examination, reports, discharge summary and nursing notes.

HOW IT IS FOUND

COLUMN NAME BR_SAUDE_TEXTO_CLINICO – Path

85% anamnese evolucao(_?(clinica|medica|enfermagem))? historia(_?(clinica|da?doenca?atual) hda exame?fisico conduta(_?medica)? plano?terapeutico sumario?(de?)?alta resumo?(de?)?alta laudo(_?(texto|descricao|medico))? prescricao?texto notas?_(clnicas|medicas|enfermagem) observac(ao|oes)?(clnicas?|medicas?|enfermagem) triagem(_?texto)? epicrise parecer?medico clinical?notes? discharge?summary

DATA TYPE BR_SAUDE_TEXTO_CLINICO – Type

Accepts text (length 30 or more); any other type rules the domain out.

HOW IT IS MASKED

The same Free Text Redaction as free text: CPF, contacts and names from the list are redacted. The clinical content stays — if it must not be in the copy, suppress the column.

BR_TEXTO_LIVRE **FREE TEXT REDACTION**

Replaces the text matching 12 patterns with [DADO], and the 3,302 terms of br-texto-nomes.txt with [NOME].

EXAMPLES

VALUE

MASKED

Paciente Souza, CPF 529.982.247-25, atendido por cefaleia.

→ Paciente [NOME], CPF [DADO], atendido por cefaleia.

11.6 Test results

BR_SAUDE_RESULTADO_EXAME

COLUMN NAME **CHARACTER MAPPING**

WHAT IT IS

Laboratory and test results — HIV, viral load, blood glucose, pregnancy, drug test —, also as numeric values.

HOW IT IS FOUND

COLUMN NAME BR_SAUDE_RESULTADO_EXAME – Path

80% resultado?(do_)?
(exame|teste|laboratorio|laboratorial|hiv|covid|pcr|sorologia|biopsia|glicemia) exame?
(resultado|laboratorial) hiv(_?(status|resultado|teste))? carga?viral cd4 glicemia
hemoglobina?glicada hba1c colesterol imc pressao?arterial pa?
(sistolica|diastolica) teste?(de_)?gravidez toxicologico antidoping sorologia lab?
results? test?results?

50% resultado teste

HOW IT IS MASKED

Character by character, keeping the format: a number stays a number.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

Reagente

→

MASKED

Suaqunwi

11.7 Blood group

BR_SAUDE_TIPO_SANGUINEO

COLUMN NAME

KNOWN VALUES

SECURE LOOKUP

WHAT IT IS

ABO group and Rh factor.

HOW IT IS FOUND

COLUMN NAME

BR_SAUDE_TIPO_SANGUINEO - Path

90%

tipo_?sangu[ií]neo grupo_?sangu[ií]neo tipagem_?sangu[ií]nea fator_?rh abo(_?rh)?
tipo_?(de_)?sangu blood_?(type|group)

KNOWN VALUES

BR_SAUDE_TIPO_SANGUINEO - List

LIST

VALUES

STRENGTH

FIRST VALUES

LIST	VALUES	STRENGTH	FIRST VALUES
br-detectar-tipos-sanguineos.txt	18	90%	A+, A-, B+, B-, AB+, AB-

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced with another group from the list.

BR_TIPO_SANGUINEO

SECURE LOOKUP

Replaces the value with one of the 8 values in `br-tipos-sanguineos.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

O+

→

MASKED

A-

11.8 Disability

BR_SAUDE_DEFICIENCIA

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The type of disability — physical, visual, hearing, intellectual, mental or psychosocial, multiple —, being a person with a disability (Lei 13.146/2015) and the disability report.

HOW IT IS FOUND

COLUMN NAME

BR_SAUDE_DEFICIENCIA - Path

90%

deficiencia[a-z]* (tp|cd|co|ds|in|fl)?deficiencia pcd pessoa_?com_?deficiencia
portador_?(de_)?(deficiencia|necessidades_?especiais) necessidades_?especia(l|is)
laudo_?pcd mobilidade_?reduzida tea autismo disabilit(y|ies)

KNOWN VALUES BR_SAUDE_DEFICIENCIA - List

LIST	VALUES	STRENGTH	FIRST VALUES
br-detectar-deficiencias.txt	29	60%	Física, Física, Visual, Auditiva, Intelectual, Mental ou psicossocial

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

A flag stays a flag and any other value is replaced with a type from the list.

BR_DEFICIENCIA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_DEFICIENCIA_LISTA .**BR_FLAG** **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt` , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-true-false.txt` , chosen by a keyed hash.

BR_FLAG_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-yes-no.txt` , chosen by a keyed hash.

BR_FLAG_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-s-n.txt` , chosen by a keyed hash.

BR_FLAG_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-y-n.txt` , chosen by a keyed hash.

BR_FLAG_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-0-1.txt` , chosen by a keyed hash.

BR_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

BR_DEFICIENCIA_LISTA **SECURE LOOKUP**

Replaces the value with one of the 7 values in `br-deficiencias.txt` , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Transtorno do espectro autista	→ Auditiva

11.9 Occupational health

BR_SAUDE_OCUPACIONAL

COLUMN NAME**KNOWN VALUES****REGEX DECOMPOSE****WHAT IT IS**

What HR keeps about an employee's health: the result of the occupational health certificate (ASO), the work accident report (CAT), medical certificates, sick leave and absences.

HOW IT IS FOUND**COLUMN NAME** BR_SAUDE_OCUPACIONAL - Path

85% `aso(_?(resultado|conclusao|parecer))?` `atestado_(de_)?saude_?ocupacional` `aptidao(_?(medica|laboral))?` `resultado_(do_)?aso` `parecer_(medico_)?ocupacional` `cat(_?(numero|tipo|acidente))?` `comunicacao_(de_)?acidente_(de_)?trabalho` `acidente_(de_)?trabalho` `afastamento_(medico|inss|doenca|motivo)` `licenca_(medica|saude|maternidade)` `atestado_?medico` `dias_(de_)?atestado` `motivo_?afastamento`

KNOWN VALUES BR_SAUDE_OCUPACIONAL – List

LIST	VALUES	STRENGTH	FIRST VALUES
br-detectar-aptidao.txt	6	60%	apto, inapto, apto com restrições, apto com restricoes, apto com restricao, inapto temporariamente

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

A fitness result is replaced with another from the list — fit, unfit —; a flag stays a flag; the ICD of the leave falls under the diagnosis domain.

BR_SAUDE_OCUPACIONAL **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_SAUDE_OCUPACIONAL_LISTA .**BR_FLAG** **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt` , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-true-false.txt` , chosen by a keyed hash.

BR_FLAG_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-yes-no.txt` , chosen by a keyed hash.

BR_FLAG_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-s-n.txt` , chosen by a keyed hash.

BR_FLAG_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-y-n.txt` , chosen by a keyed hash.

BR_FLAG_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-0-1.txt` , chosen by a keyed hash.

BR_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

BR_SAUDE_OCUPACIONAL_LISTA **SECURE LOOKUP**

Replaces the value with one of the 4 values in `br-aptidao.txt` , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Inapto temporariamente	→	Inapto

11.10 Sexual and reproductive health

BR_SAUDE_REPRODUTIVA

COLUMN NAME **REGEX DECOMPOSE****WHAT IT IS**

Pregnancy, gestational age, last menstrual period, prenatal care, childbirth, abortion, contraception, fertility, sexually transmitted infections.

HOW IT IS FOUND

COLUMN NAME BR_SAUDE_REPRODUTIVA – Path

90%
gestante gravidez gestacao idade_?gestacional semanas_?(de_)?gestacao dum data_
(da_)?ultima_?menstruacao pre_?natal aborto abortamento parto(_?tipo)? tipo_?(de_)?
parto puerperio metodo_?contraceptivo contracepcao anticoncepcao planejamento_
familiar fertilidade infertilidade fertilizacao fiv ist dst infeccoes?_
sexualmente_?transmissive(l|is) pregnan(t|cy) contracepti(on|ve)

HOW IT IS MASKED

Flags keep their shape; any other value becomes Não informado .

BR_CATEGORIA_SUPRIMIDA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: BR_SUPRIMIR .

BR_FLAG **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO **SECURE LOOKUP**

Replaces the value with one of the 2 values in br-flag-sim-nao.txt , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in br-flag-true-false.txt , chosen by a keyed hash.

BR_FLAG_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in br-flag-yes-no.txt , chosen by a keyed hash.

BR_FLAG_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in br-flag-s-n.txt , chosen by a keyed hash.

BR_FLAG_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in br-flag-y-n.txt , chosen by a keyed hash.

BR_FLAG_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in br-flag-0-1.txt , chosen by a keyed hash.

BR_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

BR_SUPRIMIR **SECURE LOOKUP**

Replaces the value with one of the 1 values in br-nao-informado.txt , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
12 semanas de gestação	→	Não informado
S	→	S

11.11 Mental health

BR_SAUDE_MENTAL

COLUMN NAME **REGEX DECOMPOSE**

WHAT IT IS

Mental disorders, psychiatric or psychological care, CAPS, suicide attempts, substance dependence.

HOW IT IS FOUND

COLUMN NAME BR_SAUDE_MENTAL – Path

90% saude_?mental transtorno[a-z]* psiquiatri[a-z]* psicolog[a-z]* depressao ansiedade
suicid[a-z]* autolesao automutilacao dependencia_?(quimica|alcool|drogas) uso_?(de_)?
(alcool|drogas|substancias) alcoolismo tabagismo caps(_?(ad|i))? internacao_
psiquiatrica mental_?health

HOW IT IS MASKED

Flags keep their shape; any other value becomes Não informado .

BR_CATEGORIA_SUPRIMIDA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: BR_SUPRIMIR .

BR_FLAG **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO **SECURE LOOKUP**

Replaces the value with one of the 2 values in br-flag-sim-nao.txt , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in br-flag-true-false.txt , chosen by a keyed hash.

BR_FLAG_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in br-flag-yes-no.txt , chosen by a keyed hash.

BR_FLAG_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in br-flag-s-n.txt , chosen by a keyed hash.

BR_FLAG_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in br-flag-y-n.txt , chosen by a keyed hash.

BR_FLAG_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in br-flag-0-1.txt , chosen by a keyed hash.

BR_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

BR_SUPRIMIR **SECURE LOOKUP**

Replaces the value with one of the 1 values in br-nao-informado.txt , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Transtorno bipolar

→ Não informado

Financial and credit data

The LGPD does not call them sensitive, but a person's financial life is protected by bank secrecy (Lei Complementar 105/2001), credit databases have a law of their own (Lei 12.414/2011) and credit protection is a legal basis of its own (art. 7, X). And several of these data reveal sensitive data: the type of an INSS benefit tells whether a person is incapacitated, has a disability or had a child, and a Bolsa Família benefit or a Cadastro Único bracket tells that they are poor. Categories are replaced within their own notation; amounts are mapped into a plausible range unrelated to the real value.

12.1 Credit and negative records

BR_FIN_CREDITO

COLUMN NAME

REGEX DECOMPOSE

WHAT IT IS

Credit score (0 to 1,000), negative record, restriction, protest, days overdue and contract status.

HOW IT IS FOUND

COLUMN NAME

BR_FIN_CREDITO – Path

90%

```
score(?:(credito|serasa|spc|boa_vista|quod|interno))? pontuacao?(de_)?credito rating_?
(de_)?credito risco?(de_)?credito nota?(de_)?credito cadastro_positivo
negativad[oa] negativacao restricao(?: (cpf|credito|spc|serasa))? protesto
inadimplen(te|cia) dias?(de_)?atraso atraso_dias situacao?(do_)?
(credito|contrato|pagamento) classificacao_risco credit_score credit_rating
```

HOW IT IS MASKED

A score gets other digits of the same length; a flag stays a flag; a status is replaced with another from the list.

BR_CREDITO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_SITUACAO_CREDITO

BR_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

BR_FLAG REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

BR_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

BR_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

BR_FLAG_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

BR_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

BR_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

BR_SITUACAO_CREDITO SECURE LOOKUP

Replaces the value with one of the 10 values in `br-situacoes-credito.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Negativado no Serasa	→	Atraso de 90 dias
745	→	471

12.2 Debt

BR_FIN_DIVIDA

COLUMN NAME

DATA TYPE

NUMERIC MAPPING

WHAT IT IS

Outstanding balance, overdue amount, instalments and credit limits.

HOW IT IS FOUND

COLUMN NAME

BR_FIN_DIVIDA - Path

85%

saldo_?devedor valor_?(da_)?(divida|debito|parcela|financiamento|emprestimo|em_?atraso|atrasado) vl_?(divida|debito|parcela|saldo_?devedor|atraso) divida(_?total)? debito_?total limite_?(de_)??credito limite_?cartao prestacao parcela(_?mensal)? loan_?(amount|balance) outstanding_?balance debt

DATA TYPE

BR_FIN_DIVIDA - Type

Accepts number; any other type rules the domain out.

HOW IT IS MASKED

Mapped to a value between R\$ 200 and R\$ 1,000,000, unrelated to the real value.

BR_VALOR_DIVIDA NUMERIC MAPPING

Maps the number to another between 200 and 1,000,000.

EXAMPLES

VALUE

MASKED

8500	→	5134
------	---	------

12.3 Income and salary

BR_FIN_RENDA

COLUMN NAME

DATA TYPE

NUMERIC MAPPING

WHAT IT IS

Salary, pay, earnings and monthly or household income.

HOW IT IS FOUND

COLUMN NAME

BR_FIN_RENDA - Path

85%

salario(_?(base|bruto|liquido|mensal|contratual))? vl_?(salario|renda|remuneracao|provento) renda(_?(mensal|familiar|bruta|liquida|per_?capita|comprovada|presumida))? remuneracao proventos vencimentos(_?basico)? faturamento_?(mensal|anual)_?(pf|pessoa) rendimento(_?(mensal|anual|tributavel))? ganho_? mensal salary income wages?

DATA TYPE

BR_FIN_RENDA - Type

Accepts number; any other type rules the domain out.

HOW IT IS MASKED

Mapped to a value between R\$ 1,500 and R\$ 50,000.

BR_RENDA NUMERIC MAPPING

Maps the number to another between 1,500 and 50,000.

EXAMPLES

VALUE		MASKED
3200	→	6046

12.4 Assets

BR_FIN_PATRIMONIO

COLUMN NAME DATA TYPE NUMERIC MAPPING

WHAT IT IS

Value of real estate and goods, account balances, savings, investments and FGTS.

HOW IT IS FOUND

COLUMN NAME BR_FIN_PATRIMONIO - Path

```
85%    patrimonio(_?(liquido|total))? valor_(do_)?  
(imovel|patrimonio|bens|veiculo|venal|investimento|aplicacao) vl_  
(imovel|patrimonio|bens|venal|investimento|aplicacao) saldo_  
(conta|poupanca|investimento|aplicacao|fgts) bens_(e_?direitos|declarados) declaracao_  
(de_)?bens saldo_?fgts net_?worth account_?balance
```

DATA TYPE BR_FIN_PATRIMONIO - Type

Accepts number; any other type rules the domain out.

HOW IT IS MASKED

Mapped to a value between R\$ 20,000 and R\$ 5,000,000.

BR_PATRIMONIO NUMERIC MAPPING

Maps the number to another between 20,000 and 5,000,000.

EXAMPLES

VALUE		MASKED
185000	→	477515

12.5 Benefit and pension amount

BR_FIN_VALOR_BENEFICIO

COLUMN NAME DATA TYPE NUMERIC MAPPING

WHAT IT IS

Amount of a retirement pension, survivor's pension, allowance, BPC and child support.

HOW IT IS FOUND

COLUMN NAME BR_FIN_VALOR_BENEFICIO - Path

```
90%    valor_(do_)?(beneficio|aposentadoria|pensao|auxilio|bolsa|bpc) vl_  
(beneficio|aposentadoria|pensao|auxilio|bpc) renda_?mensal_?inicial rmi pensao_  
alimenticia aposentadoria_?valor pension_?amount
```

DATA TYPE BR_FIN_VALOR_BENEFICIO – Type

Accepts number; any other type rules the domain out.

HOW IT IS MASKED

Mapped to a value between R\$ 1,500 and R\$ 8,000.

BR_VALOR_BENEFICIO **NUMERIC MAPPING**

Maps the number to another between 1,500 and 8,000.

EXAMPLES

VALUE		MASKED
1800	→	7193

12.6 INSS benefit type

BR_FIN_BENEFICIO_INSS

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The type of social security or welfare benefit, in words or as a code: 31 and 91 (temporary incapacity), 32 and 92 (permanent incapacity), 80 (maternity pay), 87 and 88 (BPC for people with disabilities and for the elderly), 41 and 42 (retirement), 21 (survivor's pension). Several types reveal an illness, a disability or a pregnancy.

HOW IT IS FOUND

COLUMN NAME BR_FIN_BENEFICIO_INSS – Path

90% especie_?(do_)?beneficio (cd|co|tp|ds)?especie(_?beneficio)? tipo_?(de_)?beneficio(_?(inss|previdenciario))? beneficio_?(inss|previdenciario)(_?(tipo|especie))? natureza? beneficio b(31|91|32|92|87|88|41|42)

KNOWN VALUES BR_FIN_BENEFICIO_INSS – List

LIST	VALUES	STRENGTH	FIRST VALUES
br-detectar-beneficios-inss.txt	38	80%	Aposentadoria por idade, Aposentadoria por tempo de contribuição, Aposentadoria por tempo de contribuicao, Aposentadoria especial, Aposentadoria por incapacidade permanente, Auxílio por incapacidade temporária
br-detectar-codigos-beneficios-inss.txt	14	50%	21, 31, 32, 36, 41, 42

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A code is replaced with another benefit-type code; a text with another type from the list.

BR_BENEFICIO_INSS **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_BENEFICIO_INSS_LISTA .

BR_FLAG **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO **SECURE LOOKUP**

Replaces the value with one of the 2 values in br-flag-sim-nao.txt , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in br-flag-true-false.txt , chosen by a keyed hash.

BR_FLAG_YES_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-yes-no.txt` , chosen by a keyed hash.

BR_FLAG_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-s-n.txt` , chosen by a keyed hash.

BR_FLAG_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-y-n.txt` , chosen by a keyed hash.

BR_FLAG_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `br-flag-0-1.txt` , chosen by a keyed hash.

BR_BENEFICIO_INSS_CODIGO **SECURE LOOKUP**

Replaces the value with one of the 14 values in `br-beneficios-inss-codigos.txt` , chosen by a keyed hash.

BR_BENEFICIO_INSS_LISTA **SECURE LOOKUP**

Replaces the value with one of the 11 values in `br-beneficios-inss.txt` , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Auxílio-reclusão	→	BPC ao idoso
31	→	88

12.7 Benefit number

BR_FIN_NUMERO_BENEFICIO**COLUMN NAME****VALUE FORMAT****CHARACTER MAPPING****WHAT IT IS**

The INSS benefit number (NB), 123.456.789-0 .

HOW IT IS FOUND**COLUMN NAME****BR_FIN_NUMERO_BENEFICIO** - Path**85%**

```
n(r|u|um|o|ro|umero)?(do)?beneficio nb(?(inss|beneficio))? numero?beneficio?inss
beneficio?numero matricula?(inss|beneficio)
```

VALUE FORMAT**BR_FIN_NUMERO_BENEFICIO** - Regex**PATTERN****STRENGTH****NOTES**

```
\d{3}\.\d{3}\.\d{3}-\d
```

50%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Digit by digit, keeping format and separators.

BR_CM_DIGITOS**CHARACTER MAPPING**

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
123.456.789-0	→	171.367.811-5

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Taking part in social programmes and in the Cadastro Único — Bolsa Família, Benefício de Prestação Continuada, Auxílio Gás, Pé-de-Meia, Tarifa Social de Energia, Minha Casa, Minha Vida, Garantia-Safra, Seguro-Defeso —, often as an S/N flag.

HOW IT IS FOUND

COLUMN NAME BR_FIN_PROGRAMA_SOCIAL - Path

90% bolsa_?familia beneficiario_?(bolsa|programa|bpc|auxilio) programa_?
(social|governo|transferencia_?renda) auxilio_?(brasil|emergencial|gas) cad_?unico
cadunico cadastro_?unico inscrito_?cadunico pe_?de_?meia tarifa_?social bpc loas
minha_?casa_?minha_?vida mcmv seguro_?defeso garantia_?safra transferencia_?(de_?)?
renda social_?programme

KNOWN VALUES BR_FIN_PROGRAMA_SOCIAL - List

LIST	VALUES	STRENGTH	FIRST VALUES
br-detectar-programas-sociais.txt	31	80%	Bolsa Família, Bolsa Familia, Benefício de Prestação Continuada, Beneficio de Prestacao Continuada, Auxílio Gás dos Brasileiros, Auxilio Gas dos Brasileiros
br-detectar-flag.txt	8	30%	s, n, sim, não, nao, y

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A flag stays a flag and any other value is replaced with a programme from the list.

BR_PROGRAMA_SOCIAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_PROGRAMA_SOCIAL_LISTA .

BR_FLAG REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-sim-nao.txt , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-true-false.txt , chosen by a keyed hash.

BR_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-yes-no.txt , chosen by a keyed hash.

BR_FLAG_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-s-n.txt , chosen by a keyed hash.

BR_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-y-n.txt , chosen by a keyed hash.

BR_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-0-1.txt , chosen by a keyed hash.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

BR_PROGRAMA_SOCIAL_LISTA SECURE LOOKUP

Replaces the value with one of the 9 values in br-programas-sociais.txt , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Auxílio Brasil

→

Minha Casa, Minha Vida

S

→

S

12.9 Income bracket and social class

BR_FIN_FAIXA_RENDA

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The Cadastro Único brackets (extreme poverty, poverty, low income), the Critério Brasil classes (A, B1, B2, C1, C2, DE), income deciles and quintiles.

HOW IT IS FOUND

COLUMN NAME

BR_FIN_FAIXA_RENDA - Path

90%

```
faixa_(de_)?renda(_?(familiar|per_capita|cadunico))? faixa_salarial classe_?
(social|economica|socioeconomica) criterio_brasil nivel_socioeconomico situacao_(de_)?
(pobreza|vulnerabilidade) pobreza extrema_pobreza vulnerabilidade_social renda_per_?
capita_faixa quintil decil income_(bracket|quintile|decile)
```

KNOWN VALUES

BR_FIN_FAIXA_RENDA - List

LIST

VALUES

STRENGTH

FIRST VALUES

LIST	VALUES	STRENGTH	FIRST VALUES
br-detectar-faixas-renda.txt	16	40%	extrema pobreza, pobreza, baixa renda, acima de meio salário mínimo, acima de meio salario minimo, classe a

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Replaced within its own notation: a bracket with another bracket, a class with another class, a decile with another decile.

BR_FAIXA_RENDA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches:

BR_FAIXA_CADUNICO

BR_QUINTIL SECURE LOOKUP

Replaces the value with one of the 5 values in br-quintil.txt, chosen by a keyed hash.

BR_DECIL_ALTO SECURE LOOKUP

Replaces the value with one of the 5 values in br-decil-alto.txt, chosen by a keyed hash.

BR_DECIL SECURE LOOKUP

Replaces the value with one of the 10 values in br-decil.txt, chosen by a keyed hash.

BR_CLASSE_SOCIAL SECURE LOOKUP

Replaces the value with one of the 6 values in br-classes-sociais.txt, chosen by a keyed hash.

BR_FAIXA_CADUNICO SECURE LOOKUP

Replaces the value with one of the 4 values in br-faixas-cadunico.txt, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Extrema pobreza

→

Extrema pobreza

Classe C1

→

Classe B2

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

The tenure of the dwelling – owned outright or with a mortgage, rented, lent, occupation.

HOW IT IS FOUND

COLUMN NAME

BR_FIN_MORADIA – Path

90%

condicao_?(de_)?(ocupacao|moradia) situacao_?(do_)?(domicilio|imovel|moradia) tipo_?(de_)?moradia posse_?(do_)?imovel imovel_?(proprio|alugado) reside_?em_?imovel moradia_?(propria|alugada) domicilio_?(proprio|alugado) housing_?tenure

KNOWN VALUES

BR_FIN_MORADIA – List

LIST	VALUES	STRENGTH	FIRST VALUES
br-detectar-moradia.txt	26	60%	Próprio quitado, Proprio quitado, Próprio em pagamento, Proprio em pagamento, Alugado, Cedido

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Replaced with another tenure from the list.

BR_MORADIA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: BR_MORADIA_LISTA .

BR_FLAG REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-sim-nao.txt , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-true-false.txt , chosen by a keyed hash.

BR_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-yes-no.txt , chosen by a keyed hash.

BR_FLAG_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-s-n.txt , chosen by a keyed hash.

BR_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-y-n.txt , chosen by a keyed hash.

BR_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-0-1.txt , chosen by a keyed hash.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

BR_MORADIA_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in br-condicoes-moradia.txt , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Invasão	→	Outra

Criminal records and court cases

Not on the list of art. 5, II, but among the data that most expose people to discrimination: requiring a job candidate's record without justification gives rise to presumed moral damages (TST, IRR Tema 1), and the LGPD only stays out of the way when processing is exclusively for public security or for the investigation and prosecution of criminal offences (art. 4, III). In a company, this data gets the same care as sensitive data.

13.1 Criminal records

BR_PENAL_ANTECEDENTES

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Criminal record certificates, convictions, indictment, type of crime, prison status, sentence regime, electronic monitoring, socio-educational measures and arrest warrants.

HOW IT IS FOUND

COLUMN NAME BR_PENAL_ANTECEDENTES – Path

95% antecedentes?(?|(criminais|criminal|penais))? certidao?(de_)?antecedentes atestado?(de_)?antecedentes folha?(de_)?antecedentes ficha?criminal registro?criminal condenac(ao|oes)(?|(criminal|penal))? condenado crime?(?(tipo|natureza))? tipo?(de_)?crime delito infracao?penal artigo?(penal|cp) tipificacao?penal indiciamento indiciado reincidente situacao?(prisional|carceraria|penal) regime?(prisional|de?cumprimento) preso presidiario egresso?(prisional|sistema) medida?socioeducativa tornozeleira monitoramento?eletronico mandado?(de_)?prisao criminal?record

55% situacao?processual resultado?(do_)?processo sentenca

HOW IT IS MASKED

Replaced with a generic status from the list — nothing on record, convicted, acquitted —, whatever the outcome; a flag stays a flag.

BR_ANTECEDENTES REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_ANTECEDENTES_LISTA .

BR_FLAG REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

BR_FLAG_SIM_NAO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-sim-nao.txt , chosen by a keyed hash.

BR_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-true-false.txt , chosen by a keyed hash.

BR_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-yes-no.txt , chosen by a keyed hash.

BR_FLAG_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-s-n.txt , chosen by a keyed hash.

BR_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-y-n.txt , chosen by a keyed hash.

BR_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in br-flag-0-1.txt , chosen by a keyed hash.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

BR_ANTECEDENTES_LISTA **SECURE LOOKUP**

Replaces the value with one of the 8 values in `br-antecedentes.txt`, chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Condenado por furto em 2019	→ Não informado

13.2 Case number

BR_PENAL_PROCESSO

COLUMN NAME **VALUE FORMAT** **REGEX DECOMPOSE**

WHAT IT IS

The unique case number of the Judiciary (Resolução CNJ 65/2008), `0001234-56.2023.8.26.0100`: sequence, check digits, year, branch of the judiciary, court and origin. Cases can be looked up on court portals and lead to the parties. Also police reports, inquiries and summary reports.

HOW IT IS FOUND

COLUMN NAME	BR_PENAL_PROCESSO - Path		
90%	<pre>n(r u um o ro umero)?(do_)?processo(_?(judicial criminal penal cnj trabalhista))? processo_(judicial criminal penal cnj numero trabalhista) numeracao_unica (nr nu num)? cnj n(r u um o ro umero)?(bo boletim boletim_ocorrencia inquerito ip termo_ circunstanciado tco execucao_penal mandado) boletim_(de_)?ocorrencia inquerito(_? policial)?(_?numero)? autos_(numero n)</pre>		
50%	<pre>processo protocolo_judicial</pre>		
VALUE FORMAT	BR_PENAL_PROCESSO - Regex		
PATTERN	STRENGTH	NOTES	
<pre>\d{7}-\d{2}\.\d{4}\.\d\.\d{2}\.\d{4}</pre>		90%	

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

The sequence and the check digits are mapped digit by digit; year, branch, court and origin stay. The check digits (modulus 97) are not recomputed. Other numbers are masked character by character.

BR_PROCESSO **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.

BR_CM_DIGITOS **CHARACTER MAPPING**

Maps each character to another of its group (1 groups), keeping length and format.

BR_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
0001234-56.2023.8.26.0100	→ 7273347-72.2023.8.26.0100

Check digits computed by the set

Brazilian systems validate their identifiers. An invalid CPF in a test database breaks the registration, the integration with the Receita Federal or the issuing of an invoice. So where the check-digit rule is public, the set recomputes it.

- **CPF.** Two Check Digit algorithms, one inside the other. The outer one receives the eleven digits and hands the first ten to the inner one, which hands the first nine to a Character Mapping and recomputes the first digit with weights 10 to 2; the outer one recomputes the second with weights 11 to 2. Check Digit writes a remainder of 0 or 1 as 0 — exactly the Receita's rule. Separators stay out of the computation, and a short number is padded with zeros on the left.
- **CNPJ, alphanumeric included.** The same design with weights 5 to 2 and 9 to 2, and with the character handling *ASCII value minus 48* — the rule of Instrução Normativa RFB 2.229/2024, in which A is worth 17. Only the root is mapped.
- **PIS and RENAVAM.** One Check Digit with weights 3 2 9 8 7 6 5 4 3 2.
- **Voter card.** Check Digit cannot follow the voter card's rule: the first digit is the remainder itself, it comes after the two state digits, and in São Paulo and Minas Gerais a remainder of 0 gives 1. The set computes it with tables: the sequence is mapped and the old digits become 00; a running sum travels through the value two digits per step, each step a Data Cleansing that turns *sum + two digits* into *two digits + new sum*; at the end, a table turns *sum + state + 00* into the state and the two right digits.
- **National Health Card.** The PIS inside a definitive CNS is masked by the PIS algorithm, and the same step tables carry the weighted sum to the end, where 0000 becomes 000 or 001 and the digit that makes the sum a multiple of 11.

The tables are Data Cleansing files generated with the set — 1,100 lines per step —, and the steps are Regex Decompose algorithms joined in String Algorithm Chains. Every identifier stays one-to-one: two different CPFs never become the same CPF.

Where people live: municipalities and CEP

With the resident population of the 2022 Demographic Census (IBGE) and the location of each seat, the set knows the size, state and position of the 5,570 municipalities.

- A **municipality** with fewer than 20,000 inhabitants becomes the nearest municipality of the same state with at least 20,000. There are 3,860 of them, with 32 million inhabitants — 15.8% of the population. Serra da Saudade (MG), the smallest, becomes São Gotardo; Borá (SP) becomes Paraguaçu Paulista; Uiramutã (RR) becomes Boa Vista.
- The **state** stays: the smallest, Roraima, had 636,707 inhabitants.
- Distance is measured between the coordinates of the seats, in a straight line. Fernando de Noronha, more than 350 km off the coast, becomes the nearest municipality of Pernambuco with at least 20,000.
- Two hundred and thirty-two names repeat — there are five Bom Jesus and five São Domingos —, and others differ only in an accent (Goianá and Goiana). A name on its own is generalized only when every municipality bearing it is small, and then it follows the largest of them. With the state (Bom Jesus – RS , Bom Jesus/RS) there is no ambiguity, and the municipality is always generalized.
- The **IBGE code**, of seven or six digits, follows the same table, and a person's name and code still agree.

- The **CEP** loses its last five digits: the postal sector stays, which gathers many streets and, outside the capitals, several municipalities.

Names are recognized as systems write them: with accents, without accents and in capitals. Municipality names that are also given names or surnames, and state names (`Paraná` , `Goiás`), are left out of the detection list, so that a column of surnames is not taken for municipalities.

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Limits

- **Binary columns** — images of RG and CNH, proof-of-life selfies, fingerprints, clinical PDFs in BLOBs — cannot be masked by these algorithms. Inventory them and report them as residual risk.
- **Check digits not recomputed.** CNH (the second digit depends on whether the first went past 9), RG (one rule per state), certificates, bank accounts (one rule per bank) and CNJ case numbers (modulus 97) are masked digit by digit, and applications that validate those digits may reject the values.
- **Provisional CNS and numbers with spaces.** A CNS starting with 7, 8 or 9, and a CNS or voter card written with spaces (`898 0012 3456 7890`), are masked digit by digit, without a valid digit.
- **State, DDD and bank code stay.** In a column where only state and DDD remain, they identify no one; combined with a birth date and a small municipality, those have already been generalized.
- **Data Cleansing lets through what it does not know:** a municipality written in a way the table does not have — abbreviated, misspelled — passes unchanged.
- **Free text** is redacted word by word: a phone written with spaces is not found, and a name that is also a common word (`Rosa` , `Luz` , `Paz`) is redacted wherever it appears.
- **Children and adolescents** — a birth date moves at most 60 days, so few people cross their 12th or 18th birthday, but some do. Where that matters, split the rule by age group with Multi-Column Condition.
- **Addresses** are replaced as a whole line. Street, neighbourhood, municipality and CEP in separate columns are masked independently and no longer form a coherent address — a masked CEP does not match the masked street.
- **The engine key** links masked values back to production for whoever holds both: in that case the result is pseudonymized data (art. 13, § 4), still personal data. Keep the key away from the users of the copy.

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Verification

Before publication, the set was checked outside a Masking Engine:

- The check-digit rules were checked against published examples: the CPF `529.982.247-25` , the first alphanumeric CNPJ issued by the Receita Federal, `00.000.000/E08G-12` (July 2026), and the voter card `1023 8501 0671` of OBMEP.
- Every classifier configuration was reviewed against what a Masking Engine accepts.
- Profiling was simulated on 164 columns with the whole set at its assignment threshold — well named, badly named and 33 columns that are not personal data — and again with only the classifiers of the essential pack.
- The table that generalizes municipalities was audited for all 5,570 municipalities: by name, by name with the state and by IBGE code.
- The algorithms, run with the Delphix algorithm plugin, masked each domain's sample values and placeholders such as `N/I` , checking every output against its vocabulary and every CPF, CNPJ, PIS, voter

card, CNS and RENAVAM against its check-digit rule; 150 of each for collisions and validity; and 20 people whose PIS and CNS must still agree. Every algorithm also masked its own sample value.

Every column was assigned as expected and every value masked as expected. Before using the set on real data, run a profiling job and a masking job on a representative copy.

Sources

- Lei 13.709 of 14 August 2018 — General Personal Data Protection Law, compiled text (Presidency of the Republic), with the amendments of Laws 13.853/2019, 14.010/2020, 14.460/2022 and 15.352/2026. Federal Constitution, art. 5, LXXIX (Constitutional Amendment 115/2022). Lei 15.211/2025 (Digital Statute of Children and Adolescents).
- Agência Nacional de Proteção de Dados: preliminary study *Anonimização e Pseudonimização para a Proteção de Dados Pessoais* (public consultation, January 2024); Resolução CD/ANPD 15/2024 (Security Incident Reporting Regulation); Resolução CD/ANPD 4/2023 (Sanction Calculation Regulation); Regulatory Agenda 2025–2026 progress report (second half of 2025).
- Lei Complementar 105/2001 (bank secrecy). Lei 12.414/2011 (positive credit register). Lei 14.534/2023 (CPF as the single number). Lei 13.146/2015 (Brazilian Inclusion Law). Superior Labour Court, IRR Tema 1 (requiring criminal record certificates).
- Resident population by municipality, 2022 Demographic Census (IBGE, SIDRA, table 4709). Most frequent given names and surnames — *Nomes no Brasil*, 2022 Census (IBGE). Colour or race, religion and indigenous peoples — 2022 Census (IBGE). Coordinates of municipal seats — [kelvins/municipios-brasileiros](https://kelvins.com.br/municipios-brasileiros) (MIT licence).
- Registered political parties — Superior Electoral Court, September 2026. Voter card check digits — OBMEP Math Clubs. Alphanumeric CNPJ — Receita Federal, Instrução Normativa RFB 2.229/2024. CEP structure — Correios. National Health Card — Ministry of Health. Unique case numbering — Resolução CNJ 65/2008. Brazilian Classification of Occupations — Ministry of Labour and Employment.

APPENDIX A

Algorithms

Every algorithm of the set in alphabetical order, with its configuration. The one a domain names is the top of its tree; the others are building blocks.

BR_ANO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_L2_ANO_NASCIMENTO

PATTERNS, TRIED IN ORDER

(\d{3})(\d)	keep · BR_CM_DIGITOS
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_ANTECEDENTES REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_ANTECEDENTES_LISTA .

Used by: BR_PENAL_ANTECEDENTES

PATTERNS, TRIED IN ORDER

(?i)(sim n[aa]o true false yes no [sny01])	BR_FLAG
(\d{1,6})	BR_CM_ALFANUM
otherwise	BR_ANTECEDENTES_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

BR_ANTECEDENTES_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in br-antecedentes.txt , chosen by a keyed hash.

Used by: BR_ANTECEDENTES

CONFIGURATION

lookupFile	br-antecedentes.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_BAIRRO SECURE LOOKUP

Replaces the value with one of the 40 values in br-bairros.txt , chosen by a keyed hash.

Used by: BR_L2_BAIRRO

CONFIGURATION

lookupFile	br-bairros.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_BENEFICIO_INSS REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_BENEFICIO_INSS_LISTA .

Used by: BR_FIN_BENEFICIO_INSS

PATTERNS, TRIED IN ORDER

(?i)(sim n[aã]o true false yes no [sny01])	BR_FLAG
(\d{1,6})	BR_BENEFICIO_INSS_CODIGO
otherwise	BR_BENEFICIO_INSS_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

BR_BENEFICIO_INSS_CODIGO SECURE LOOKUP

Replaces the value with one of the 14 values in br-beneficios-inss-codigos.txt , chosen by a keyed hash.

Used by: BR_BENEFICIO_INSS

CONFIGURATION

lookupFile	br-beneficios-inss-codigos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_BENEFICIO_INSS_LISTA SECURE LOOKUP

Replaces the value with one of the 11 values in br-beneficios-inss.txt , chosen by a keyed hash.

Used by: BR_BENEFICIO_INSS

CONFIGURATION

lookupFile	br-beneficios-inss.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_CARTAO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_L1_CARTAO

PATTERNS, TRIED IN ORDER

(\d[\d\s\~]{11,22}\d)	BR_CARTAO_LUHN
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_CARTAO_LUHN PAYMENT CARD

Replaces the card number with another that passes the Luhn check.

Used by: BR_CARTAO

CONFIGURATION

minMaskedPositions	6
preserve	0

BR_CATEGORIA_SUPRIMIDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: BR_SUPRIMIR .

Used by: BR_L3_CONVICCAO_ORGANIZACAO , BR_L3_FILIACAO_SINDICAL , BR_L3_VIDA_SEXUAL , BR_SAUDE_REPRODUTIVA , BR_SAUDE_MENTAL

PATTERNS, TRIED IN ORDER

(?i)(sim n[aã]o true false yes no [sny01])	BR_FLAG
(\d{1,6})	BR_CM_ALFANUM
otherwise	BR_SUPRIMIR

CONFIGURATION

trimInput	yes
requireMask	no

BR_CBO SECURE LOOKUP

Replaces the value with one of the 34 values in br-cbo.txt , chosen by a keyed hash.

Used by: BR_OCUPACAO_OU_CBO

CONFIGURATION

lookupFile	br-cbo.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_CEP REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: BR_L2_CEP

PATTERNS, TRIED IN ORDER

(\d{2})(\.\?)(\d)(\d{2})(-?)(\d{3})	keep · keep · keep · replace with 00 · keep · replace with 000
(\d{2})(\d{2})(\d{3})	keep · replace with 00 · replace with 000
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_CHAVE_PIX REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: BR_CM_ALFANUM .

Used by: BR_L1_CHAVE_PIX

PATTERNS, TRIED IN ORDER

([^\s]+@[^\s]+)	BR_EMAIL
(\+55\d{10,11})	BR_TELEFONE
(\d{3}\.\d{3}\.\d{3}-\d{2} \d{11})	BR_CPF
([0-9A-Z]{2}\.[0-9A-Z]{3}\.[0-9A-Z]{3}/[0-9A-Z]{4}-\d{2} [0-9A-Z]{12}\d{2})	BR_CNPJ
([0-9a-fA-F]{8}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{12})	BR_CM_HEX
otherwise	BR_CM_ALFANUM

CONFIGURATION

trimInput	yes
requireMask	no

BR_CID10 SECURE LOOKUP

Replaces the value with one of the 37 values in br-cid10.txt , chosen by a keyed hash.

Used by: BR_DIAGNOSTICO

CONFIGURATION

lookupFile	br-cid10.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_CID10_DECIMAL SECURE LOOKUP

Replaces the value with one of the 33 values in br-cid10-decimal.txt , chosen by a keyed hash.

Used by: BR_DIAGNOSTICO

CONFIGURATION

lookupFile	br-cid10-decimal.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_CLASSE_SOCIAL SECURE LOOKUP

Replaces the value with one of the 6 values in br-classes-sociais.txt , chosen by a keyed hash.

Used by: BR_FAIXA_RENDA

CONFIGURATION

lookupFile	br-classes-sociais.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

Used by: BR_L1_CNH, BR_L1_CTPS, BR_L1_PASSAPORTE, BR_L1_DOCUMENTO_ESTRANGEIRO, BR_L1_CERTIDAO, BR_L1_ENDERECO_COMPLEMENTO, BR_L1_CHASSI, BR_L1_IMOVEL, BR_L1_CONTRATO, BR_L1_USUARIO, BR_SAUDE_IDENTIFICADOR, BR_SAUDE_RESULTADO_EXAME, BR_CATEGORIA_SUPRIMIDA, BR_NOMES, BR_SOBRENOMES, BR_NOME_COMPLETO, BR_EMAIL, BR_CHAVE_PIX, BR_PLACA, BR_ETNIA, BR_RELIGIAO, BR_OPINIAO_POLITICA, BR_ORIENTACAO_SEXUAL, BR_IDENTIDADE_GENERO, BR_MEDICAMENTO, BR_DEFICIENCIA, BR_SAUDE_OCUPACIONAL, BR_PROGRAMA_SOCIAL, BR_MORADIA, BR_ANTECEDENTES, BR_PROCESSO

CONFIGURATION

characterGroups	0123456789 AEIOU BCDFGHJKLMNPQRSTUVWXYZ aeiou bcd fghjklmnpqrstvwxyz ÁÉÍÓÚÂÊÔÃÖÀ áéíóúâêôãöà
caseSensitive	yes
minMaskedPositions	0

BR_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

Used by: BR_L1_RG, BR_L1_CONTA_BANCARIA, BR_L2_ZONA_SECAO_ELEITORAL, BR_FIN_NUMERO_BENEFICIO, BR_CPF_DV1, BR_CPF, BR_CNPJ, BR_TITULO_CORPO, BR_TITULO, BR_PIS_CORPO, BR_PIS, BR_REGISTRO_PROFISSIONAL, BR_TELEFONE, BR_PLACA, BR_RENAVAM_VALIDO, BR_RENAVAM, BR_COORDENADA, BR_ANO, BR_IDADE, BR_OCUPACAO_OU_CBO, BR_CNS, BR_CREDITO, BR_PROCESSO

CONFIGURATION

characterGroups	0123456789
caseSensitive	yes
minMaskedPositions	0

BR_CM_DIGITOS_LETRAS CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

Used by: BR_CNPJ_RAIZ, BR_PLACA

CONFIGURATION

characterGroups	0123456789 ABCDEFGHIJKLMNOPQRSTUVWXYZ
caseSensitive	yes
minMaskedPositions	0

BR_CM_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

Used by: BR_L1_DISPOSITIVO, BR_CHAVE_PIX, BR_IP

CONFIGURATION

characterGroups	0123456789 abcdef ABCDEF ghijklmnopqrstuvwxyz GHIJKLMNOPQRSTUVWXYZ
-----------------	---

caseSensitive	yes
minMaskedPositions	0

BR_CNPJ REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

Used by: BR_L1_CNPJ , BR_CHAVE_PIX

PATTERNS, TRIED IN ORDER

([0-9A-Z]{2}\. [0-9A-Z]{3}\. [0-9A-Z]{3}/ [0-9A-Z]{4}-\d{2})	BR_CNPJ_VALID0
([0-9A-Z]{12}\d{2})	BR_CNPJ_VALID0
(\d{3}\.\d{3}\.\d{3}-\d{2} \d{9}-\d{2})	BR_CPF_VALID0
(\d{3,13})	BR_CNPJ_VALID0_ZER0S
otherwise	BR_CM_DIGITOS

CONFIGURATION

trimInput	yes
requireMask	no

BR_CNPJ_DV1 CHECK DIGIT

Masks the digits with BR_CNPJ_RAIZ and recomputes a modulus 11 check digit with weights 5 4 3 2 9 8 7 6 5 4 3 2 .

Used by: BR_CNPJ_VALID0 , BR_CNPJ_VALID0_ZER0S

CONFIGURATION

weightList	5 4 3 2 9 8 7 6 5 4 3 2
modulusNumber	11
checkDigitIndex	12
numDigitsForCheckdigitCalculation	12
calculateChecksumRightToLeft	no
numericAlgorithm	BR_CNPJ_RAIZ
alphaNumericAlgorithm	BR_CNPJ_RAIZ
preserveRegex	[.\-/]
inputHandlingConfig.characterHandling	ASCII_VALUE_MINUS_48
inputHandlingConfig.invalidInputHandling	ERROR
inputHandlingConfig.shortInputHandling	FALLBACK
inputHandlingConfig.padCharacter	0
inputHandlingConfig.trimWhitespace	yes

BR_CNPJ_RAIZ REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_CNPJ_DV1

PATTERNS, TRIED IN ORDER

([0-9A-Z]{2}\.?[0-9A-Z]{3}\.?[0-9A-Z]{3})(/[? [0-9A-Z]{4})	BR_CM_DIGITOS_LETRAS · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_CNPJ_VALIDO CHECK DIGIT

Masks the digits with BR_CNPJ_DV1 and recomputes a modulus 11 check digit with weights 6 5 4 3 2 9 8 7 6 5 4 3 2 .

Used by: BR_CPF , BR_CNPJ

CONFIGURATION

weightList	6 5 4 3 2 9 8 7 6 5 4 3 2
modulusNumber	11
checkDigitIndex	13
numDigitsForCheckdigitCalculation	13
calculateChecksumRightToLeft	no
numericAlgorithm	BR_CNPJ_DV1
alphaNumericAlgorithm	BR_CNPJ_DV1
preserveRegex	[.\-\/]
inputHandlingConfig.characterHandling	ASCII_VALUE_MINUS_48
inputHandlingConfig.invalidInputHandling	ERROR
inputHandlingConfig.shortInputHandling	FALLBACK
inputHandlingConfig.padCharacter	0
inputHandlingConfig.trimWhitespace	yes

BR_CNPJ_VALIDO_ZEROS CHECK DIGIT

Masks the digits with BR_CNPJ_DV1 and recomputes a modulus 11 check digit with weights 6 5 4 3 2 9 8 7 6 5 4 3 2 .

Used by: BR_CPF , BR_CNPJ

CONFIGURATION

weightList	6 5 4 3 2 9 8 7 6 5 4 3 2
modulusNumber	11
checkDigitIndex	13
numDigitsForCheckdigitCalculation	13
calculateChecksumRightToLeft	no
numericAlgorithm	BR_CNPJ_DV1
alphaNumericAlgorithm	BR_CNPJ_DV1
preserveRegex	[.\-\/]
inputHandlingConfig.characterHandling	ASCII_VALUE_MINUS_48
inputHandlingConfig.invalidInputHandling	ERROR
inputHandlingConfig.shortInputHandling	PAD_LEFT
inputHandlingConfig.padCharacter	0
inputHandlingConfig.trimWhitespace	yes

BR_CNS REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

Used by: BR_SAUDE_CNS

PATTERNS, TRIED IN ORDER

<code>([12]\d{14})</code>	<code>BR_CNS_VALIDO</code>
<code>([789])(\d{14})</code>	<code>keep · BR_CM_DIGITOS</code>
<code>otherwise</code>	<code>BR_CM_DIGITOS</code>

CONFIGURATION

<code>trimInput</code>	<code>yes</code>
<code>requireMask</code>	<code>no</code>

BR_CNS_CORPO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `BR_CNS_VALIDO`

PATTERNS, TRIED IN ORDER

<code>(\d{11})(\d{4})</code>	<code>BR_PIS_VALIDO · replace with 0000</code>
<code>otherwise</code>	<code>keep</code>

CONFIGURATION

<code>trimInput</code>	<code>yes</code>
<code>requireMask</code>	<code>no</code>

BR_CNS_DV STRING ALGORITHM CHAIN

Applies `BR_CNS_DV_P1` → `BR_CNS_DV_P2` → `BR_CNS_DV_P3` → `BR_CNS_DV_P4` → `BR_CNS_DV_P5` → `BR_CNS_DV_P6` , in this order.

Used by: `BR_CNS_VALIDO`

CONFIGURATION

<code>algorithmReferences[1]</code>	<code>BR_CNS_DV_P1</code>
<code>algorithmReferences[2]</code>	<code>BR_CNS_DV_P2</code>
<code>algorithmReferences[3]</code>	<code>BR_CNS_DV_P3</code>
<code>algorithmReferences[4]</code>	<code>BR_CNS_DV_P4</code>
<code>algorithmReferences[5]</code>	<code>BR_CNS_DV_P5</code>
<code>algorithmReferences[6]</code>	<code>BR_CNS_DV_P6</code>

BR_CNS_DV_P1 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `BR_CNS_DV`

PATTERNS, TRIED IN ORDER

<code>(.{0})(.{2})(.*)</code>	<code>keep · BR_CNS_DV_T1 · keep</code>
<code>otherwise</code>	<code>keep</code>

CONFIGURATION

<code>trimInput</code>	<code>yes</code>
<code>requireMask</code>	<code>no</code>

BR_CNS_DV_P2 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `BR_CNS_DV`

PATTERNS, TRIED IN ORDER

(.{2})(.{3})(.*)	keep · BR_CNS_DV_T2 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_CNS_DV_P3 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_CNS_DV

PATTERNS, TRIED IN ORDER

(.{4})(.{3})(.*)	keep · BR_CNS_DV_T3 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_CNS_DV_P4 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_CNS_DV

PATTERNS, TRIED IN ORDER

(.{6})(.{3})(.*)	keep · BR_CNS_DV_T4 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_CNS_DV_P5 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_CNS_DV

PATTERNS, TRIED IN ORDER

(.{8})(.{3})(.*)	keep · BR_CNS_DV_T5 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_CNS_DV_P6 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_CNS_DV

PATTERNS, TRIED IN ORDER

(.{10})(.{6})(.*)	keep · BR_CNS_DV_T6 · keep
-------------------	----------------------------

otherwise	keep
CONFIGURATION	
trimInput	yes
requireMask	no

BR_CNS_DV_T1 DATA CLEANSING

Replaces each value found in `br-cns-dv-1.txt` (100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_CNS_DV_P1`

CONFIGURATION	
lookupFile	<code>br-cns-dv-1.txt</code>
delimiter	,
caseSensitive	yes
trimWhitespace	yes

BR_CNS_DV_T2 DATA CLEANSING

Replaces each value found in `br-cns-dv-2.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_CNS_DV_P2`

CONFIGURATION	
lookupFile	<code>br-cns-dv-2.txt</code>
delimiter	,
caseSensitive	yes
trimWhitespace	yes

BR_CNS_DV_T3 DATA CLEANSING

Replaces each value found in `br-cns-dv-3.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_CNS_DV_P3`

CONFIGURATION	
lookupFile	<code>br-cns-dv-3.txt</code>
delimiter	,
caseSensitive	yes
trimWhitespace	yes

BR_CNS_DV_T4 DATA CLEANSING

Replaces each value found in `br-cns-dv-4.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_CNS_DV_P4`

CONFIGURATION	
lookupFile	<code>br-cns-dv-4.txt</code>
delimiter	,
caseSensitive	yes
trimWhitespace	yes

BR_CNS_DV_T5 DATA CLEANSING

Replaces each value found in `br-cns-dv-5.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_CNS_DV_P5`

CONFIGURATION

lookupFile	<code>br-cns-dv-5.txt</code>
delimiter	,
caseSensitive	yes
trimWhitespace	yes

BR_CNS_DV_T6 DATA CLEANSING

Replaces each value found in `br-cns-dv-6.txt` (110 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_CNS_DV_P6`

CONFIGURATION

lookupFile	<code>br-cns-dv-6.txt</code>
delimiter	,
caseSensitive	yes
trimWhitespace	yes

BR_CNS_VALIDO STRING ALGORITHM CHAIN

Applies `BR_CNS_CORPO` → `BR_CNS_DV`, in this order.

Used by: `BR_CNS`

CONFIGURATION

algorithmReferences[1]	<code>BR_CNS_CORPO</code>
algorithmReferences[2]	<code>BR_CNS_DV</code>

BR_CODIGO_MUNICIPIO DATA CLEANSING

Replaces each value found in `br-codigos-municipio-generalizados.txt` (7,720 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_L2_CODIGO_MUNICIPIO`

CONFIGURATION

lookupFile	<code>br-codigos-municipio-generalizados.txt</code>
delimiter	,
caseSensitive	yes
trimWhitespace	yes

BR_CONTAGEM_LIMITADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `BR_L2_DEPENDENTES`

PATTERNS, TRIED IN ORDER

<code>([0-5])</code>	keep
<code>(\d+)</code>	replace with <code>5</code>
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_COORDENADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: BR_L1_GEOLOCALIZACAO

PATTERNS, TRIED IN ORDER

(-?\d{1,3}[.,]\d)(\d+)\s*([,;])\s*(-?\d{1,3}[.,]\d)(\d+)	keep · BR_CM_DIGITOS · keep · keep · BR_CM_DIGITOS
(-?\d{1,3}[.,]\d)(\d+)	keep · BR_CM_DIGITOS
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_CPF REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

Used by: BR_L1_CPF , BR_CHAVE_PIX

PATTERNS, TRIED IN ORDER

(\d{3}\.\?\d{3}\.\?\d{3}-?\d{2})	BR_CPF_VALIDO
(\d{3,10})	BR_CPF_VALIDO_ZEROS
([0-9A-Z]{2}\.[0-9A-Z]{3}\.[0-9A-Z]{3}/[0-9A-Z]{4}-\d{2})	BR_CNPJ_VALIDO
([0-9A-Z]{12}\d{2})	BR_CNPJ_VALIDO
(\d{12,13})	BR_CNPJ_VALIDO_ZEROS
otherwise	BR_CM_DIGITOS

CONFIGURATION

trimInput	yes
requireMask	no

BR_CPF_DV1 CHECK DIGIT

Masks the digits with BR_CM_DIGITOS and recomputes a modulus 11 check digit with weights 10 9 8 7 6 5 4 3 2 .

Used by: BR_CPF_VALIDO , BR_CPF_VALIDO_ZEROS

CONFIGURATION

weightList	10 9 8 7 6 5 4 3 2
modulusNumber	11
checkDigitIndex	9
numDigitsForCheckdigitCalculation	9
calculateChecksumRightToLeft	no
numericAlgorithm	BR_CM_DIGITOS
alphaNumericAlgorithm	BR_CM_DIGITOS
preserveRegex	[.\-/]

inputHandlingConfig.characterHandling	STANDARD
inputHandlingConfig.invalidInputHandling	ERROR
inputHandlingConfig.shortInputHandling	FALLBACK
inputHandlingConfig.padCharacter	0
inputHandlingConfig.trimWhitespace	yes

BR_CPF_VALIDO CHECK DIGIT

Masks the digits with `BR_CPF_DV1` and recomputes a modulus 11 check digit with weights `11 10 9 8 7 6 5 4 3 2`.

Used by: `BR_CPF`, `BR_CNPJ`

CONFIGURATION

weightList	11 10 9 8 7 6 5 4 3 2
modulusNumber	11
checkDigitIndex	10
numDigitsForCheckdigitCalculation	10
calculateChecksumRightToLeft	no
numericAlgorithm	BR_CPF_DV1
alphaNumericAlgorithm	BR_CPF_DV1
preserveRegex	[.\-/]
inputHandlingConfig.characterHandling	STANDARD
inputHandlingConfig.invalidInputHandling	ERROR
inputHandlingConfig.shortInputHandling	FALLBACK
inputHandlingConfig.padCharacter	0
inputHandlingConfig.trimWhitespace	yes

BR_CPF_VALIDO_ZEROS CHECK DIGIT

Masks the digits with `BR_CPF_DV1` and recomputes a modulus 11 check digit with weights `11 10 9 8 7 6 5 4 3 2`.

Used by: `BR_CPF`

CONFIGURATION

weightList	11 10 9 8 7 6 5 4 3 2
modulusNumber	11
checkDigitIndex	10
numDigitsForCheckdigitCalculation	10
calculateChecksumRightToLeft	no
numericAlgorithm	BR_CPF_DV1
alphaNumericAlgorithm	BR_CPF_DV1
preserveRegex	[.\-/]
inputHandlingConfig.characterHandling	STANDARD
inputHandlingConfig.invalidInputHandling	ERROR
inputHandlingConfig.shortInputHandling	PAD_LEFT
inputHandlingConfig.padCharacter	0
inputHandlingConfig.trimWhitespace	yes

BR_CREDITO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_SITUACAO_CREDITO .

Used by: BR_FIN_CREDITO

PATTERNS, TRIED IN ORDER

(\d{1,4})	BR_CM_DIGITOS
(?i)(sim n[ãõo] true false yes no [sny01])	BR_FLAG
otherwise	BR_SITUACAO_CREDITO

CONFIGURATION

trimInput	yes
requireMask	no

BR_DATA_EVENTO DATE SHIFT

Moves the date by -30 to 30 days.

Used by: BR_L2_DATA_EVENTO

CONFIGURATION

minRange	-30
maxRange	30
unit	DAYS
roll	no

BR_DATA_NASCIMENTO DATE SHIFT

Moves the date by -60 to 60 days.

Used by: BR_L2_DATA_NASCIMENTO

CONFIGURATION

minRange	-60
maxRange	60
unit	DAYS
roll	no

BR_DECIL SECURE LOOKUP

Replaces the value with one of the 10 values in br-decil.txt , chosen by a keyed hash.

Used by: BR_FAIXA_RENDA

CONFIGURATION

lookupFile	br-decil.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_DECIL_ALTO SECURE LOOKUP

Replaces the value with one of the 5 values in br-decil-alto.txt , chosen by a keyed hash.

Used by: BR_FAIXA_RENDA

CONFIGURATION

lookupFile	br-decil-alto.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_DEFICIENCIA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_DEFICIENCIA_LISTA .

Used by: BR_SAUDE_DEFICIENCIA

PATTERNS, TRIED IN ORDER

(?i)(sim n[aã]o true false yes no [sny01])	BR_FLAG
(\d{1,6})	BR_CM_ALFANUM
otherwise	BR_DEFICIENCIA_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

BR_DEFICIENCIA_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in br-deficiencias.txt , chosen by a keyed hash.

Used by: BR_DEFICIENCIA

CONFIGURATION

lookupFile	br-deficiencias.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_DIAGNOSTICO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches:

BR_DIAGNOSTICO_TEXTO .

Used by: BR_SAUDE_DIAGNOSTICO

PATTERNS, TRIED IN ORDER

([A-TV-Za-tv-z]\d{2}\.\d{1,2})	BR_CID10_DECIMAL
([A-TV-Za-tv-z]\d{2,3})	BR_CID10
(?i)(sim n[aã]o true false yes no [sny01])	BR_FLAG
otherwise	BR_DIAGNOSTICO_TEXTO

CONFIGURATION

trimInput	yes
requireMask	no

BR_DIAGNOSTICO_TEXTO SECURE LOOKUP

Replaces the value with one of the 30 values in `br-diagnostics.txt` , chosen by a keyed hash.

Used by: `BR_DIAGNOSTICO`

CONFIGURATION

lookupFile	br-diagnostics.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_EMAIL REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM` .

Used by: `BR_L1_EMAIL` , `BR_CHAVE_PIX`

PATTERNS, TRIED IN ORDER

<code>(([^@\\s]+)@([^\s]+)\.([A-Za-z]{2,24})(?:\.[A-Za-z]{2})?)</code>	<code>BR_CM_ALFANUM</code> · <code>BR_CM_ALFANUM</code> · replace with <code>test</code>
otherwise	<code>BR_CM_ALFANUM</code>

CONFIGURATION

trimInput	yes
requireMask	no

BR_EMPREGADOR SECURE LOOKUP

Replaces the value with one of the 24 values in `br-empregadores.txt` , chosen by a keyed hash.

Used by: `BR_L2_EMPREGADOR`

CONFIGURATION

lookupFile	br-empregadores.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_ENDERECO SECURE LOOKUP

Replaces the value with one of the 4,000 values in `br-enderecos.txt` , chosen by a keyed hash.

Used by: `BR_L1_ENDERECO`

CONFIGURATION

lookupFile	br-enderecos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_ESCOLA SECURE LOOKUP

Replaces the value with one of the 12 values in `br-escolas.txt` , chosen by a keyed hash.

Used by: `BR_L2_ESCOLA`

CONFIGURATION

lookupFile	br-escolas.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_ESCOLARIDADE REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`BR_ESCOLARIDADE_LISTA` .

Used by: `BR_L2_ESCOLARIDADE`

PATTERNS, TRIED IN ORDER

<code>(?i)(sim n[ãõo] true false yes no [sny01])</code>	<code>BR_FLAG</code>
<code>(\d{1,6})</code>	<code>BR_ESCOLARIDADE_CODIGO</code>
otherwise	<code>BR_ESCOLARIDADE_LISTA</code>

CONFIGURATION

trimInput	yes
requireMask	no

BR_ESCOLARIDADE_CODIGO SECURE LOOKUP

Replaces the value with one of the 12 values in `br-escolaridade-codigos.txt` , chosen by a keyed hash.

Used by: `BR_ESCOLARIDADE`

CONFIGURATION

lookupFile	br-escolaridade-codigos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_ESCOLARIDADE_LISTA SECURE LOOKUP

Replaces the value with one of the 10 values in `br-escolaridade.txt` , chosen by a keyed hash.

Used by: `BR_ESCOLARIDADE`

CONFIGURATION

lookupFile	br-escolaridade.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes

trimWhitespaceInLookupFile	yes
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BR_ESTADO_CIVIL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_ESTADO_CIVIL_LISTA .

Used by: BR_L2_ESTADO_CIVIL

PATTERNS, TRIED IN ORDER

(?i)(sim n[aã]o true false yes no [sny01])	BR_FLAG
(\d{1,6})	BR_ESTADO_CIVIL_CODIGO
otherwise	BR_ESTADO_CIVIL_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

BR_ESTADO_CIVIL_CODIGO SECURE LOOKUP

Replaces the value with one of the 5 values in br-estado-civil-codigos.txt , chosen by a keyed hash.

Used by: BR_ESTADO_CIVIL

CONFIGURATION

lookupFile	br-estado-civil-codigos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_ESTADO_CIVIL_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in br-estado-civil.txt , chosen by a keyed hash.

Used by: BR_ESTADO_CIVIL

CONFIGURATION

lookupFile	br-estado-civil.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_ETNIA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: BR_ETNIA_LISTA .

Used by: BR_L3_ETNIA

PATTERNS, TRIED IN ORDER

(?i)(sim n[aã]o true false yes no [sny01])	BR_FLAG
(\d{1,6})	BR_CM_ALFANUM
otherwise	BR_ETNIA_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

BR_ETNIA_LISTA SECURE LOOKUP

Replaces the value with one of the 30 values in `br-etnias.txt` , chosen by a keyed hash.

Used by: `BR_ETNIA`

CONFIGURATION

lookupFile	<code>br-etnias.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_FAIXA_CADUNICO SECURE LOOKUP

Replaces the value with one of the 4 values in `br-faixas-cadunico.txt` , chosen by a keyed hash.

Used by: `BR_FAIXA_RENDA`

CONFIGURATION

lookupFile	<code>br-faixas-cadunico.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_LOOKUP_FILE</code>
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_FAIXA_RENDA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: `BR_FAIXA_CADUNICO` .

Used by: `BR_FIN_FAIXA_RENDA`

PATTERNS, TRIED IN ORDER

<code>([1-5])</code>	<code>BR_QUINTIL</code>
<code>([6-9] 10)</code>	<code>BR_DECIL_ALTO</code>
<code>(?i)(decil\s*)(\d{1,2})</code>	<code>keep · BR_DECIL</code>
<code>(?i)(quintil\s*)([1-5])</code>	<code>keep · BR_QUINTIL</code>
<code>(?i)(classe\s*)?(A B1 B2 C1 C2 D-E DE D E)</code>	<code>keep · BR_CLASSE_SOCIAL</code>
otherwise	<code>BR_FAIXA_CADUNICO</code>

CONFIGURATION

trimInput	yes
requireMask	no

BR_FLAG REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: `keep`.

Used by: `BR_CATEGORIA_SUPRIMIDA` , `BR_ESTADO_CIVIL` , `BR_ESCOLARIDADE` , `BR_RACA_COR` , `BR_ETNIA` , `BR_RELIGIAO` , `BR_OPINIAO_POLITICA` , `BR_PARTIDO` , `BR_ORIENTACAO_SEXUAL` , `BR_IDENTIDADE_GENERO` , `BR_DIAGNOSTICO` , `BR_MEDICAMENTO` ,

BR_DEFICIENCIA , BR_SAUDE_OCUPACIONAL , BR_CREDITO , BR_BENEFICIO_INSS , BR_PROGRAMA_SOCIAL , BR_MORADIA , BR_ANTECEDENTES

PATTERNS, TRIED IN ORDER

(?i)(sim n[aã]o)	BR_FLAG_SIM_NAO
(?i)(true false)	BR_FLAG_TRUE_FALSE
(?i)(yes no)	BR_FLAG_YES_NO
(?i)([sn])	BR_FLAG_S_N
(?i)([yn])	BR_FLAG_Y_N
([01])	BR_FLAG_0_1
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt` , chosen by a keyed hash.

Used by: `BR_FLAG`

CONFIGURATION

lookupFile	<code>br-flag-0-1.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_FLAG_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt` , chosen by a keyed hash.

Used by: `BR_FLAG`

CONFIGURATION

lookupFile	<code>br-flag-s-n.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_FLAG_SIM_NAO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt` , chosen by a keyed hash.

Used by: `BR_FLAG`

CONFIGURATION

lookupFile	<code>br-flag-sim-nao.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no

trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt` , chosen by a keyed hash.

Used by: `BR_FLAG`

CONFIGURATION

lookupFile	<code>br-flag-true-false.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt` , chosen by a keyed hash.

Used by: `BR_FLAG`

CONFIGURATION

lookupFile	<code>br-flag-y-n.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt` , chosen by a keyed hash.

Used by: `BR_FLAG`

CONFIGURATION

lookupFile	<code>br-flag-yes-no.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_IDADE REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `BR_L2_IDADE`

PATTERNS, TRIED IN ORDER

<code>(\d{3,})</code>	replace with <code>90</code>
<code>([1-9])(\d)([.,]\d+)</code>	keep · <code>BR_CM_DIGITOS</code> · keep
<code>([1-9])(\d)</code>	keep · <code>BR_CM_DIGITOS</code>
<code>(\d)</code>	<code>BR_CM_DIGITOS</code>

otherwise	keep
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CONFIGURATION

trimInput	yes
requireMask	no

BR_IDENTIDADE_GENERO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_IDENTIDADE_GENERO_LISTA .

Used by: BR_L3_IDENTIDADE_GENERO

PATTERNS, TRIED IN ORDER

(?i)(sim n[aã]o true false yes no [sny01])	BR_FLAG
(\d{1,6})	BR_CM_ALFANUM
otherwise	BR_IDENTIDADE_GENERO_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

BR_IDENTIDADE_GENERO_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in br-identidades-genero.txt , chosen by a keyed hash.

Used by: BR_IDENTIDADE_GENERO

CONFIGURATION

lookupFile	br-identidades-genero.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_IP REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: BR_CM_HEX .

Used by: BR_L1_IP

PATTERNS, TRIED IN ORDER

(\d{1,3})\.\d{1,3}\.\d{1,3}\.\d{1,3}	BR_OCTETO · BR_OCTETO · BR_OCTETO · BR_OCTETO
otherwise	BR_CM_HEX

CONFIGURATION

trimInput	yes
requireMask	no

BR_MEDICAMENTO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_MEDICAMENTO_LISTA .

Used by: BR_SAUDE_MEDICAMENTO

PATTERNS, TRIED IN ORDER

(?i)(sim n[aã]o true false yes no [sny01])	BR_FLAG
(\d{1,6})	BR_CM_ALFANUM
otherwise	BR_MEDICAMENTO_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

BR_MEDICAMENTO_LISTA SECURE LOOKUP

Replaces the value with one of the 25 values in `br-medicamentos.txt` , chosen by a keyed hash.

Used by: `BR_MEDICAMENTO`

CONFIGURATION

lookupFile	br-medicamentos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_MORADIA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_MORADIA_LISTA` .

Used by: `BR_FIN_MORADIA`

PATTERNS, TRIED IN ORDER

(?i)(sim n[aã]o true false yes no [sny01])	BR_FLAG
(\d{1,6})	BR_CM_ALFANUM
otherwise	BR_MORADIA_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

BR_MORADIA_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in `br-condicoes-moradia.txt` , chosen by a keyed hash.

Used by: `BR_MORADIA`

CONFIGURATION

lookupFile	br-condicoes-moradia.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_MUNICIPIO DATA CLEANSING

Replaces each value found in `br-municipios-generalizados.txt` (55,203 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_L2_MUNICIPIO`

CONFIGURATION

lookupFile	br-municipios-generalizados.txt
delimiter	
caseSensitive	yes
trimWhitespace	yes

BR_NACIONALIDADE SECURE LOOKUP

Replaces the value with one of the 16 values in `br-nacionalidades.txt`, chosen by a keyed hash.

Used by: `BR_L2_NACIONALIDADE`

CONFIGURATION

lookupFile	br-nacionalidades.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_NOME NAME

Replaces the name with one of the 895 names in `br-nomes.txt`.

Used by: `BR_PALAVRA_NOME`

CONFIGURATION

lookupFile	br-nomes.txt
maskedValueCase	PRESERVE_INPUT
filterAccent	yes
inputCaseSensitive	no

BR_NOME_COMPLETO REGEX DECOMPOSE

Tries 7 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.

Used by: `BR_L1_NOME_COMPLETO`

PATTERNS, TRIED IN ORDER

<code>([^,]+), \s*(. +)</code>	<code>BR_SOBRENOMES</code> . <code>BR_NOMES</code>
<code>(\S+)</code>	<code>BR_PALAVRA_NOME</code>
<code>(\S+)\s+(\S+)</code>	<code>BR_PALAVRA_NOME</code> . <code>BR_PALAVRA_SOBRENOME</code>
<code>(\S+)\s+(\S+)\s+(\S+)</code>	<code>BR_PALAVRA_NOME</code> . <code>BR_PALAVRA_SOBRENOME</code> . <code>BR_PALAVRA_SOBRENOME</code>
<code>(\S+)\s+(\S+)\s+(\S+)\s+(\S+)</code>	<code>BR_PALAVRA_NOME</code> . <code>BR_PALAVRA_NOME</code> . <code>BR_PALAVRA_SOBRENOME</code> . <code>BR_PALAVRA_SOBRENOME</code>
<code>(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)</code>	<code>BR_PALAVRA_NOME</code> . <code>BR_PALAVRA_NOME</code> . <code>BR_PALAVRA_SOBRENOME</code> . <code>BR_PALAVRA_SOBRENOME</code> . <code>BR_PALAVRA_SOBRENOME</code>
<code>(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)</code>	<code>BR_PALAVRA_NOME</code> . <code>BR_PALAVRA_NOME</code> . <code>BR_PALAVRA_SOBRENOME</code> . <code>BR_PALAVRA_SOBRENOME</code> . <code>BR_PALAVRA_SOBRENOME</code> . <code>BR_PALAVRA_SOBRENOME</code>
otherwise	<code>BR_CM_ALFANUM</code>

CONFIGURATION

trimInput	yes
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requireMask	no
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BR_NOMES REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: BR_CM_ALFANUM .

Used by: BR_L1_NOME , BR_NOME_COMPLETO

PATTERNS, TRIED IN ORDER

(\S+)	BR_PALAVRA_NOME
(\S+)\s+(\S+)	BR_PALAVRA_NOME · BR_PALAVRA_NOME
(\S+)\s+(\S+)\s+(\S+)	BR_PALAVRA_NOME · BR_PALAVRA_NOME · BR_PALAVRA_NOME
otherwise	BR_CM_ALFANUM

CONFIGURATION

trimInput	yes
requireMask	no

BR_OCTETO NUMERIC MAPPING

Maps the number to another between 1 and 254.

Used by: BR_IP

CONFIGURATION

minValue	1
maxValue	254

BR_OCUPACAO SECURE LOOKUP

Replaces the value with one of the 40 values in br-ocupacoes.txt , chosen by a keyed hash.

Used by: BR_OCUPACAO_OU_CB0

CONFIGURATION

lookupFile	br-ocupacoes.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_OCUPACAO_OU_CB0 REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: BR_OCUPACAO .

Used by: BR_L2_OCUPACAO

PATTERNS, TRIED IN ORDER

(\d{6})	BR_CB0
(\d{4})-(\d{2})	BR_CM_DIGITOS · BR_CM_DIGITOS
otherwise	BR_OCUPACAO

CONFIGURATION

trimInput	yes
requireMask	no

BR_OPINIAO_POLITICA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_OPINIAO_POLITICA_LISTA .

Used by: BR_L3_OPINIAO_POLITICA

PATTERNS, TRIED IN ORDER

(?i)(sim n[aã]o true false yes no [sny01])	BR_FLAG
(\d{1,6})	BR_CM_ALFANUM
otherwise	BR_OPINIAO_POLITICA_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

BR_OPINIAO_POLITICA_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in br-opinioes-politicas.txt , chosen by a keyed hash.

Used by: BR_OPINIAO_POLITICA

CONFIGURATION

lookupFile	br-opinioes-politicas.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_ORIENTACAO_SEXUAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_ORIENTACAO_SEXUAL_LISTA .

Used by: BR_L3_ORIENTACAO_SEXUAL

PATTERNS, TRIED IN ORDER

(?i)(sim n[aã]o true false yes no [sny01])	BR_FLAG
(\d{1,6})	BR_CM_ALFANUM
otherwise	BR_ORIENTACAO_SEXUAL_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

BR_ORIENTACAO_SEXUAL_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in br-orientacoes-sexuais.txt , chosen by a keyed hash.

Used by: BR_ORIENTACAO_SEXUAL

CONFIGURATION

lookupFile	br-orientacoes-sexuais.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no

trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_PALAVRA_NOME REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: **BR_NOME** .

Used by: **BR_NOMES** , **BR_NOME_COMPLETO**

PATTERNS, TRIED IN ORDER

(?i)(d[aeo]s? e di del della van von der du le la)	keep
(?i)(j[uú]nior jr\. filho filha neto neta sobrinho segundo terceiro)	keep
otherwise	BR_NOME

CONFIGURATION

trimInput	yes
requireMask	no

BR_PALAVRA_SOBRENOME REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: **BR_SOBRENOME** .

Used by: **BR_SOBRENOMES** , **BR_NOME_COMPLETO**

PATTERNS, TRIED IN ORDER

(?i)(d[aeo]s? e di del della van von der du le la)	keep
(?i)(j[uú]nior jr\. filho filha neto neta sobrinho segundo terceiro)	keep
otherwise	BR_SOBRENOME

CONFIGURATION

trimInput	yes
requireMask	no

BR_PARTIDO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: **BR_PARTIDO_NOME** .

Used by: **BR_L3_FILIACAO_PARTIDARIA**

PATTERNS, TRIED IN ORDER

(?i)(sim n[aa]o true false yes no [sny01])	BR_FLAG
([1-9]\d)	BR_PARTIDO_NUMERO
(?i) (MDB PDT PT PCdoB PSB PSDB AGIR MOBILIZA CIDA DANIA PV AVANTE PP PSTU PCB PRTB DC PCO PODE REPUBLICANOS PSOL PL PSD SOLIDARIEDADE NOVO R EDE DEMOCRATA UP UNIAO PRD MISSAO UNIÃO MISSÃ O PFL DEM PMDB PSL PTB PROS PSC PMN PTC PHS P PS PR PRB)	BR_PARTIDO_SIGLA
otherwise	BR_PARTIDO_NOME

CONFIGURATION

trimInput	yes
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requireMask	no
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BR_PARTIDO_NOME SECURE LOOKUP

Replaces the value with one of the 31 values in `br-partidos-nomes.txt` , chosen by a keyed hash.

Used by: `BR_PARTIDO`

CONFIGURATION

lookupFile	<code>br-partidos-nomes.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_PARTIDO_NUMERO SECURE LOOKUP

Replaces the value with one of the 30 values in `br-partidos-numeros.txt` , chosen by a keyed hash.

Used by: `BR_PARTIDO`

CONFIGURATION

lookupFile	<code>br-partidos-numeros.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_PARTIDO_SIGLA SECURE LOOKUP

Replaces the value with one of the 30 values in `br-partidos-siglas.txt` , chosen by a keyed hash.

Used by: `BR_PARTIDO`

CONFIGURATION

lookupFile	<code>br-partidos-siglas.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_LOOKUP_FILE</code>
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_PATRIMONIO NUMERIC MAPPING

Maps the number to another between 20,000 and 5,000,000.

Used by: `BR_FIN_PATRIMONIO`

CONFIGURATION

minValue	<code>20000</code>
maxValue	<code>5000000</code>

BR_PIS REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS` .

Used by: `BR_L1_PIS_NIS`

PATTERNS, TRIED IN ORDER

(\d{3}\.?\d{5}\.?\d{2}-?\d)	BR_PIS_VALIDO
(\d{9,10})	BR_PIS_VALIDO_ZEROS
otherwise	BR_CM_DIGITOS

CONFIGURATION

trimInput	yes
requireMask	no

BR_PIS_CORPO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_PIS_VALIDO , BR_PIS_VALIDO_ZEROS

PATTERNS, TRIED IN ORDER

(\d)(.*)	keep · BR_CM_DIGITOS
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_PIS_VALIDO CHECK DIGIT

Masks the digits with BR_PIS_CORPO and recomputes a modulus 11 check digit with weights 3 2 9 8 7 6 5 4 3 2 .

Used by: BR_PIS , BR_CNS_CORPO

CONFIGURATION

weightList	3 2 9 8 7 6 5 4 3 2
modulusNumber	11
checkDigitIndex	10
numDigitsForCheckdigitCalculation	10
calculateChecksumRightToLeft	no
numericAlgorithm	BR_PIS_CORPO
alphaNumericAlgorithm	BR_PIS_CORPO
preserveRegex	[\.\-\/]
inputHandlingConfig.characterHandling	STANDARD
inputHandlingConfig.invalidInputHandling	ERROR
inputHandlingConfig.shortInputHandling	FALLBACK
inputHandlingConfig.padCharacter	0
inputHandlingConfig.trimWhitespace	yes

BR_PIS_VALIDO_ZEROS CHECK DIGIT

Masks the digits with BR_PIS_CORPO and recomputes a modulus 11 check digit with weights 3 2 9 8 7 6 5 4 3 2 .

Used by: BR_PIS

CONFIGURATION

weightList	3 2 9 8 7 6 5 4 3 2
modulusNumber	11

checkDigitIndex	10
numDigitsForCheckdigitCalculation	10
calculateChecksumRightToLeft	no
numericAlgorithm	BR_PIS_CORPO
alphaNumericAlgorithm	BR_PIS_CORPO
preserveRegex	[.\-/]
inputHandlingConfig.characterHandling	STANDARD
inputHandlingConfig.invalidInputHandling	ERROR
inputHandlingConfig.shortInputHandling	PAD_LEFT
inputHandlingConfig.padCharacter	0
inputHandlingConfig.trimWhitespace	yes

BR_PLACA REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: BR_CM_ALFANUM .

Used by: BR_L1_PLACA

PATTERNS, TRIED IN ORDER

([A-Za-z]{3})(-?)(\d)([A-Za-z0-9])(\d{2})	BR_CM_DIGITOS_LETRAS · keep · BR_CM_DIGITOS · BR_CM_DIGITOS_LETRAS · BR_CM_DIGITOS
otherwise	BR_CM_ALFANUM

CONFIGURATION

trimInput	yes
requireMask	no

BR_PROCESSO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: BR_CM_ALFANUM .

Used by: BR_PENAL_PROCESSO

PATTERNS, TRIED IN ORDER

(\d{7})(-?)(\d{2})(\.\?\d{4}\.\?\d\.\?\d{2}\.\?\d{4})	BR_CM_DIGITOS · keep · BR_CM_DIGITOS · keep \d{4})
otherwise	BR_CM_ALFANUM

CONFIGURATION

trimInput	yes
requireMask	no

BR_PROGRAMA_SOCIAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_PROGRAMA_SOCIAL_LISTA .

Used by: BR_FIN_PROGRAMA_SOCIAL

PATTERNS, TRIED IN ORDER

(?i)(sim n[aã]o true false yes no [sny01])	BR_FLAG
(\d{1,6})	BR_CM_ALFANUM
otherwise	BR_PROGRAMA_SOCIAL_LISTA

CONFIGURATION

trimInput	yes
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requireMask	no
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BR_PROGRAMA_SOCIAL_LISTA SECURE LOOKUP

Replaces the value with one of the 9 values in `br-programas-sociais.txt` , chosen by a keyed hash.

Used by: `BR_PROGRAMA_SOCIAL`

CONFIGURATION

lookupFile	<code>br-programas-sociais.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_QUINTIL SECURE LOOKUP

Replaces the value with one of the 5 values in `br-quintil.txt` , chosen by a keyed hash.

Used by: `BR_FAIXA_RENDA`

CONFIGURATION

lookupFile	<code>br-quintil.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_RACA_COR REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_RACA_COR_LISTA` .

Used by: `BR_L3_RACA_COR`

PATTERNS, TRIED IN ORDER

<code>(?i)(sim n[aa]o true false yes no [sny01])</code>	<code>BR_FLAG</code>
<code>(\d{1,6})</code>	<code>BR_RACA_COR_CODIGO</code>
otherwise	<code>BR_RACA_COR_LISTA</code>

CONFIGURATION

trimInput	yes
requireMask	no

BR_RACA_COR_CODIGO SECURE LOOKUP

Replaces the value with one of the 6 values in `br-raca-cor-codigos.txt` , chosen by a keyed hash.

Used by: `BR_RACA_COR`

CONFIGURATION

lookupFile	<code>br-raca-cor-codigos.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	no
trimWhitespaceFromInput	yes

trimWhitespaceInLookupFile	yes
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BR_RACA_COR_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in `br-raca-cor.txt` , chosen by a keyed hash.

Used by: `BR_RACA_COR`

CONFIGURATION

lookupFile	<code>br-raca-cor.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	<code>no</code>
trimWhitespaceFromInput	<code>yes</code>
trimWhitespaceInLookupFile	<code>yes</code>

BR_REDIGIR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `BR_L1_CREDENCIAL` , `BR_L3_GENETICO` , `BR_L3_BIOMETRICO`

PATTERNS, TRIED IN ORDER

<code>(.+)</code>	each character → <code>X</code>
otherwise	<code>keep</code>

CONFIGURATION

trimInput	<code>yes</code>
requireMask	<code>no</code>

BR_REGISTRO_PROFISSIONAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS` .

Used by: `BR_L1_REGISTRO_PROFISSIONAL`

PATTERNS, TRIED IN ORDER

<code>(?i)([A-Z]{2,6}\s*[-/]? \s*[A-Z]{2}\s*[-/]? \s*)(\d[\d.\-]*) (.*)</code>	<code>keep · BR_CM_DIGITOS · keep</code>
<code>(\d[\d.\-]*) (\s*[-/]? \s*[A-Za-z]{2})</code>	<code>BR_CM_DIGITOS · keep</code>
otherwise	<code>BR_CM_DIGITOS</code>

CONFIGURATION

trimInput	<code>yes</code>
requireMask	<code>no</code>

BR_RELIGIAO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_RELIGIAO_LISTA` .

Used by: `BR_L3_RELIGIAO`

PATTERNS, TRIED IN ORDER

<code>(?i)(sim n[aa]o true false yes no [sny01])</code>	<code>BR_FLAG</code>
<code>(\d{1,6})</code>	<code>BR_CM_ALFANUM</code>
otherwise	<code>BR_RELIGIAO_LISTA</code>

CONFIGURATION

trimInput	yes
requireMask	no

BR_RELIGIAO_LISTA SECURE LOOKUP

Replaces the value with one of the 13 values in br-religioes.txt , chosen by a keyed hash.

Used by: BR_RELIGIAO

CONFIGURATION

lookupFile	br-religioes.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_RENAVAM REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

Used by: BR_L1_RENAVAM

PATTERNS, TRIED IN ORDER

(\d{9,11})	BR_RENAVAM_VALIDO
otherwise	BR_CM_DIGITOS

CONFIGURATION

trimInput	yes
requireMask	no

BR_RENAVAM_VALIDO CHECK DIGIT

Masks the digits with BR_CM_DIGITOS and recomputes a modulus 11 check digit with weights 3 2 9 8 7 6 5 4 3 2 .

Used by: BR_RENAVAM

CONFIGURATION

weightList	3 2 9 8 7 6 5 4 3 2
modulusNumber	11
checkDigitIndex	10
numDigitsForCheckdigitCalculation	10
calculateChecksumRightToLeft	no
numericAlgorithm	BR_CM_DIGITOS
alphaNumericAlgorithm	BR_CM_DIGITOS
preserveRegex	[.\-/]
inputHandlingConfig.characterHandling	STANDARD
inputHandlingConfig.invalidInputHandling	ERROR
inputHandlingConfig.shortInputHandling	PAD_LEFT
inputHandlingConfig.padCharacter	0
inputHandlingConfig.trimWhitespace	yes

BR_RENDA NUMERIC MAPPING

Maps the number to another between 1,500 and 50,000.

Used by: BR_FIN_RENDA

CONFIGURATION

minValue	1500
maxValue	50000

BR_SAUDE_OCUPACIONAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

BR_SAUDE_OCUPACIONAL_LISTA .

Used by: BR_SAUDE_OCUPACIONAL

PATTERNS, TRIED IN ORDER

(?i)(sim n[aã]o true false yes no [sny01])	BR_FLAG
(\d{1,6})	BR_CM_ALFANUM
otherwise	BR_SAUDE_OCUPACIONAL_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

BR_SAUDE_OCUPACIONAL_LISTA SECURE LOOKUP

Replaces the value with one of the 4 values in br-aptidao.txt , chosen by a keyed hash.

Used by: BR_SAUDE_OCUPACIONAL

CONFIGURATION

lookupFile	br-aptidao.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_SEXO REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: BR_L2_SEX0

PATTERNS, TRIED IN ORDER

(?i)(feminino masculino)	BR_SEXO_FEMININO_MASCULINO
(?i)(mulher homem)	BR_SEXO_MULHER_HOMEM
(?i)(fem masc)	BR_SEXO_FEM_MASC
(?i)(female male)	BR_SEXO_FEMALE_MALE
(?i)([fm])	BR_SEXO_F_M
([12])	BR_SEXO_1_2
otherwise	keep

CONFIGURATION

trimInput	yes
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requireMask	no
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BR_SEX0_1_2 SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-1-2.txt` , chosen by a keyed hash.

Used by: `BR_SEX0`

CONFIGURATION

lookupFile	<code>br-sexo-1-2.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_SEX0_F_M SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-f-m.txt` , chosen by a keyed hash.

Used by: `BR_SEX0`

CONFIGURATION

lookupFile	<code>br-sexo-f-m.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_SEX0_FEM_MASC SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-fem-masc.txt` , chosen by a keyed hash.

Used by: `BR_SEX0`

CONFIGURATION

lookupFile	<code>br-sexo-fem-masc.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_SEX0_FEMALE_MALE SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-female-male.txt` , chosen by a keyed hash.

Used by: `BR_SEX0`

CONFIGURATION

lookupFile	<code>br-sexo-female-male.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes

trimWhitespaceInLookupFile	yes
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BR_SEXO_FEMININO_MASCULINO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-feminino-masculino.txt` , chosen by a keyed hash.

Used by: `BR_SEX0`

CONFIGURATION

lookupFile	<code>br-sexo-feminino-masculino.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	<code>no</code>
trimWhitespaceFromInput	<code>yes</code>
trimWhitespaceInLookupFile	<code>yes</code>

BR_SEXO_MULHER_HOMEM SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-mulher-homem.txt` , chosen by a keyed hash.

Used by: `BR_SEX0`

CONFIGURATION

lookupFile	<code>br-sexo-mulher-homem.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	<code>no</code>
trimWhitespaceFromInput	<code>yes</code>
trimWhitespaceInLookupFile	<code>yes</code>

BR_SITUACAO_CREDITO SECURE LOOKUP

Replaces the value with one of the 10 values in `br-situacoes-credito.txt` , chosen by a keyed hash.

Used by: `BR_CREDITO`

CONFIGURATION

lookupFile	<code>br-situacoes-credito.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	<code>no</code>
trimWhitespaceFromInput	<code>yes</code>
trimWhitespaceInLookupFile	<code>yes</code>

BR_SOBRENOME NAME

Replaces the name with one of the 535 names in `br-sobrenomes.txt` .

Used by: `BR_PALAVRA_SOBRENOME`

CONFIGURATION

lookupFile	<code>br-sobrenomes.txt</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
filterAccent	<code>yes</code>
inputCaseSensitive	<code>no</code>

BR_SOBRENOMES REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: BR_CM_ALFANUM .

Used by: BR_L1_SOBRENOME , BR_NOME_COMPLETO

PATTERNS, TRIED IN ORDER

(\S+)	BR_PALAVRA_SOBRENOME
(\S+)\s+(\S+)	BR_PALAVRA_SOBRENOME · BR_PALAVRA_SOBRENOME
(\S+)\s+(\S+)\s+(\S+)	BR_PALAVRA_SOBRENOME · BR_PALAVRA_SOBRENOME · BR_PALAVRA_SOBRENOME
(\S+)\s+(\S+)\s+(\S+)\s+(\S+)	BR_PALAVRA_SOBRENOME · BR_PALAVRA_SOBRENOME · BR_PALAVRA_SOBRENOME · BR_PALAVRA_SOBRENOME
otherwise	BR_CM_ALFANUM

CONFIGURATION

trimInput	yes
requireMask	no

BR_SUPRIMIR SECURE LOOKUP

Replaces the value with one of the 1 values in br-nao-informado.txt , chosen by a keyed hash.

Used by: BR_CATEGORIA_SUPRIMIDA

CONFIGURATION

lookupFile	br-nao-informado.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_TELEFONE REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

Used by: BR_L1_TELEFONE , BR_CHAVE_PIX

PATTERNS, TRIED IN ORDER

(\+?55\s?)(\(?0?\d{2}\)\s?)(9)(\d{4})([\s-]?) (\d{4})	keep · keep · keep · BR_CM_DIGITOS · keep · BR_CM_DIGITOS
(\+?55\s?)(\(?0?\d{2}\)\s?)([2-5])(\d{3})([\s-]?) (\d{4})	keep · keep · keep · BR_CM_DIGITOS · keep · BR_CM_DIGITOS
(9)(\d{4})([\s-]?) (\d{4})	keep · BR_CM_DIGITOS · keep · BR_CM_DIGITOS
([2-5])(\d{3})([\s-]?) (\d{4})	keep · BR_CM_DIGITOS · keep · BR_CM_DIGITOS
otherwise	BR_CM_DIGITOS

CONFIGURATION

trimInput	yes
requireMask	no

BR_TEXTO_LIVRE FREE TEXT REDACTION

Replaces the text matching 12 patterns with [DADO] , and the 3,302 terms of br-texto-nomes.txt with [NOME] .

Used by: BR_L1_TEXTO_LIVRE , BR_SAUDE_TEXTO_CLINICO

PATTERNS, TRIED IN ORDER

(?<![\d.-])\d{3}\.\d{3}\.\d{3}-\d{2}(?![\d\-])
(?<!\d)\d{11}(?!\\d)
(?<![\dA-Z.])[0-9A-Z]{2}\.[0-9A-Z]{3}\.[0-9A-Z]{3}/[0-9A-Z]{4}-\d{2}(?!\\d)
(?<!\d)\d{14}(?!\\d)
[\\w.+\\-]+@[\\w\\-]+(\\. [\\w\\-]+)+
(?<![\\d\\-])\\((?\\d{2}\\\\)?-?9?\\d{4}-\\d{4}(?! [\\d\\-])
(?<![\\d\\-])9\\d{4}-\\d{4}(?! [\\d\\-])
(?<!\d)(\\+?55)?\\d{10,11}(?!\\d)
(?<!\d)\d{13,19}(?!\\d)
(?<!\d)\d{5}-\d{3}(?!\\d)
(?<![A-Za-z])[A-Z]{3}-?\\d[A-Z0-9]\d{2}(?! [A-Za-z\\d])
(?<!\d)(\\d{1,3}\\\\.){3}\\d{1,3}(?!\\d)

CONFIGURATION

regExRedactValue	[DADO]
lookupFile	br-texto-nomes.txt
lookupFileRedactValue	[NOME]
isDenyList	yes

BR_TIPO_SANGUINEO SECURE LOOKUP

Replaces the value with one of the 8 values in br-tipos-sanguineos.txt , chosen by a keyed hash.

Used by: BR_SAUDE_TIPO_SANGUINEO

CONFIGURATION

lookupFile	br-tipos-sanguineos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

BR_TITULO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

Used by: BR_L1_TITULO_ELEITOR

PATTERNS, TRIED IN ORDER

(\\d{12})	BR_TITULO_VALIDO
(\\d{11})	BR_TITULO_VALIDO_1_ZERO
(\\d{10})	BR_TITULO_VALIDO_2_ZEROS
otherwise	BR_CM_DIGITOS

CONFIGURATION

trimInput	yes
requireMask	no

BR_TITULO_1_ZERO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_TITULO_VALIDO_1_ZERO

PATTERNS, TRIED IN ORDER

(\d)(\d{10})	BR_TITULO_1_ZERO_TABELA · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_TITULO_1_ZERO_TABELA DATA CLEANSING

Replaces each value found in br-zeros-1.txt (10 pairs) by its counterpart; values not in the file stay as they are.

Used by: BR_TITULO_1_ZERO

CONFIGURATION

lookupFile	br-zeros-1.txt
delimiter	,
caseSensitive	yes
trimWhitespace	yes

BR_TITULO_2_ZEROS REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_TITULO_VALIDO_2_ZEROS

PATTERNS, TRIED IN ORDER

(\d)(\d{9})	BR_TITULO_2_ZEROS_TABELA · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_TITULO_2_ZEROS_TABELA DATA CLEANSING

Replaces each value found in br-zeros-2.txt (10 pairs) by its counterpart; values not in the file stay as they are.

Used by: BR_TITULO_2_ZEROS

CONFIGURATION

lookupFile	br-zeros-2.txt
delimiter	,
caseSensitive	yes
trimWhitespace	yes

BR_TITULO_CORPO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_TITULO_VALIDO

PATTERNS, TRIED IN ORDER

(\d{8})(\d{2})(\d{2})	BR_CM_DIGITOS · keep · replace with 00
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_TITULO_DV STRING ALGORITHM CHAIN

Applies BR_TITULO_DV_P1 → BR_TITULO_DV_P2 → BR_TITULO_DV_P3 → BR_TITULO_DV_P4 → BR_TITULO_DV_P5 , in this order.

Used by: BR_TITULO_VALIDO

CONFIGURATION

algorithmReferences[1]	BR_TITULO_DV_P1
algorithmReferences[2]	BR_TITULO_DV_P2
algorithmReferences[3]	BR_TITULO_DV_P3
algorithmReferences[4]	BR_TITULO_DV_P4
algorithmReferences[5]	BR_TITULO_DV_P5

BR_TITULO_DV_P1 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_TITULO_DV

PATTERNS, TRIED IN ORDER

(.{0})(.{2})(.*)	keep · BR_TITULO_DV_T1 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_TITULO_DV_P2 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_TITULO_DV

PATTERNS, TRIED IN ORDER

(.{2})(.{3})(.*)	keep · BR_TITULO_DV_T2 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

BR_TITULO_DV_P3 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_TITULO_DV

PATTERNS, TRIED IN ORDER

(.{4})(.{3})(.*)	keep · BR_TITULO_DV_T3 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
-----------	-----

requireMask	no
BR_TITULO_DV_P4 REGEX DECOMPOSE	
Matches one pattern and gives each part of the value its own action; when none matches: keep.	
Used by: BR_TITULO_DV	
PATTERNS, TRIED IN ORDER	
(.{6})(.{3})(.*)	keep · BR_TITULO_DV_T4 · keep
otherwise	keep
CONFIGURATION	
trimInput	yes
requireMask	no
BR_TITULO_DV_P5 REGEX DECOMPOSE	
Matches one pattern and gives each part of the value its own action; when none matches: keep.	
Used by: BR_TITULO_DV	
PATTERNS, TRIED IN ORDER	
(.{8})(.{5})(.*)	keep · BR_TITULO_DV_T5 · keep
otherwise	keep
CONFIGURATION	
trimInput	yes
requireMask	no
BR_TITULO_DV_T1 DATA CLEANSING	
Replaces each value found in br-titulo-dv-1.txt (100 pairs) by its counterpart; values not in the file stay as they are.	
Used by: BR_TITULO_DV_P1	
CONFIGURATION	
lookupFile	br-titulo-dv-1.txt
delimiter	,
caseSensitive	yes
trimWhitespace	yes
BR_TITULO_DV_T2 DATA CLEANSING	
Replaces each value found in br-titulo-dv-2.txt (1,100 pairs) by its counterpart; values not in the file stay as they are.	
Used by: BR_TITULO_DV_P2	
CONFIGURATION	
lookupFile	br-titulo-dv-2.txt
delimiter	,
caseSensitive	yes
trimWhitespace	yes

BR_TITULO_DV_T3 DATA CLEANSING

Replaces each value found in `br-titulo-dv-3.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_TITULO_DV_P3`

CONFIGURATION

lookupFile	<code>br-titulo-dv-3.txt</code>
delimiter	,
caseSensitive	yes
trimWhitespace	yes

BR_TITULO_DV_T4 DATA CLEANSING

Replaces each value found in `br-titulo-dv-4.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_TITULO_DV_P4`

CONFIGURATION

lookupFile	<code>br-titulo-dv-4.txt</code>
delimiter	,
caseSensitive	yes
trimWhitespace	yes

BR_TITULO_DV_T5 DATA CLEANSING

Replaces each value found in `br-titulo-dv-5.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_TITULO_DV_P5`

CONFIGURATION

lookupFile	<code>br-titulo-dv-5.txt</code>
delimiter	,
caseSensitive	yes
trimWhitespace	yes

BR_TITULO_VALIDO STRING ALGORITHM CHAIN

Applies `BR_TITULO_CORPO` → `BR_TITULO_DV` , in this order.

Used by: `BR_TITULO_VALIDO_1_ZERO` , `BR_TITULO_VALIDO_2_ZEROS` , `BR_TITULO`

CONFIGURATION

algorithmReferences[1]	<code>BR_TITULO_CORPO</code>
algorithmReferences[2]	<code>BR_TITULO_DV</code>

BR_TITULO_VALIDO_1_ZERO STRING ALGORITHM CHAIN

Applies `BR_TITULO_1_ZERO` → `BR_TITULO_VALIDO` , in this order.

Used by: `BR_TITULO`

CONFIGURATION

algorithmReferences[1]	<code>BR_TITULO_1_ZERO</code>
algorithmReferences[2]	<code>BR_TITULO_VALIDO</code>

BR_TITULO_VALIDO_2_ZEROS STRING ALGORITHM CHAIN

Applies BR_TITULO_2_ZEROS → BR_TITULO_VALIDO , in this order.

Used by: BR_TITULO

CONFIGURATION

algorithmReferences[1]	BR_TITULO_2_ZEROS
algorithmReferences[2]	BR_TITULO_VALIDO

BR_VALOR_BENEFICIO NUMERIC MAPPING

Maps the number to another between 1,500 and 8,000.

Used by: BR_FIN_VALOR_BENEFICIO

CONFIGURATION

minValue	1500
maxValue	8000

BR_VALOR_DIVIDA NUMERIC MAPPING

Maps the number to another between 200 and 1,000,000.

Used by: BR_FIN_DIVIDA

CONFIGURATION

minValue	200
maxValue	1000000

APPENDIX B

Files

The lookup and detection files the set reads. On a Masking Engine each one is uploaded with the object that reads it — the lookup file of a Secure Lookup or Data Cleansing algorithm, the value list of a List classifier — and referenced by that object's configuration.

FILE	LINES	USED BY	FIRST LINES
br- antecedentes.txt	8	BR_ANTECEDENTES_LISTA	Nada consta · Consta registro · Indiciado · Denunciado
br-aptidao.txt	4	BR_SAUDE_OCUPACIONAL_LISTA	Apto · Apto com restrições · Inapto · Não avaliado
br-bairros.txt	40	BR_BAIRRO	Centro · Jardim América · Vila Nova · Jardim Primavera
br-beneficios- inss-codigos.txt	14	BR_BENEFICIO_INSS_CODIGO	21 · 31 · 32 · 36
br-beneficios- inss.txt	11	BR_BENEFICIO_INSS_LISTA	Aposentadoria por idade · Aposentadoria por tempo de contribuição · Aposentadoria especial · Aposentadoria por incapacidade permanente
br-cbo.txt	34	BR_CBO	411005 · 411010 · 521110 · 421125
br-cid10- decimal.txt	33	BR_CID10_DECIMAL	E11.9 · E78.5 · J45.9 · J06.9
br-cid10.txt	37	BR_CID10	I10 · E11 · E78 · J45
br-classes- sociais.txt	6	BR_CLASSE_SOCIAL	A · B1 · B2 · C1
br-cns-dv-1.txt	100	BR_CNS_DV_T1	00,000 · 01,013 · 02,026 · 03,039
br-cns-dv-2.txt	1,100	BR_CNS_DV_T2	000,000 · 001,011 · 002,022 · 003,033
br-cns-dv-3.txt	1,100	BR_CNS_DV_T3	000,000 · 001,01A · 002,029 · 003,038
br-cns-dv-4.txt	1,100	BR_CNS_DV_T4	000,000 · 001,018 · 002,025 · 003,032
br-cns-dv-5.txt	1,100	BR_CNS_DV_T5	000,000 · 001,016 · 002,021 · 003,037
br-cns-dv-6.txt	110	BR_CNS_DV_T6	000000,00000 · 010000,10006 · 020000,20001 · 030000,30007
br-codigos- municipio- generalizados.txt	7,720	BR_CODIGO_MUNICIPIO	1200013,1200450 · 120001,120045 · 1200054,1200104 · 120005,120010
br-condicoes- moradia.txt	6	BR_MORADIA_LISTA	Próprio quitado · Próprio em pagamento · Alugado · Cedido
br-decil-alto.txt	5	BR_DECIL_ALTO	6 · 7 · 8 · 9
br-decil.txt	10	BR_DECIL	1 · 2 · 3 · 4
br- deficiencias.txt	7	BR_DEFICIENCIA_LISTA	Física · Visual · Auditiva · Intelectual
br-detectar- aptidao.txt	6	BR_SAUDE_OCUPACIONAL - List	apto · inapto · apto com restrições · apto com restricoes
br-detectar- beneficios- inss.txt	38	BR_FIN_BENEFICIO_INSS - List	Aposentadoria por idade · Aposentadoria por tempo de contribuição · Aposentadoria por tempo de contribuicao · Aposentadoria especial

FILE	LINES	USED BY	FIRST LINES
br-detectar-codigos-beneficios-inss.txt	14	BR_FIN_BENEFICIO_INSS – List	21 · 31 · 32 · 36
br-detectar-codigos-raca-cor.txt	7	BR_L3_RACA_COR – List	1 · 2 · 3 · 4
br-detectar-deficiencias.txt	29	BR_SAUDE_DEFICIENCIA – List	Física · Física · Visual · Auditiva
br-detectar-escolaridade.txt	32	BR_L2_ESCOLARIDADE – List	Sem instrução · Sem instrucao · Fundamental incompleto · Fundamental completo
br-detectar-estado-civil.txt	23	BR_L2_ESTADO_CIVIL – List	Solteiro(a) · Casado(a) · União estável · Uniao estavel
br-detectar-etnias.txt	63	BR_L3_ETNIA – List	Tikúna · Tikuna · Kokama · Makuxí
br-detectar-faixas-renda.txt	16	BR_FIN_FAIXA_RENDA – List	extrema pobreza · pobreza · baixa renda · acima de meio salário mínimo
br-detectar-flag.txt	8	BR_L3_ETNIA – List BR_FIN_PROGRAMA_SOCIAL – List	s · n · sim · não
br-detectar-identidades-genero.txt	26	BR_L3_IDENTIDADE_GENERO – List	Mulher cisgênero · Mulher cisgenero · Homem cisgênero · Homem cisgenero
br-detectar-medicamentos.txt	61	BR_SAUDE_MEDICAMENTO – List	Losartana potássica · Losartana potassica · Metformina · Omeprazol
br-detectar-moradia.txt	26	BR_FIN_MORADIA – List	Próprio quitado · Proprio quitado · Próprio em pagamento · Proprio em pagamento
br-detectar-municipios.txt	7,431	BR_L2_MUNICIPIO – List	Acrelândia · Acrelandia · Assis Brasil · Brasília
br-detectar-nacionalidades.txt	58	BR_L2_NACIONALIDADE – List	Brasileira · Venezuelana · Haitiana · Boliviana
br-detectar-nomes.txt	1,046	BR_L1_NOME – List BR_L1_NOME_COMPLETO – List	Maria · Ana · Francisca · Júlia
br-detectar-opinioes-politicas.txt	23	BR_L3_OPINIAO_POLITICA – List	Esquerda · Centro-esquerda · Centro · Centro-direita
br-detectar-orientacoes-sexuais.txt	18	BR_L3_ORIENTACAO_SEXUAL – List	Heterossexual · Gay · Lésbica · Lesbica
br-detectar-partidos.txt	81	BR_L3_FILIACAO_PARTIDARIA – List	MDB · Movimento Democrático Brasileiro · Movimento Democratico Brasileiro · PDT
br-detectar-programas-sociais.txt	31	BR_FIN_PROGRAMA_SOCIAL – List	Bolsa Família · Bolsa Familia · Benefício de Prestação Continuada · Beneficio de Prestacao Continuada
br-detectar-raca-cor.txt	23	BR_L3_RACA_COR – List	Branca · Preta · Parda · Amarela

FILE	LINES	USED BY	FIRST LINES
br-detectar-religioes.txt	67	BR_L3_RELIGIAO - List	Católica Apostólica Romana · Catolica Apostolica Romana · Evangélica · Evangelica
br-detectar-sexo.txt	10	BR_L2_SEX0 - List	f · m · feminino · masculino
br-detectar-sindicatos.txt	29	BR_L3_FILIACAO_SINDICAL - List	CUT · Força Sindical · Forca Sindical · UGT
br-detectar-sobrenomes.txt	636	BR_L1_SOBRENOME - List BR_L1_NOME_COMPLETO - List	Silva · Santos · Oliveira · Souza
br-detectar-tipos-sanguineos.txt	18	BR_SAUDE_TIPO_SANGUINEO - List	A+ · A- · B+ · B-
br-diagnosticos.txt	30	BR_DIAGNOSTICO_TEXTO	Hipertensão essencial · Diabetes mellitus tipo 2 · Asma · Lombalgia
br-empregadores.txt	24	BR_EMPREGADOR	Comercial Horizonte Ltda. · Construtora Vale Verde Ltda. · Transportes Rota Sul Ltda. · Supermercados Bom Preço Ltda.
br-enderecos.txt	4,000	BR_ENDERECO	Avenida Juscelino Kubitschek, 1340 - Jardim Brasil · Rua Santa Catarina, 936 · Rua Minas Gerais, 16 · Avenida Beira-Rio, 551, Apto 172
br-escolaridade-codigos.txt	12	BR_ESCOLARIDADE_CODIGO	01 · 02 · 03 · 04
br-escolaridade.txt	10	BR_ESCOLARIDADE_LISTA	Sem instrução · Fundamental incompleto · Fundamental completo · Médio incompleto
br-escolas.txt	12	BR_ESCOLA	Escola Estadual Professora Maria José · Escola Municipal Monteiro Lobato · Colégio Estadual Rui Barbosa · Escola Estadual Castro Alves
br-estado-civil-codigos.txt	5	BR_ESTADO_CIVIL_CODIGO	1 · 2 · 3 · 4
br-estado-civil.txt	6	BR_ESTADO_CIVIL_LISTA	Solteiro(a) · Casado(a) · União estável · Divorciado(a)
br-etnias.txt	30	BR_ETNIA_LISTA	Tikúna · Kokama · Makuxí · Guarani Kaiowá
br-faixas-cadunico.txt	4	BR_FAIXA_CADUNICO	Extrema pobreza · Pobreza · Baixa renda · Acima de meio salário mínimo
br-flag-0-1.txt	2	BR_FLAG_0_1	0 · 1
br-flag-s-n.txt	2	BR_FLAG_S_N	S · N
br-flag-sim-nao.txt	2	BR_FLAG_SIM_NAO	Sim · Não
br-flag-true-false.txt	2	BR_FLAG_TRUE_FALSE	true · false
br-flag-y-n.txt	2	BR_FLAG_Y_N	Y · N
br-flag-yes-no.txt	2	BR_FLAG_YES_NO	Yes · No
br-identidades-genero.txt	8	BR_IDENTIDADE_GENERO_LISTA	Mulher cisgênero · Homem cisgênero · Mulher trans · Homem trans
br-medicamentos.txt	25	BR_MEDICAMENTO_LISTA	Losartana potássica · Metformina · Omeprazol · Sinvastatina
br-municipios-generalizados.txt	55,203	BR_MUNICIPIO	Acrelândia Senador Guiomard · ACRELÂNDIA SENADOR GUIOMARD · Acrelandia Senador Guiomard · ACRELANDIA SENADOR GUIOMARD

FILE	LINES	USED BY	FIRST LINES
br-nacionalidades.txt	16	BR_NACIONALIDADE	Brasileira · Venezuelana · Haitiana · Boliviana
br-nao-informado.txt	1	BR_SUPRIMIR	Não informado
br-nomes.txt	895	BR_NOME	Maria · Ana · Francisca · Júlia
br-ocupacoes.txt	40	BR_OCUPACAO	Auxiliar administrativo · Assistente administrativo · Vendedor de comércio varejista · Operador de caixa
br-opinioes-politicas.txt	7	BR_OPINIAO_POLITICA_LISTA	Esquerda · Centro-esquerda · Centro · Centro-direita
br-orientacoes-sexuais.txt	8	BR_ORIENTACAO_SEXUAL_LISTA	Heterossexual · Gay · Lésbica · Bissexual
br-partidos-nomes.txt	31	BR_PARTIDO_NOME	Movimento Democrático Brasileiro · Partido Democrático Trabalhista · Partido dos Trabalhadores · Partido Comunista do Brasil
br-partidos-numeros.txt	30	BR_PARTIDO_NUMERO	15 · 12 · 13 · 65
br-partidos-siglas.txt	30	BR_PARTIDO_SIGLA	MDB · PDT · PT · PCdoB
br-programas-sociais.txt	9	BR_PROGRAMA_SOCIAL_LISTA	Bolsa Família · Benefício de Prestação Continuada · Auxílio Gás dos Brasileiros · Pé-de-Meia
br-quintil.txt	5	BR_QUINTIL	1 · 2 · 3 · 4
br-raca-cor-codigos.txt	6	BR_RACA_COR_CODIGO	1 · 2 · 3 · 4
br-raca-cor.txt	6	BR_RACA_COR_LISTA	Branca · Preta · Parda · Amarela
br-religioes.txt	13	BR_RELIGIAO_LISTA	Católica Apostólica Romana · Evangélica · Evangélica pentecostal · Evangélica de missão
br-sexo-1-2.txt	2	BR_SEXO_1_2	1 · 2
br-sexo-f-m.txt	2	BR_SEXO_F_M	F · M
br-sexo-fem-masc.txt	2	BR_SEXO_FEM_MASC	Fem · Masc
br-sexo-female-male.txt	2	BR_SEXO_FEMALE_MALE	Female · Male
br-sexo-feminino-masculino.txt	2	BR_SEXO_FEMININO_MASCULINO	Feminino · Masculino
br-sexo-mulher-homem.txt	2	BR_SEXO_MULHER_HOMEM	Mulher · Homem
br-situacoes-credito.txt	10	BR_SITUACAO_CREDITO	Em dia · Atraso de 30 dias · Atraso de 60 dias · Atraso de 90 dias
br-sobrenomes.txt	535	BR_SOBRENOME	Silva · Santos · Oliveira · Souza
br-texto-nomes.txt	3,302	BR_TEXTO_LIVRE	Maria · MARIA · Ana · ANA
br-tipos-sanguineos.txt	8	BR_TIPO_SANGUINEO	A+ · A- · B+ · B-
br-titulo-dv-1.txt	100	BR_TITULO_DV_T1	00,000 · 01,013 · 02,026 · 03,039

FILE	LINES	USED BY	FIRST LINES
br-titulo-dv-2.txt	1,100	BR_TITULO_DV_T2	000,000 · 001,015 · 002,02A · 003,034
br-titulo-dv-3.txt	1,100	BR_TITULO_DV_T3	000,000 · 001,017 · 002,023 · 003,03A
br-titulo-dv-4.txt	1,100	BR_TITULO_DV_T4	000,000 · 001,019 · 002,027 · 003,035
br-titulo-dv-5.txt	1,100	BR_TITULO_DV_T5	00000,0000 · 00100,0116 · 00200,0213 · 00300,0302
br-zeros-1.txt	10	BR_TITULO_1_ZERO_TABEL A	0,00 · 1,01 · 2,02 · 3,03
br-zeros-2.txt	10	BR_TITULO_2_ZEROS_TABEL A	0,000 · 1,001 · 2,002 · 3,003

Exclusion lists

Words that keep a column-name rule from matching when they follow it — so that `NOMBRE_PRODUCTO` is not taken for a person's name.

E1 produto prod item artigo empresa emp fantasia razao social comercial arquivo arq rua logradouro lograd bairro municipio mun cidade estado uf pais banco agencia plano campanha projeto servico tabela coluna campo usuario host servidor dominio marca modelo categoria tipo escola colegio instituicao curso disciplina materia documento doc centro unidade clinica hospital area departamento depto setor cargo funcao sistema app aplicacao grupo lista relatorio processo evento tarefa perfil papel classe objeto conta medicamento remedio diagnostico estudo programa beneficio partido sindicato religiao etnia povo idioma lingua lugar local loja fornecedor forn convenio contrato apolice seguradora operadora org organizacao orgao entidade parametro variavel metodo indice job script fila recurso rede dispositivo template imagem foto pagina site modulo menu opcao acao permissao imposto tributo moeda emissor nota fatura boleto pagamento formulario zona rota linha predio edificio tela chave tag rotulo titulo codigo config empregador imovel filial matriz loja evento exame procedimento vacina turma sala equipe time projeto natureza motivo situacao status fase etapa origem destino canal campanha arquivo anexo file table column field user server product company brand category type school course document department system group report role account drug place store supplier contract policy device image page module action key label code org entity project city country street bank cliente cli paciente pac funcionario func empregado colaborador servidor segurado beneficiario benef titular dependente dep mae pai responsavel resp conjugue companheiro contato avalista fiador devedor tomador requerente solicitante testemunha representante procurador proprietario locatario locador inquilino vitima reu autor acusado investigado doador herdeiro pessoa pes motorista condutor cidadao eleitor contribuinte aluno estudante crianca medico enfermeiro professor socio tutor curador pagador favorecido destinatario remetente candidato associado membro cooperado consumidor comprador corretor interessado parte advogado usuario atendente vendedor gerente supervisor avo filho filha irmao parente customer patient employee member holder mother father spouse contact person student driver owner completo social civil sobrenome ultimo abreviado reduzido guerra usuario

E2 produto prod item artigo empresa emp fantasia razao social comercial arquivo arq rua logradouro
lograd bairro municipio mun cidade estado uf pais banco agencia plano campanha projeto servico
tabela coluna campo usuario host servidor dominio marca modelo categoria tipo escola colegio
instituicao curso disciplina materia documento doc centro unidade clinica hospital area
departamento depto setor cargo funcao sistema app aplicacao grupo lista relatorio processo evento
tarefa perfil papel classe objeto conta medicamento remedio diagnostico estudo programa beneficio
partido sindicato religiao etnia povo idioma lingua lugar local loja fornecedor forn convenio
contrato apolice seguradora operadora org organizacao orgao entidade parametro variavel metodo
indice job script fila recurso rede dispositivo template imagem foto pagina site modulo menu opcao
acao permissao imposto tributo moeda emissor nota fatura boleto pagamento formulario zona rota
linha predio edificio tela chave tag rotulo titulo codigo config empregador imovel filial matriz
loja evento exame procedimento vacina turma sala equipe time projeto natureza motivo situacao
status fase etapa origem destino canal campanha arquivo anexo file table column field user server
product company brand category type school course document department system group report role
account drug place store supplier contract policy device image page module action key label code
org entity project city country street bank cliente cli paciente pac funcionario func empregado
colaborador servidor segurado beneficiario benef titular dependente dep mae pai responsavel resp
conjuge companheiro contato avalista fiador devedor tomador requerente solicitante testemunha
representante procurador proprietario locatario locador inquilino vitima reu autor acusado
investigado doador herdeiro pessoa pes motorista condutor cidadao eleitor contribuinte aluno
estudante crianca medico enfermeiro professor socio tutor curador pagador favorecido destinatario
remetente candidato associado membro cooperado consumidor comprador corretor interessado parte
advogado usuario atendente vendedor gerente supervisor avo filho filha irmao parente customer
patient employee member holder mother father spouse contact person student driver owner

E3 produto prod item artigo empresa emp fantasia razao social comercial arquivo arq rua logradouro
lograd bairro municipio mun cidade estado uf pais banco agencia plano campanha projeto servico
tabela coluna campo usuario host servidor dominio marca modelo categoria tipo escola colegio
instituicao curso disciplina materia documento doc centro unidade clinica hospital area
departamento depto setor cargo funcao sistema app aplicacao grupo lista relatorio processo evento
tarefa perfil papel classe objeto conta medicamento remedio diagnostico estudo programa beneficio
partido sindicato religiao etnia povo idioma lingua lugar local loja fornecedor forn convenio
contrato apolice seguradora operadora org organizacao orgao entidade parametro variavel metodo
indice job script fila recurso rede dispositivo template imagem foto pagina site modulo menu opcao
acao permissao imposto tributo moeda emissor nota fatura boleto pagamento formulario zona rota
linha predio edificio tela chave tag rotulo titulo codigo config empregador imovel filial matriz
loja evento exame procedimento vacina turma sala equipe time projeto natureza motivo situacao
status fase etapa origem destino canal campanha arquivo anexo file table column field user server
product company brand category type school course document department system group report role
account drug place store supplier contract policy device image page module action key label code
org entity project city country street bank cliente cli paciente pac funcionario func empregado
colaborador servidor segurado beneficiario benef titular dependente dep mae pai responsavel resp
conjuge companheiro contato avalista fiador devedor tomador requerente solicitante testemunha
representante procurador proprietario locatario locador inquilino vitima reu autor acusado
investigado doador herdeiro pessoa pes motorista condutor cidadao eleitor contribuinte aluno
estudante crianca medico enfermeiro professor socio tutor curador pagador favorecido destinatario
remetente candidato associado membro cooperado consumidor comprador corretor interessado parte
advogado usuario atendente vendedor gerente supervisor avo filho filha irmao parente customer
patient employee member holder mother father spouse contact person student driver owner full
complete last first

E4 ip mac email e_?mail eletronico web url tipo id cobranca_?id

E5 ip mac email e_mail web url type id

E6 id cd cod codigo tp tipo dt data st status qt nr nu

E7 id tipo codigo data status fila log

APPENDIX D

Profile set and classifiers

The profile set `BR - LGPD - v1` runs these 148 classifiers, with an assignment threshold of 60%.

Identificadores diretos, quase-identificadores, dados pessoais sensíveis (art. 5º, II), dados de saúde, dados financeiros e de crédito, e antecedentes criminais, segundo a Lei Geral de Proteção de Dados Pessoais (Lei 13.709/2018).

CLASSIFIER	KIND	DOMAIN
BR_L1_CPF - Path	COLUMN NAME	BR_L1_CPF
BR_L1_CPF - Type	DATA TYPE	BR_L1_CPF
BR_L1_CPF - Regex	VALUE FORMAT	BR_L1_CPF
BR_L1_CNPJ - Path	COLUMN NAME	BR_L1_CNPJ
BR_L1_CNPJ - Type	DATA TYPE	BR_L1_CNPJ
BR_L1_CNPJ - Regex	VALUE FORMAT	BR_L1_CNPJ
BR_L1_RG - Path	COLUMN NAME	BR_L1_RG
BR_L1_RG - Regex	VALUE FORMAT	BR_L1_RG
BR_L1_CNH - Path	COLUMN NAME	BR_L1_CNH
BR_L1_TITULO_ELEITOR - Path	COLUMN NAME	BR_L1_TITULO_ELEITOR
BR_L1_TITULO_ELEITOR - Regex	VALUE FORMAT	BR_L1_TITULO_ELEITOR
BR_L1_PIS_NIS - Path	COLUMN NAME	BR_L1_PIS_NIS
BR_L1_PIS_NIS - Regex	VALUE FORMAT	BR_L1_PIS_NIS
BR_L1_CTPS - Path	COLUMN NAME	BR_L1_CTPS
BR_L1_PASSAPORTE - Path	COLUMN NAME	BR_L1_PASSAPORTE
BR_L1_PASSAPORTE - Regex	VALUE FORMAT	BR_L1_PASSAPORTE
BR_L1_DOCUMENTO_ESTRANGEIRO - Path	COLUMN NAME	BR_L1_DOCUMENTO_ESTRANGEIRO
BR_L1_CERTIDAO - Path	COLUMN NAME	BR_L1_CERTIDAO
BR_L1_CERTIDAO - Regex	VALUE FORMAT	BR_L1_CERTIDAO
BR_L1_REGISTRO_PROFISSIONAL - Path	COLUMN NAME	BR_L1_REGISTRO_PROFISSIONAL
BR_L1_NOME - Path	COLUMN NAME	BR_L1_NOME
BR_L1_NOME - List	KNOWN VALUES	BR_L1_NOME
BR_L1_SOBRENOME - Path	COLUMN NAME	BR_L1_SOBRENOME
BR_L1_SOBRENOME - List	KNOWN VALUES	BR_L1_SOBRENOME
BR_L1_NOME_COMPLETO - Path	COLUMN NAME	BR_L1_NOME_COMPLETO
BR_L1_NOME_COMPLETO - List	KNOWN VALUES	BR_L1_NOME_COMPLETO
BR_L1_EMAIL - Path	COLUMN NAME	BR_L1_EMAIL
BR_L1_EMAIL - Type	DATA TYPE	BR_L1_EMAIL
BR_L1_EMAIL - Regex	VALUE FORMAT	BR_L1_EMAIL
BR_L1_TELEFONE - Path	COLUMN NAME	BR_L1_TELEFONE

CLASSIFIER	KIND	DOMAIN
BR_L1_TELEFONE – Regex	VALUE FORMAT	BR_L1_TELEFONE
BR_L1_ENDERECO – Path	COLUMN NAME	BR_L1_ENDERECO
BR_L1_ENDERECO – Type	DATA TYPE	BR_L1_ENDERECO
BR_L1_ENDERECO – Regex	VALUE FORMAT	BR_L1_ENDERECO
BR_L1_ENDERECO_COMPLEMENTO – Path	COLUMN NAME	BR_L1_ENDERECO_COMPLEMENTO
BR_L1_CONTA_BANCARIA – Path	COLUMN NAME	BR_L1_CONTA_BANCARIA
BR_L1_CARTAO – Path	COLUMN NAME	BR_L1_CARTAO
BR_L1_CARTAO – Regex	VALUE FORMAT	BR_L1_CARTAO
BR_L1_CHAVE_PIX – Path	COLUMN NAME	BR_L1_CHAVE_PIX
BR_L1_CHAVE_PIX – Regex	VALUE FORMAT	BR_L1_CHAVE_PIX
BR_L1_IP – Path	COLUMN NAME	BR_L1_IP
BR_L1_IP – Regex	VALUE FORMAT	BR_L1_IP
BR_L1_DISPOSITIVO – Path	COLUMN NAME	BR_L1_DISPOSITIVO
BR_L1_DISPOSITIVO – Regex	VALUE FORMAT	BR_L1_DISPOSITIVO
BR_L1_PLACA – Path	COLUMN NAME	BR_L1_PLACA
BR_L1_PLACA – Regex	VALUE FORMAT	BR_L1_PLACA
BR_L1_RENAVAM – Path	COLUMN NAME	BR_L1_RENAVAM
BR_L1_CHASSI – Path	COLUMN NAME	BR_L1_CHASSI
BR_L1_CHASSI – Regex	VALUE FORMAT	BR_L1_CHASSI
BR_L1_IMOVEL – Path	COLUMN NAME	BR_L1_IMOVEL
BR_L1_CONTRATO – Path	COLUMN NAME	BR_L1_CONTRATO
BR_L1_USUARIO – Path	COLUMN NAME	BR_L1_USUARIO
BR_L1_CREDENCIAL – Path	COLUMN NAME	BR_L1_CREDENCIAL
BR_L1_GEOLOCALIZACAO – Path	COLUMN NAME	BR_L1_GEOLOCALIZACAO
BR_L1_GEOLOCALIZACAO – Regex	VALUE FORMAT	BR_L1_GEOLOCALIZACAO
BR_L1_TEXTO_LIVRE – Path	COLUMN NAME	BR_L1_TEXTO_LIVRE
BR_L1_TEXTO_LIVRE – Type	DATA TYPE	BR_L1_TEXTO_LIVRE
BR_L1_TEXTO_LIVRE – Regex	VALUE FORMAT	BR_L1_TEXTO_LIVRE
BR_L2_DATA_NASCIMENTO – Path	COLUMN NAME	BR_L2_DATA_NASCIMENTO
BR_L2_DATA_NASCIMENTO – Type	DATA TYPE	BR_L2_DATA_NASCIMENTO
BR_L2_ANO_NASCIMENTO – Path	COLUMN NAME	BR_L2_ANO_NASCIMENTO
BR_L2_ANO_NASCIMENTO – Type	DATA TYPE	BR_L2_ANO_NASCIMENTO
BR_L2_IDADE – Path	COLUMN NAME	BR_L2_IDADE
BR_L2_IDADE – Type	DATA TYPE	BR_L2_IDADE
BR_L2_DATA_EVENTO – Path	COLUMN NAME	BR_L2_DATA_EVENTO
BR_L2_DATA_EVENTO – Type	DATA TYPE	BR_L2_DATA_EVENTO
BR_L2_SEX0 – Path	COLUMN NAME	BR_L2_SEX0

CLASSIFIER	KIND	DOMAIN
BR_L2_SEX0 - List	KNOWN VALUES	BR_L2_SEX0
BR_L2_MUNICIPIO - Path	COLUMN NAME	BR_L2_MUNICIPIO
BR_L2_MUNICIPIO - List	KNOWN VALUES	BR_L2_MUNICIPIO
BR_L2_CODIGO_MUNICIPIO - Path	COLUMN NAME	BR_L2_CODIGO_MUNICIPIO
BR_L2_CODIGO_MUNICIPIO - Regex	VALUE FORMAT	BR_L2_CODIGO_MUNICIPIO
BR_L2_CEP - Path	COLUMN NAME	BR_L2_CEP
BR_L2_CEP - Regex	VALUE FORMAT	BR_L2_CEP
BR_L2_BAIRRO - Path	COLUMN NAME	BR_L2_BAIRRO
BR_L2_ZONA_SECAO_ELEITORAL - Path	COLUMN NAME	BR_L2_ZONA_SECAO_ELEITORAL
BR_L2_NACIONALIDADE - Path	COLUMN NAME	BR_L2_NACIONALIDADE
BR_L2_NACIONALIDADE - List	KNOWN VALUES	BR_L2_NACIONALIDADE
BR_L2_ESTADO_CIVIL - Path	COLUMN NAME	BR_L2_ESTADO_CIVIL
BR_L2_ESTADO_CIVIL - List	KNOWN VALUES	BR_L2_ESTADO_CIVIL
BR_L2_OCUPACAO - Path	COLUMN NAME	BR_L2_OCUPACAO
BR_L2_EMPREGADOR - Path	COLUMN NAME	BR_L2_EMPREGADOR
BR_L2_ESCOLARIDADE - Path	COLUMN NAME	BR_L2_ESCOLARIDADE
BR_L2_ESCOLARIDADE - List	KNOWN VALUES	BR_L2_ESCOLARIDADE
BR_L2_ESCOLA - Path	COLUMN NAME	BR_L2_ESCOLA
BR_L2_DEPENDENTES - Path	COLUMN NAME	BR_L2_DEPENDENTES
BR_L2_DEPENDENTES - Type	DATA TYPE	BR_L2_DEPENDENTES
BR_L3_RACA_COR - Path	COLUMN NAME	BR_L3_RACA_COR
BR_L3_RACA_COR - List	KNOWN VALUES	BR_L3_RACA_COR
BR_L3_ETNIA - Path	COLUMN NAME	BR_L3_ETNIA
BR_L3_ETNIA - List	KNOWN VALUES	BR_L3_ETNIA
BR_L3_RELIGIAO - Path	COLUMN NAME	BR_L3_RELIGIAO
BR_L3_RELIGIAO - List	KNOWN VALUES	BR_L3_RELIGIAO
BR_L3_CONVICCAO_ORGANIZACAO - Path	COLUMN NAME	BR_L3_CONVICCAO_ORGANIZACAO
BR_L3_OPINIAO_POLITICA - Path	COLUMN NAME	BR_L3_OPINIAO_POLITICA
BR_L3_OPINIAO_POLITICA - List	KNOWN VALUES	BR_L3_OPINIAO_POLITICA
BR_L3_FILIACAO_PARTIDARIA - Path	COLUMN NAME	BR_L3_FILIACAO_PARTIDARIA
BR_L3_FILIACAO_PARTIDARIA - List	KNOWN VALUES	BR_L3_FILIACAO_PARTIDARIA
BR_L3_FILIACAO_SINDICAL - Path	COLUMN NAME	BR_L3_FILIACAO_SINDICAL
BR_L3_FILIACAO_SINDICAL - List	KNOWN VALUES	BR_L3_FILIACAO_SINDICAL
BR_L3_GENETICO - Path	COLUMN NAME	BR_L3_GENETICO
BR_L3_BIOMETRICO - Path	COLUMN NAME	BR_L3_BIOMETRICO
BR_L3_BIOMETRICO - Type	DATA TYPE	BR_L3_BIOMETRICO
BR_L3_ORIENTACAO_SEXUAL - Path	COLUMN NAME	BR_L3_ORIENTACAO_SEXUAL

CLASSIFIER	KIND	DOMAIN
BR_L3_ORIENTACAO_SEXUAL - List	KNOWN VALUES	BR_L3_ORIENTACAO_SEXUAL
BR_L3_VIDA_SEXUAL - Path	COLUMN NAME	BR_L3_VIDA_SEXUAL
BR_L3_IDENTIDADE_GENERO - Path	COLUMN NAME	BR_L3_IDENTIDADE_GENERO
BR_L3_IDENTIDADE_GENERO - List	KNOWN VALUES	BR_L3_IDENTIDADE_GENERO
BR_SAUDE_CNS - Path	COLUMN NAME	BR_SAUDE_CNS
BR_SAUDE_CNS - Regex	VALUE FORMAT	BR_SAUDE_CNS
BR_SAUDE_IDENTIFICADOR - Path	COLUMN NAME	BR_SAUDE_IDENTIFICADOR
BR_SAUDE_DIAGNOSTICO - Path	COLUMN NAME	BR_SAUDE_DIAGNOSTICO
BR_SAUDE_DIAGNOSTICO - Regex	VALUE FORMAT	BR_SAUDE_DIAGNOSTICO
BR_SAUDE_MEDICAMENTO - Path	COLUMN NAME	BR_SAUDE_MEDICAMENTO
BR_SAUDE_MEDICAMENTO - List	KNOWN VALUES	BR_SAUDE_MEDICAMENTO
BR_SAUDE_TEXTO_CLINICO - Path	COLUMN NAME	BR_SAUDE_TEXTO_CLINICO
BR_SAUDE_TEXTO_CLINICO - Type	DATA TYPE	BR_SAUDE_TEXTO_CLINICO
BR_SAUDE_RESULTADO_EXAME - Path	COLUMN NAME	BR_SAUDE_RESULTADO_EXAME
BR_SAUDE_TIPO_SANGUINEO - Path	COLUMN NAME	BR_SAUDE_TIPO_SANGUINEO
BR_SAUDE_TIPO_SANGUINEO - List	KNOWN VALUES	BR_SAUDE_TIPO_SANGUINEO
BR_SAUDE_DEFICIENCIA - Path	COLUMN NAME	BR_SAUDE_DEFICIENCIA
BR_SAUDE_DEFICIENCIA - List	KNOWN VALUES	BR_SAUDE_DEFICIENCIA
BR_SAUDE_OCUPACIONAL - Path	COLUMN NAME	BR_SAUDE_OCUPACIONAL
BR_SAUDE_OCUPACIONAL - List	KNOWN VALUES	BR_SAUDE_OCUPACIONAL
BR_SAUDE_REPRODUTIVA - Path	COLUMN NAME	BR_SAUDE_REPRODUTIVA
BR_SAUDE_MENTAL - Path	COLUMN NAME	BR_SAUDE_MENTAL
BR_FIN_CREDITO - Path	COLUMN NAME	BR_FIN_CREDITO
BR_FIN_DIVIDA - Path	COLUMN NAME	BR_FIN_DIVIDA
BR_FIN_DIVIDA - Type	DATA TYPE	BR_FIN_DIVIDA
BR_FIN_RENDA - Path	COLUMN NAME	BR_FIN_RENDA
BR_FIN_RENDA - Type	DATA TYPE	BR_FIN_RENDA
BR_FIN_PATRIMONIO - Path	COLUMN NAME	BR_FIN_PATRIMONIO
BR_FIN_PATRIMONIO - Type	DATA TYPE	BR_FIN_PATRIMONIO
BR_FIN_VALOR_BENEFICIO - Path	COLUMN NAME	BR_FIN_VALOR_BENEFICIO
BR_FIN_VALOR_BENEFICIO - Type	DATA TYPE	BR_FIN_VALOR_BENEFICIO
BR_FIN_BENEFICIO_INSS - Path	COLUMN NAME	BR_FIN_BENEFICIO_INSS
BR_FIN_BENEFICIO_INSS - List	KNOWN VALUES	BR_FIN_BENEFICIO_INSS
BR_FIN_NUMERO_BENEFICIO - Path	COLUMN NAME	BR_FIN_NUMERO_BENEFICIO
BR_FIN_NUMERO_BENEFICIO - Regex	VALUE FORMAT	BR_FIN_NUMERO_BENEFICIO
BR_FIN_PROGRAMA_SOCIAL - Path	COLUMN NAME	BR_FIN_PROGRAMA_SOCIAL
BR_FIN_PROGRAMA_SOCIAL - List	KNOWN VALUES	BR_FIN_PROGRAMA_SOCIAL

CLASSIFIER	KIND	DOMAIN
BR_FIN_FAIXA_REnda - Path	COLUMN NAME	BR_FIN_FAIXA_REnda
BR_FIN_FAIXA_REnda - List	KNOWN VALUES	BR_FIN_FAIXA_REnda
BR_FIN_MORADIA - Path	COLUMN NAME	BR_FIN_MORADIA
BR_FIN_MORADIA - List	KNOWN VALUES	BR_FIN_MORADIA
BR_PENAL_ANTECEDENTES - Path	COLUMN NAME	BR_PENAL_ANTECEDENTES
BR_PENAL_PROCESSO - Path	COLUMN NAME	BR_PENAL_PROCESSO
BR_PENAL_PROCESSO - Regex	VALUE FORMAT	BR_PENAL_PROCESSO